The working (6,7): a complete named derivation

90 graphical steps | Four explicit sphere moves | Portrait US Letter

VERY BRIEF SUMMARY. Remove the incident half twists, use a five-point disk, turn the remaining positive factors into FOUR exterior twists by PC lanterns, roses, daisies and Hurwitz transports, then cap their fourth power.

The previously missing geometric boundary closure is now supplied. The final equality is obtained from actual named substitutions and factor transports, rather than a whole-product action recognition. All four sphere moves retain their disk discrepancies, exact sphere certificates and separately checked genus-two lift.

What is still open. This detailed route has 90 retained graphical steps: one fewer than the previous91-step proof and eleven fewer than the101-step version and shorter than the earlier 177-step archive. It does not yet meet the goal of fewer than twenty meaningful operations. Reusable named patterns and fixed-factor drags are identified, with all adjacent checks retained in the certificates.

How to read it. Read each factorization downward. Curves are paired with the braid representing that exact support; blue marked-point dots continue at braid separators. The unnumbered support points use the same size and spacing in every factor. Only the changed interval is printed at a move, with the unchanged factors counted above and below. Every actual SPHERE MOVE has a yellow banner. Individual supports are not cumulative joint cut-system certificates.

Logical scope. Common frames and cyclic starts are explicit conjugations. The terminal four-exterior product is central, so reversing those conjugations recovers the source equality without assuming it was central. Marked points are not framed boundaries; signed framing balances are retained for PC substitutions. A single five-point cap has lift -I; the sixth- and fourth-power caps used here have +I.

Contents

Full diagrams, named sub-derivations and sphere discrepancies.

Complete named diagram sequence

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90 graphical steps retained. Four sphere moves labeled; full discrepancies in Section IX.

What the steps will do

1. Reach the five-point chart. Split the required squares, match and absorb the incident half twists, change the exterior point explicitly, and retain then separately cap the six framing corrections.

2. Use the existing roots. Borrow Q14 from the actual four-point daisy, transport it to match the original half twists, return the same Q14 and remove the last half pair. A cyclic start after the square and outer bypasses a standalone commutation; the short-side drag crosses that outer and later rejoins the same Q15 state. No auxiliary half twists are introduced.

3. Extract two exterior twists. Actual cabled lanterns and borrowed-petal configurations expose two visible exterior twists. Keep both while closing the residual factors.

4. Return the loan as six actual pairs. Use the positive T45 common frame, commute the central exterior and transport the same outer upward by inverse Hurwitz moves. The actual six pairs match the expanded PC rose without a preliminary petal reorder or a temporary three-point root.

5. Carry the first outer upward. Choose the cyclic cut before gathering the first daisy, then carry the same outer UP through its petals. This changes the actual route and avoids dragging it through the second outer. Return the extra square unit; all five inputs match the transported PC daisy.

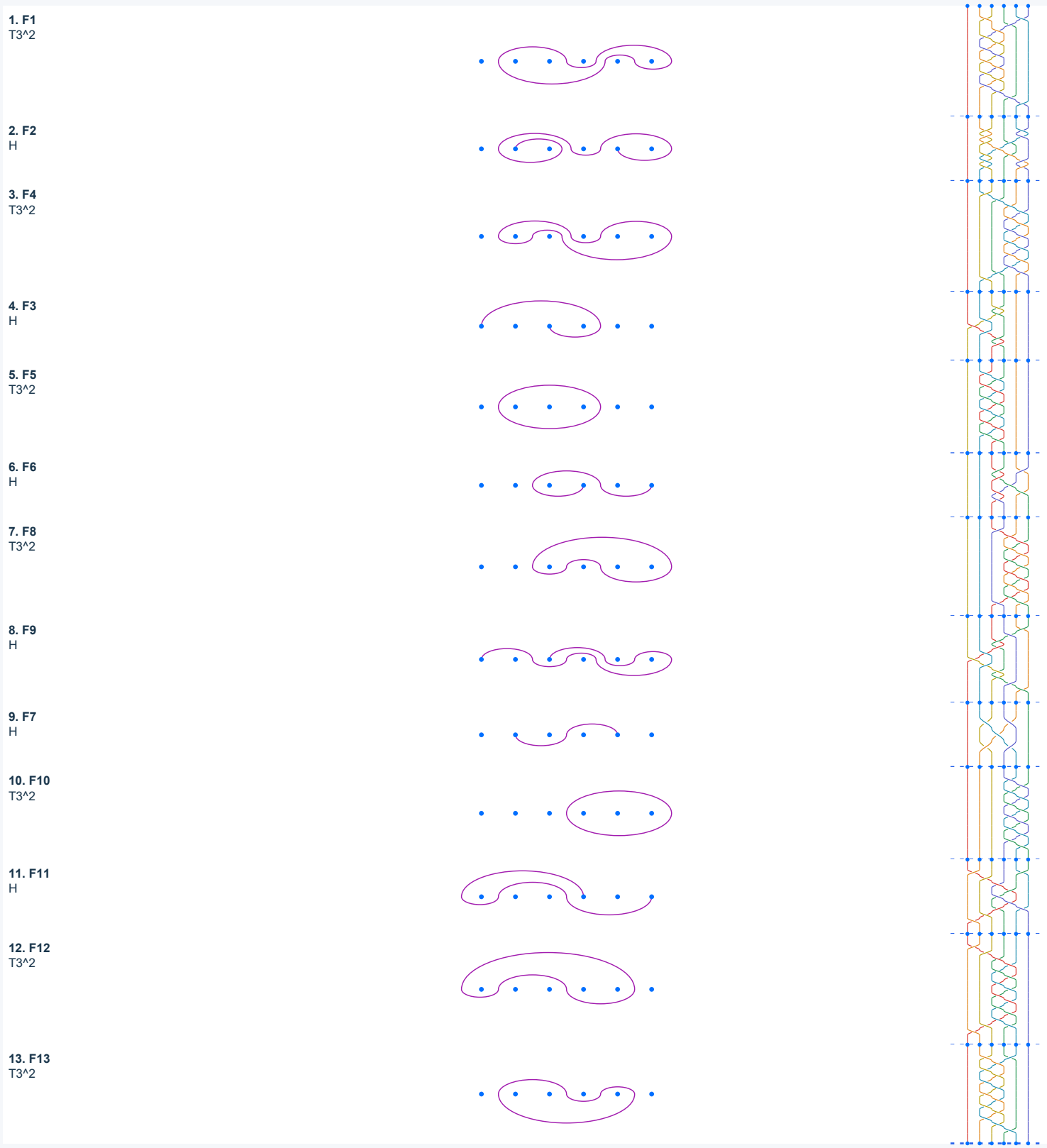
6. Close the second daisy in its cyclic order. Start at the new exterior, move the extra petal past only two central exteriors, and match the actual cyclic PC daisy. All individual inputs, seven internal interchanges, the internal cyclic conjugation and complete nonabelian kernels are retained. Four exterior twists remain visible, then combine and undergo the separate fourth-power sphere cap with +I lift.

I. The fixed-point boundary chart

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before The fixed-point boundary chart

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.



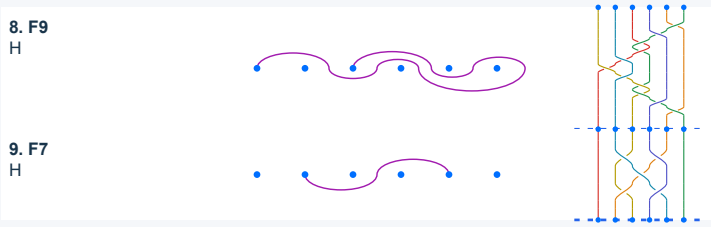
I - step 001

Change the working ordering by an inverse Hurwitz move

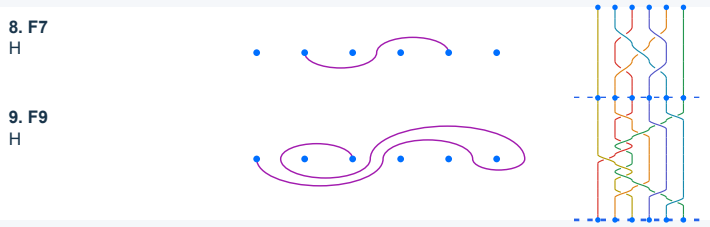
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



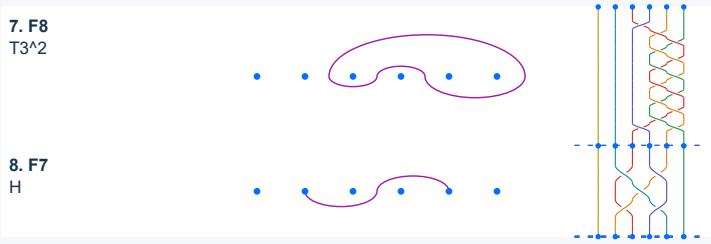
I - step 002

Change the working ordering by an inverse Hurwitz move

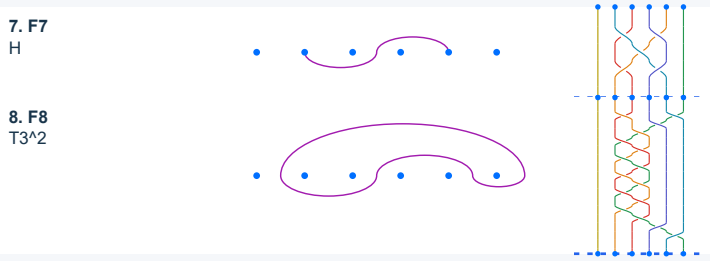
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



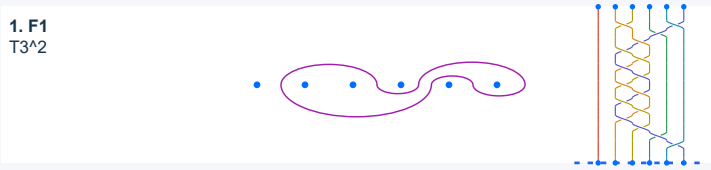
I - step 003

Split F1 into ordinary roots

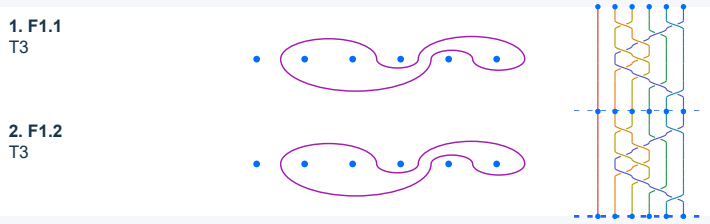
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



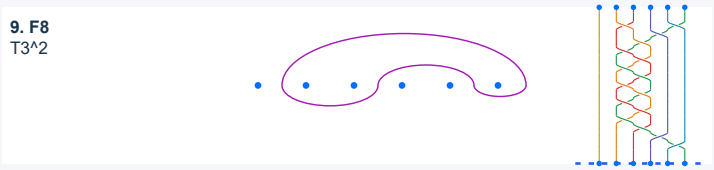
I - step 004

Split F8 into ordinary roots

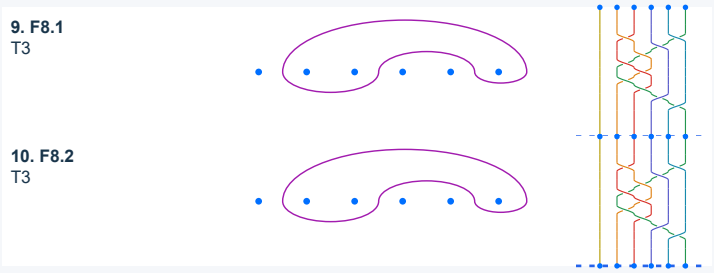
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



I - step 005

Split F10 into ordinary roots

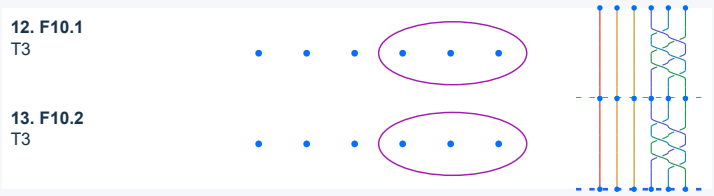
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



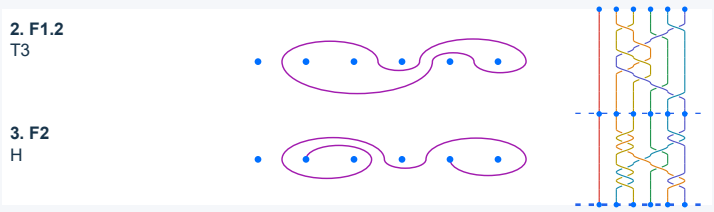
I - step 006

F1.2 down past F2

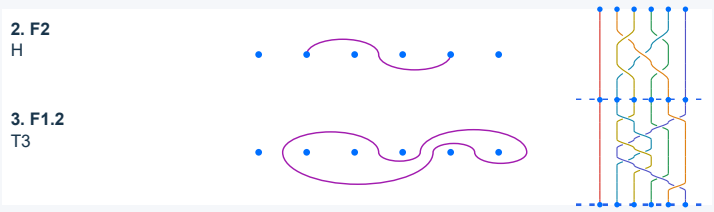
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



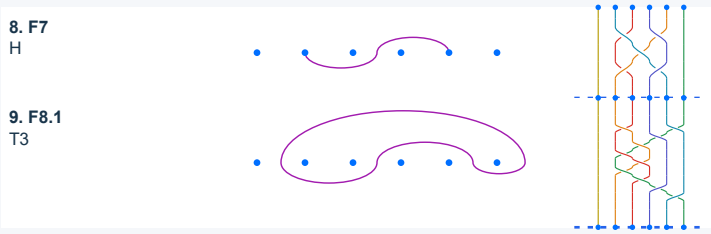
I - step 007

F8.1 up past F7

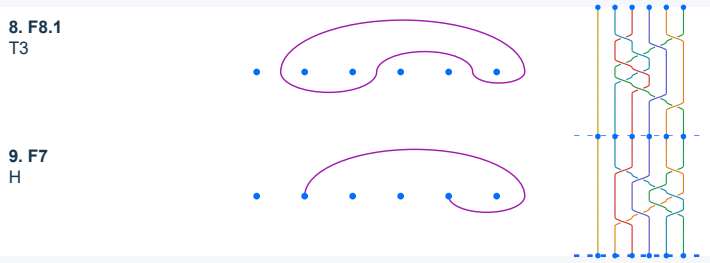
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



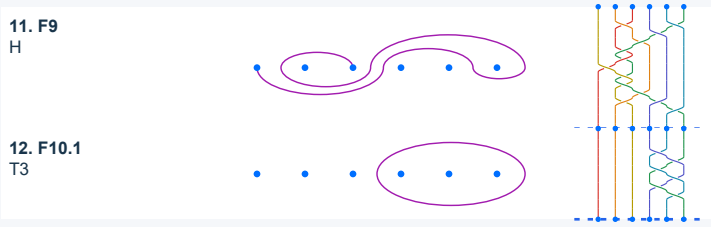
I - step 008

F10.1 up past F9

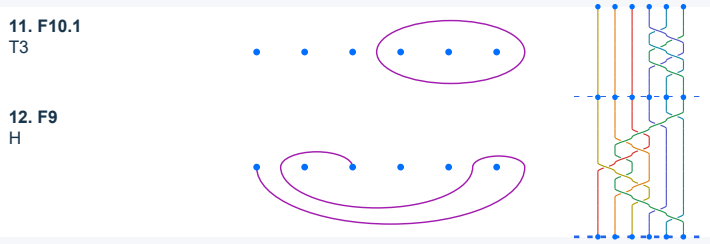
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



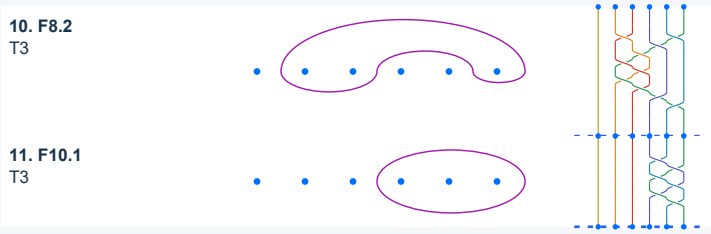
I - step 009

F10.1 up past F8.2

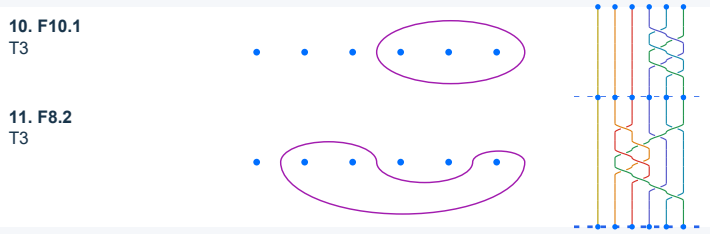
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



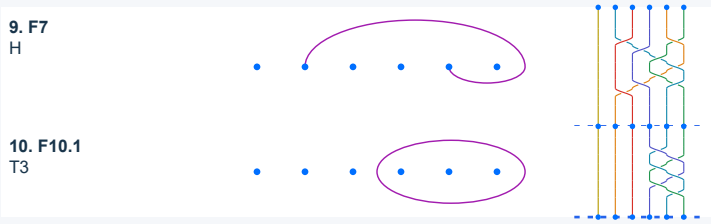
I - step 010

F10.1 up past F7

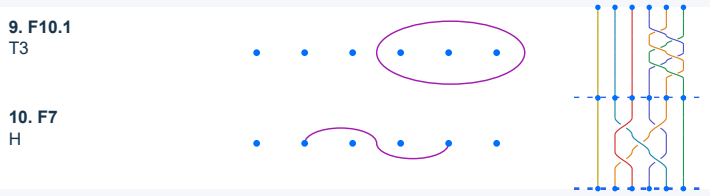
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



I - step 011

Use the exact disk equality of the matched arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



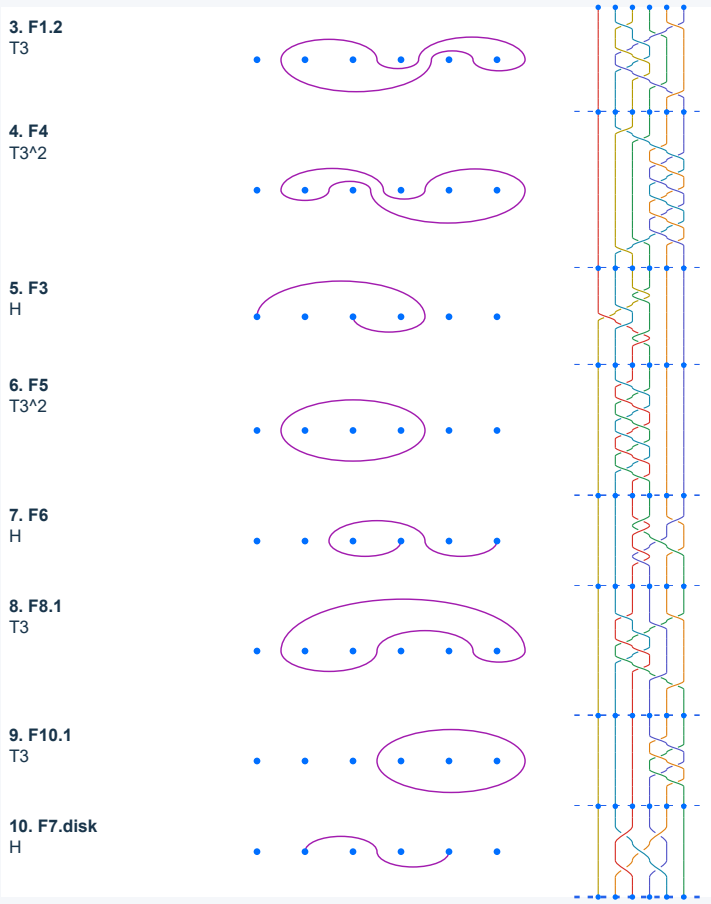
I - step 012

Drag the matched arc up across SEVEN factors (one Hurwitz drag)

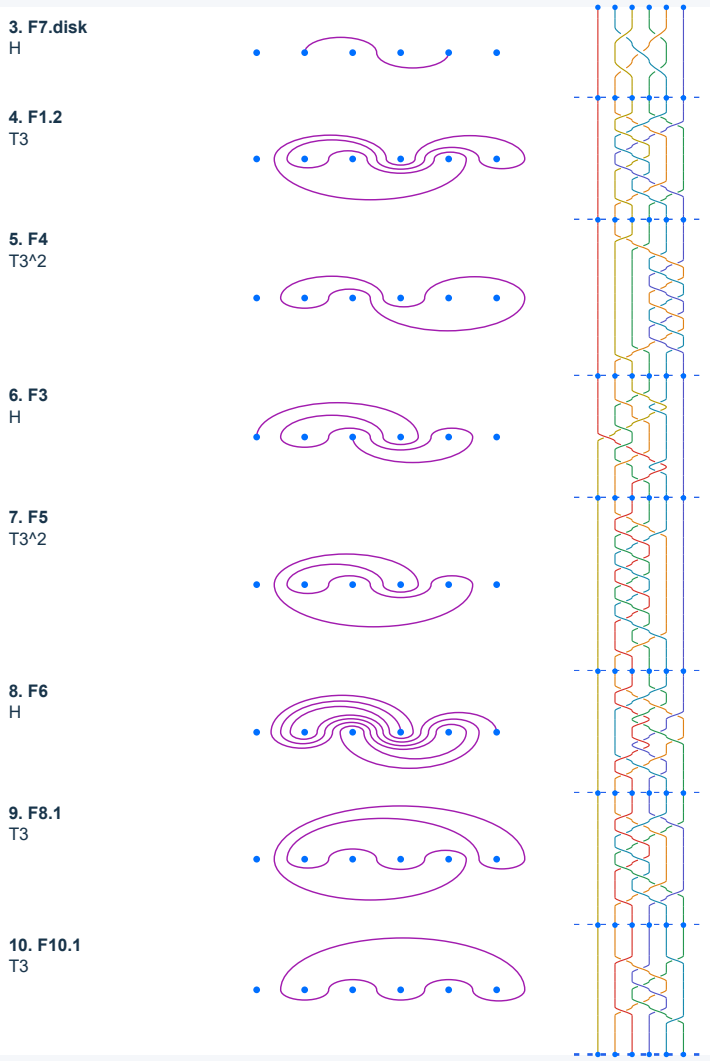
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 7.

Before



After



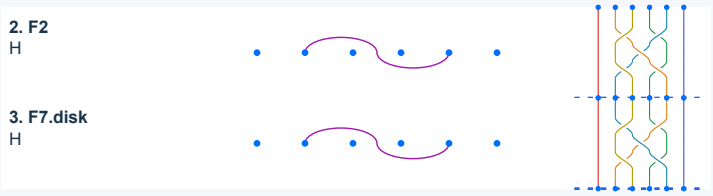
I - step 013

Combine the two matched half twists

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



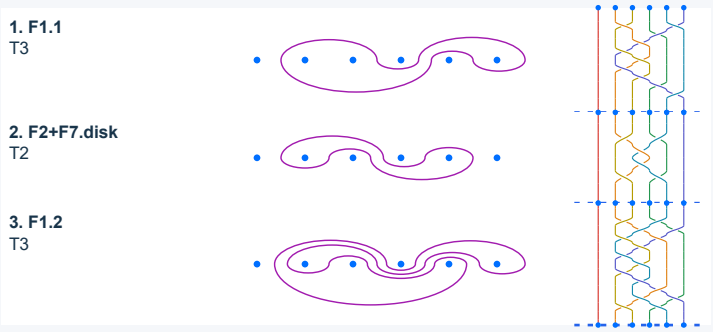
I - step 014

SPHERE LANTERN: absorb the matched pair and the two prefix roots

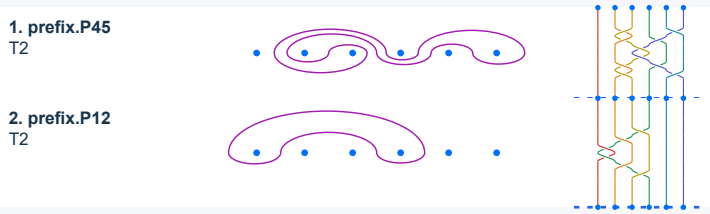
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



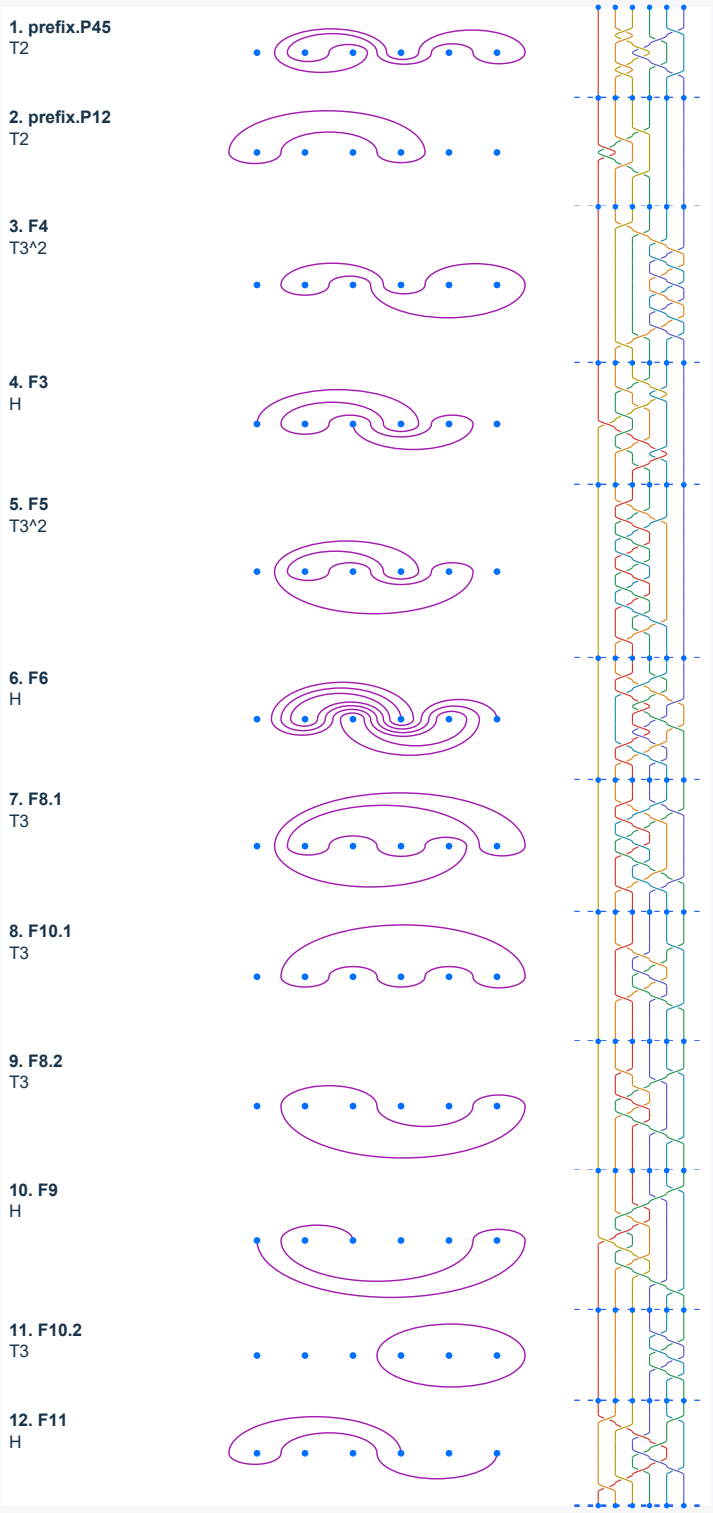
I - step 015

Global frame: fixed working point 2 becomes displayed point 6

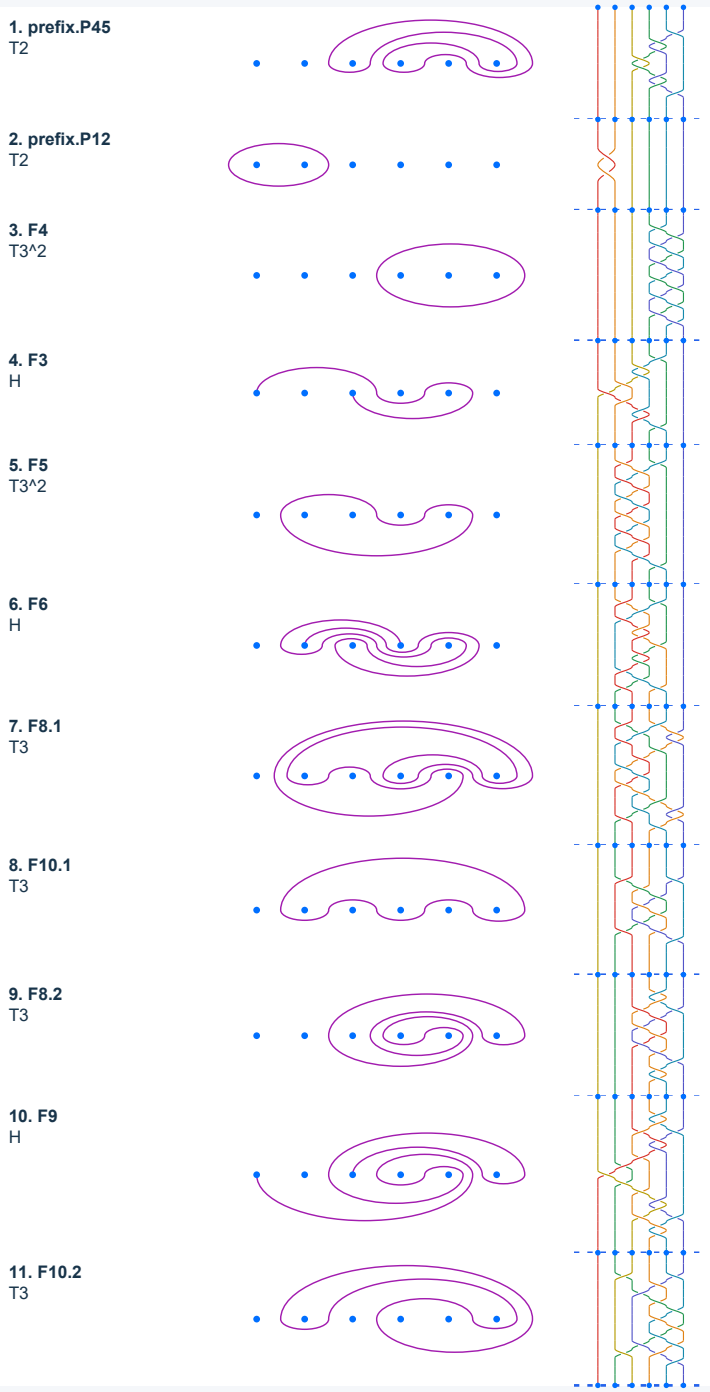
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



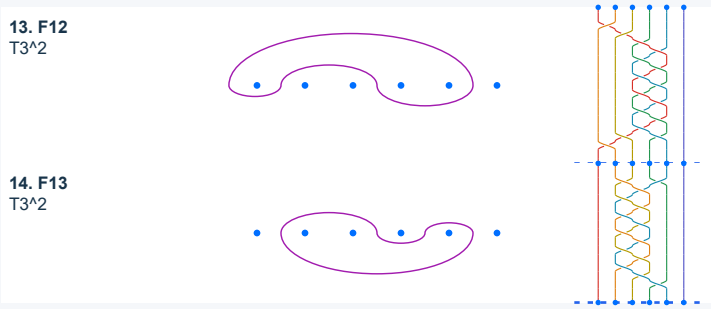
I - step 015 - continued

Global frame: fixed working point 2 becomes displayed point 6

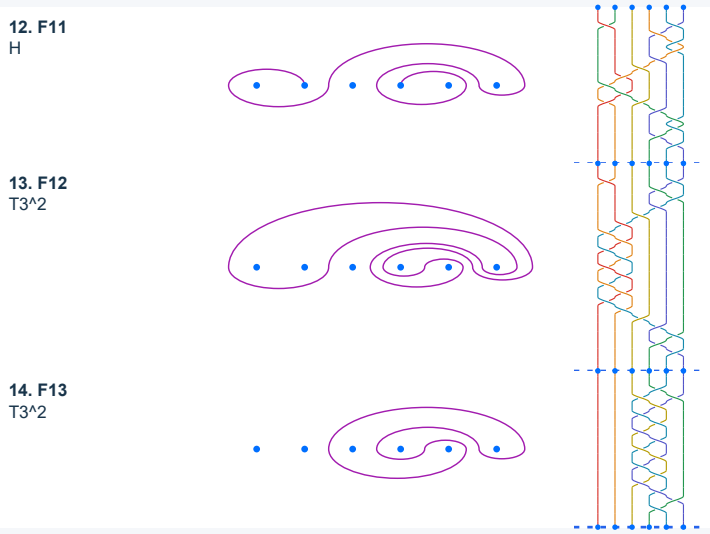
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



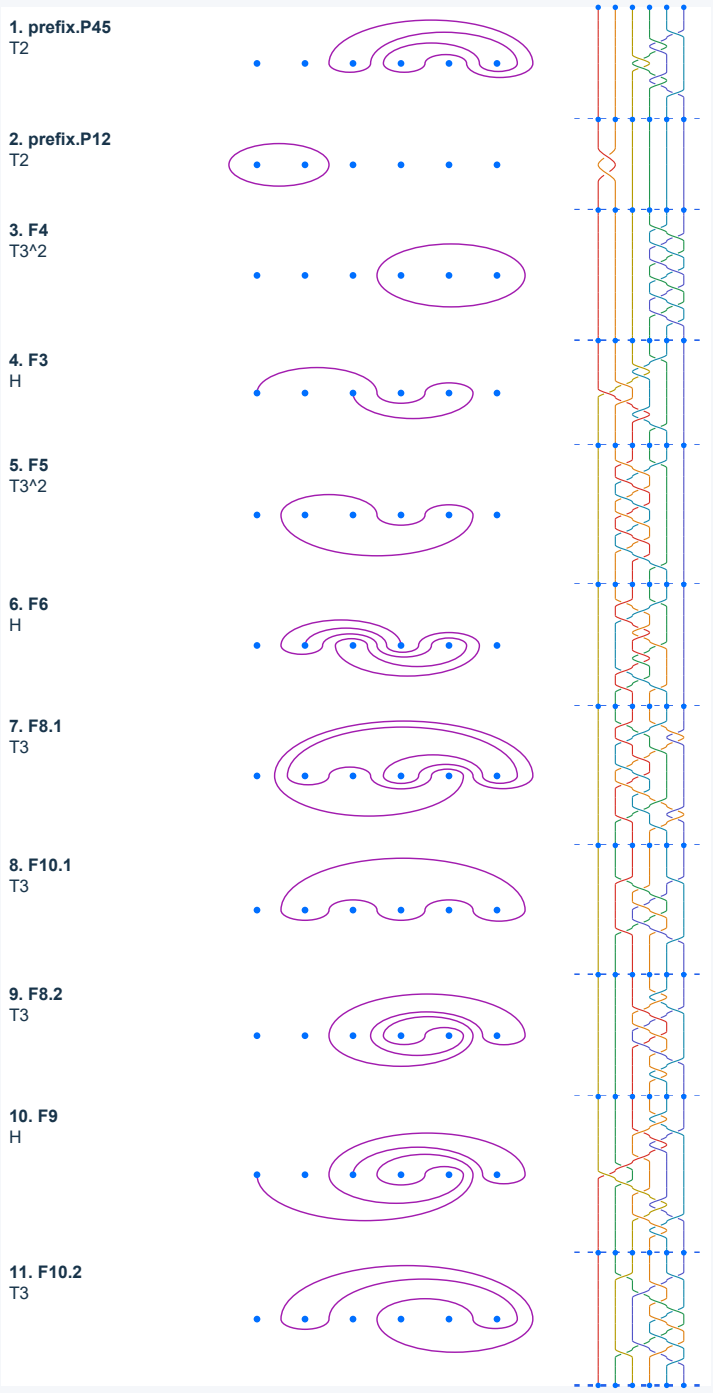
I - step 016

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

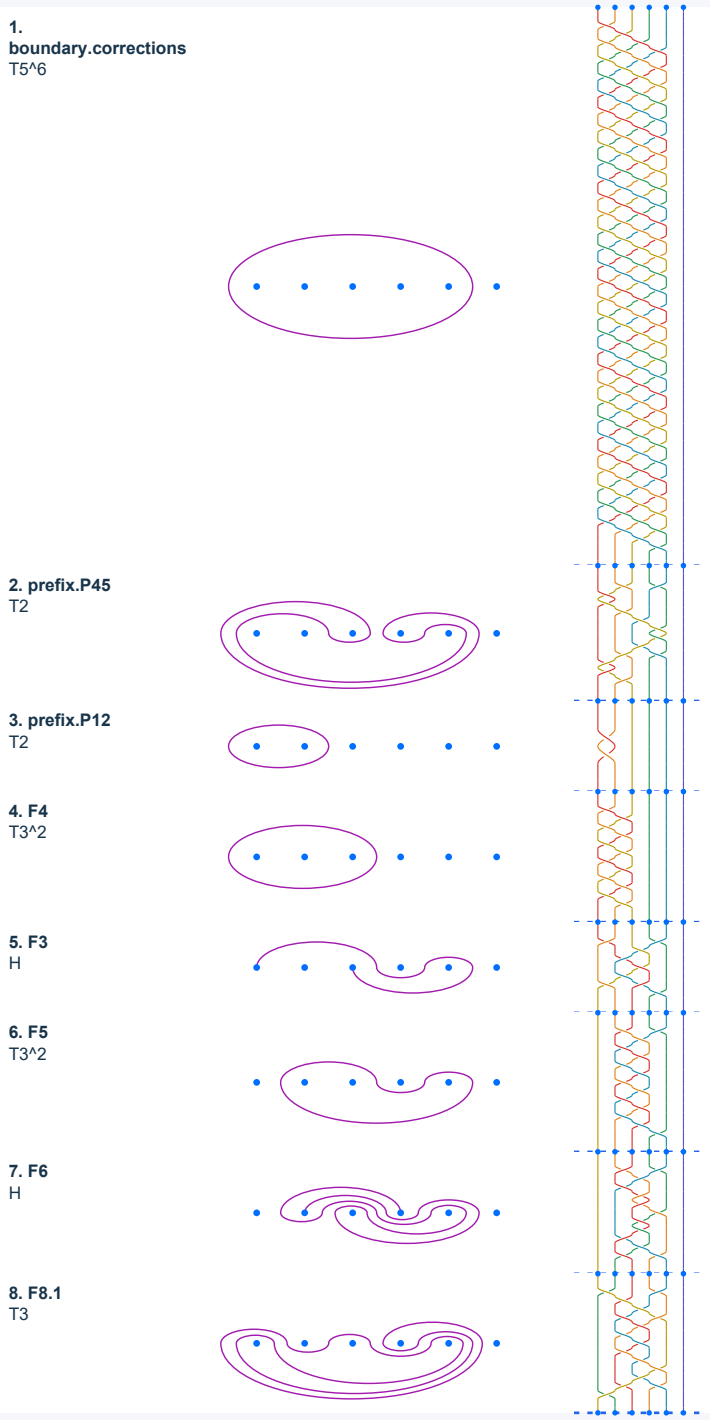
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



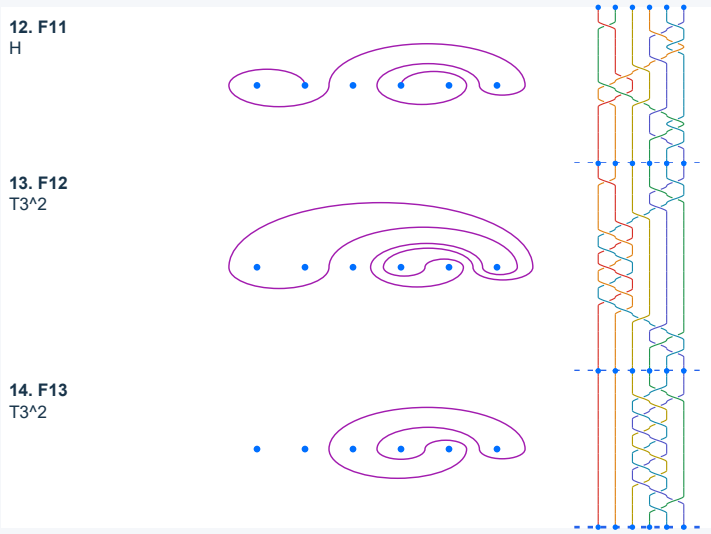
I - step 016 - continued

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

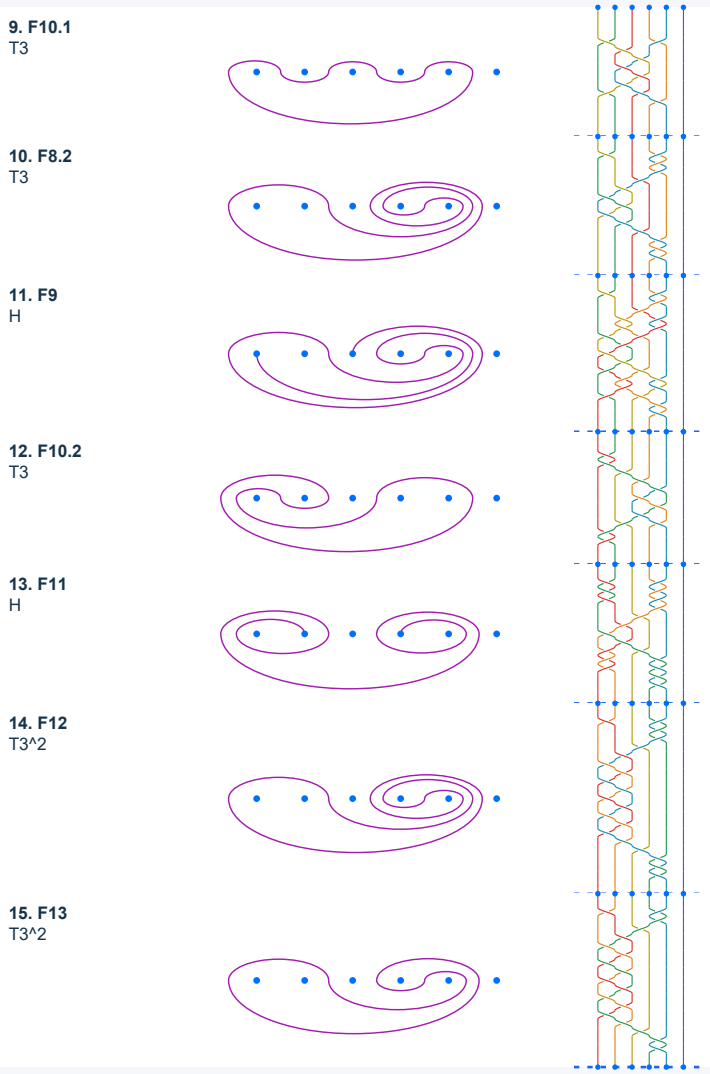
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



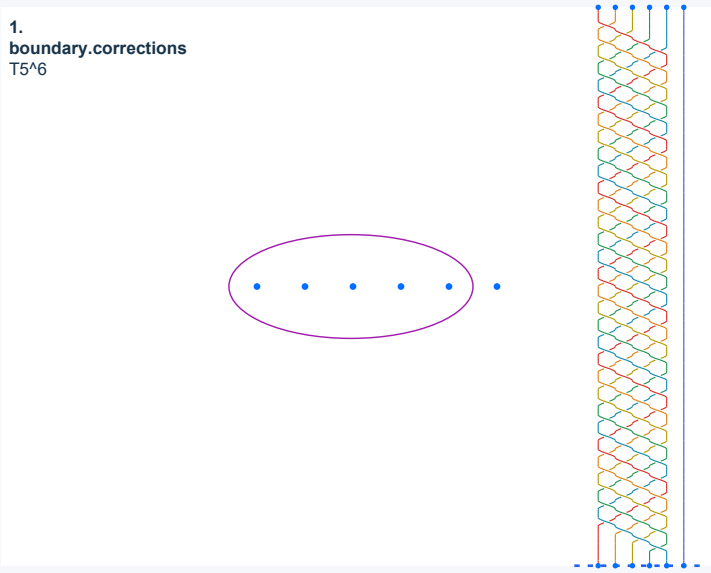
I - step 017

SPHERE MOVE: cap the SIX framing corrections

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

II. Existing roots remove all half twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

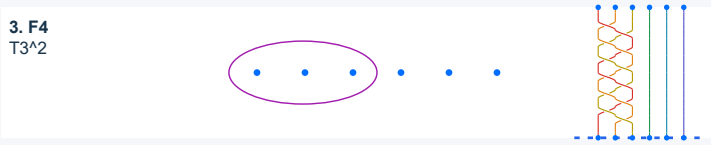
II - step 018

Split F4 into its two positive units

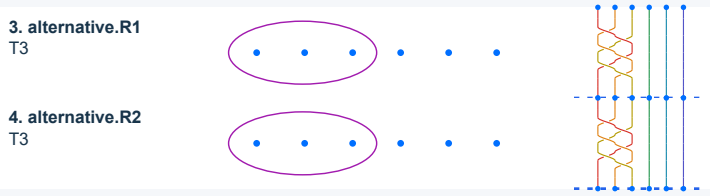
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



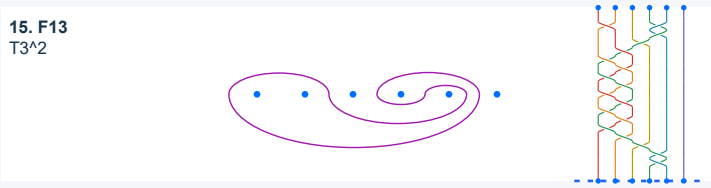
II - step 019

Split F13 into its two positive units

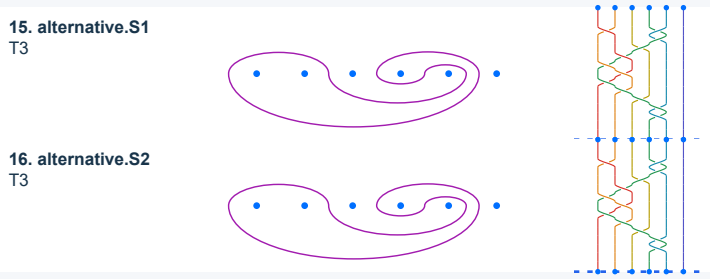
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



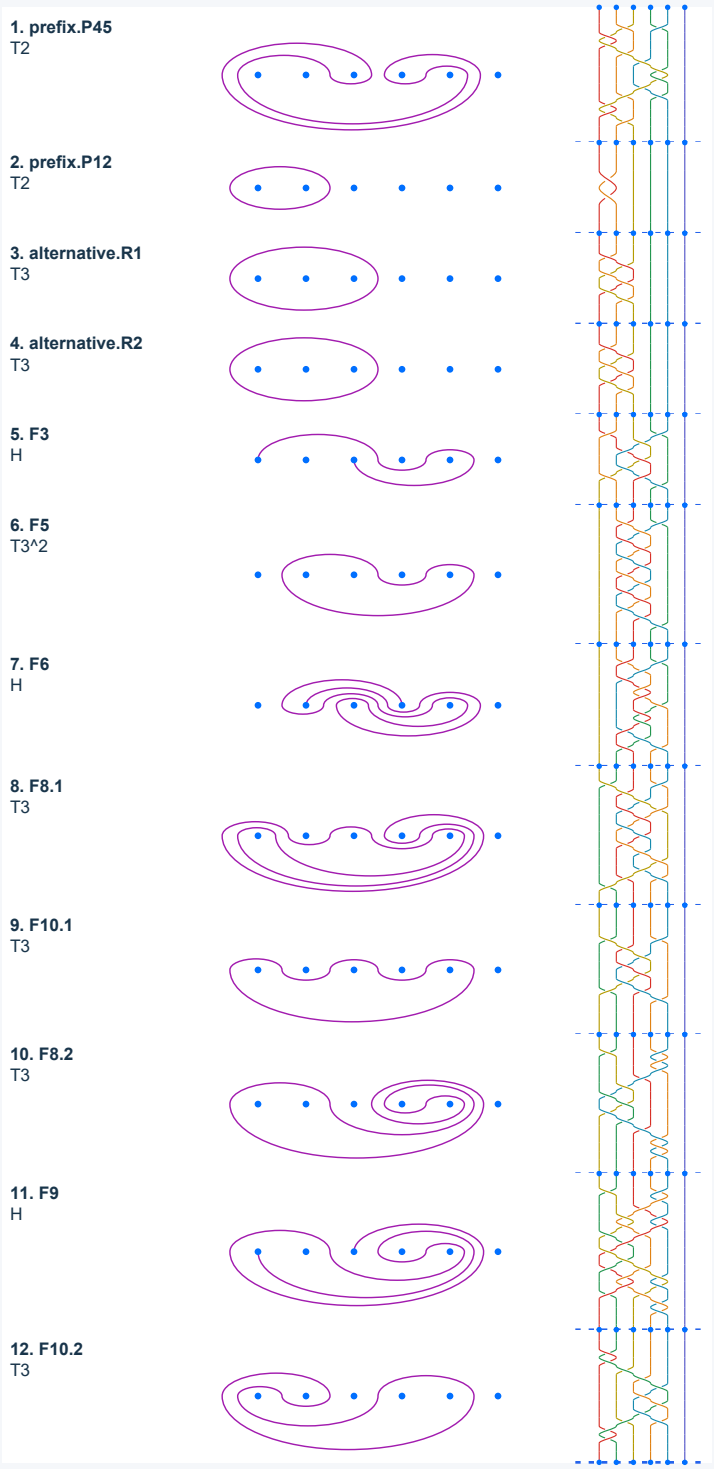
II - step 020

Choose the cyclic start at the existing 124 root; its whole common frame is retained

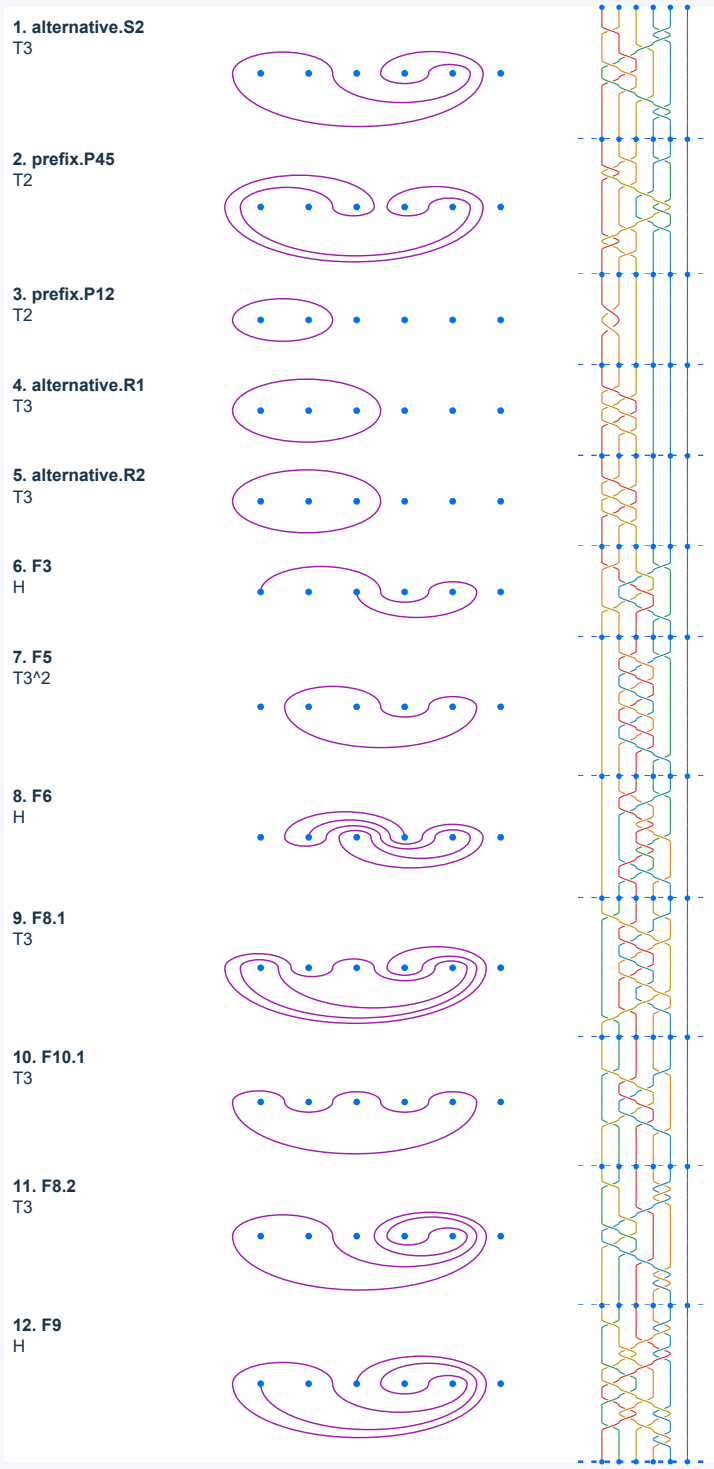
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



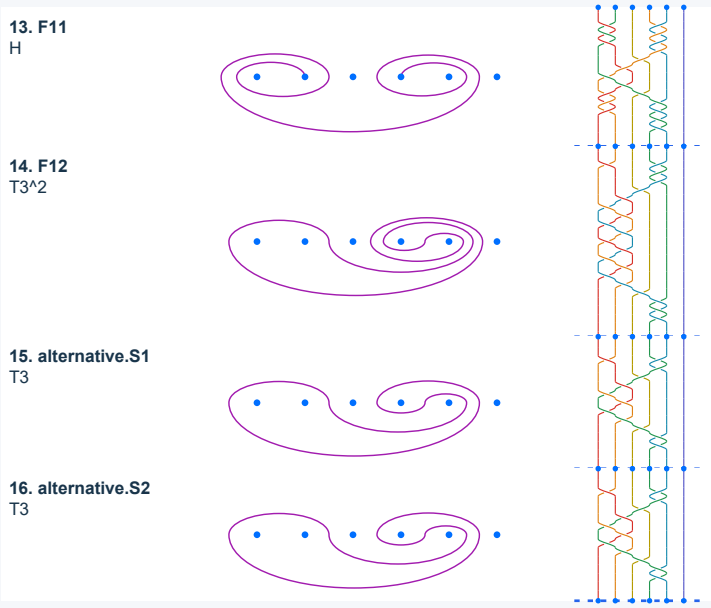
II - step 020 - continued

Choose the cyclic start at the existing 124 root; its whole common frame is retained

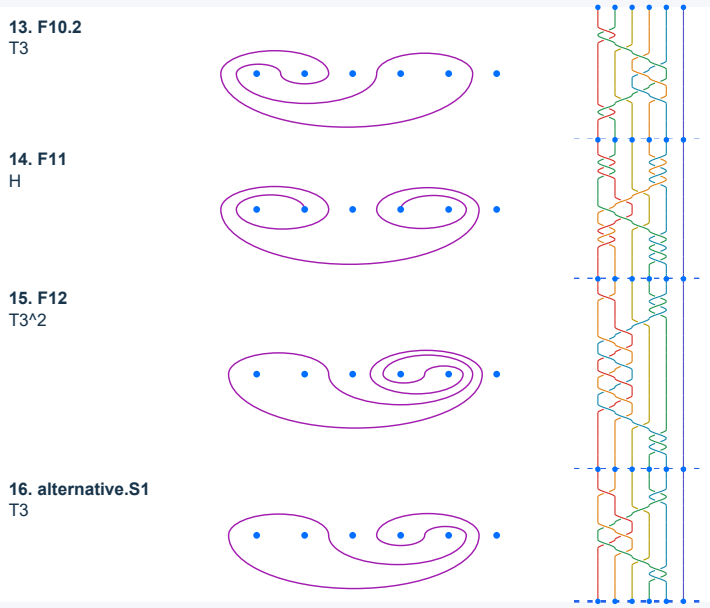
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



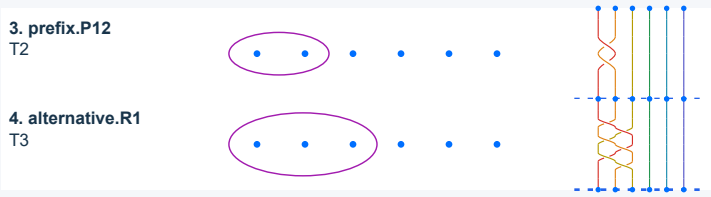
II - step 021

Commute one round T123 root above the round T12

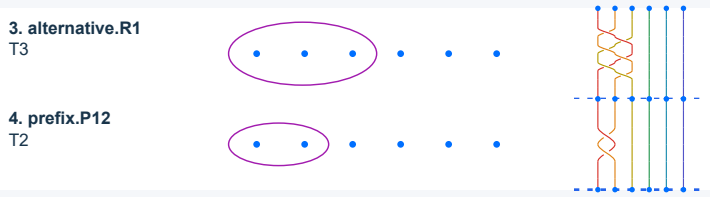
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 12 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



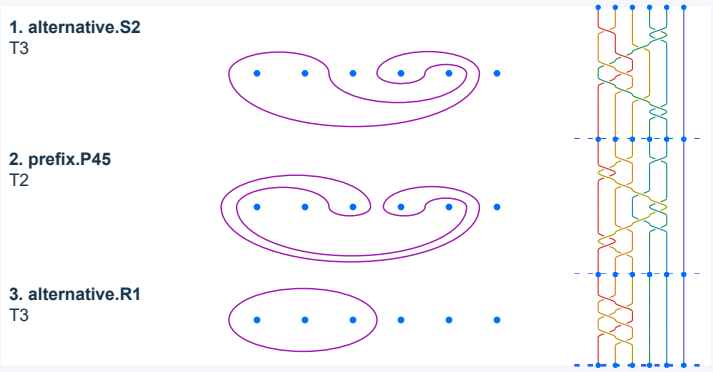
II - step 022

PC cluster lantern on the ACTUAL 124, 34 and 123 curves

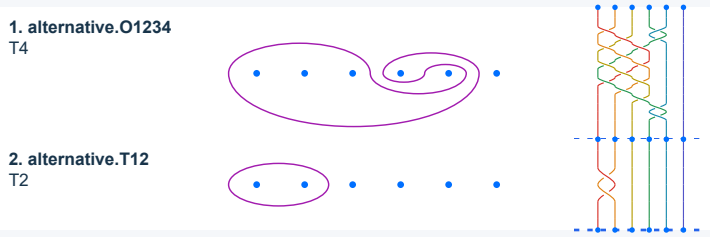
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



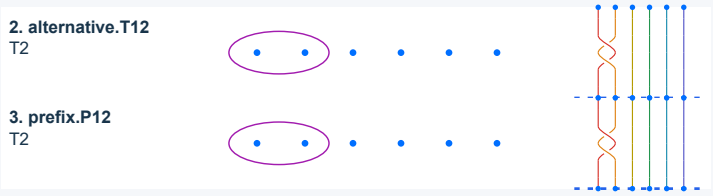
II - step 023

Combine the matching alternative.T12 and prefix.P12

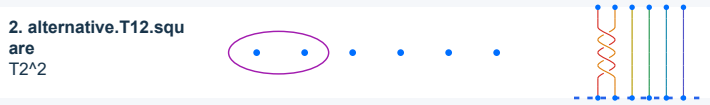
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



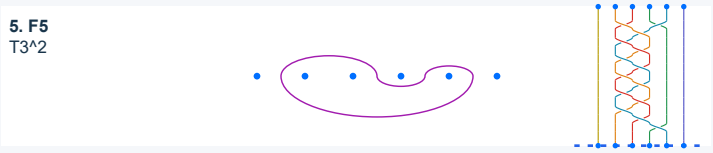
II - step 024

Split F5 into its two positive units

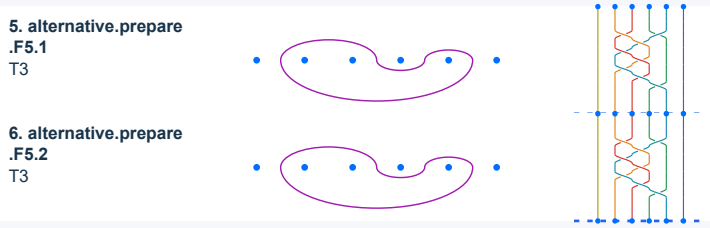
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



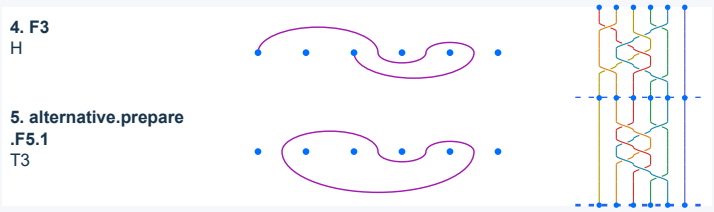
II - step 025

One 235 root up over F3

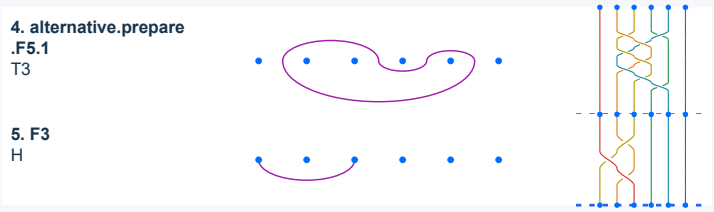
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



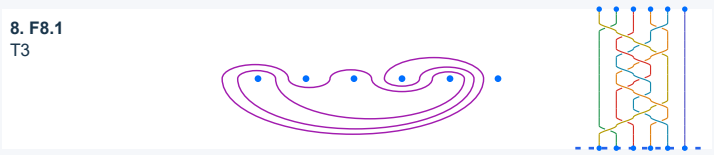
II - step 026

Original 345 daisy supplies actual P35

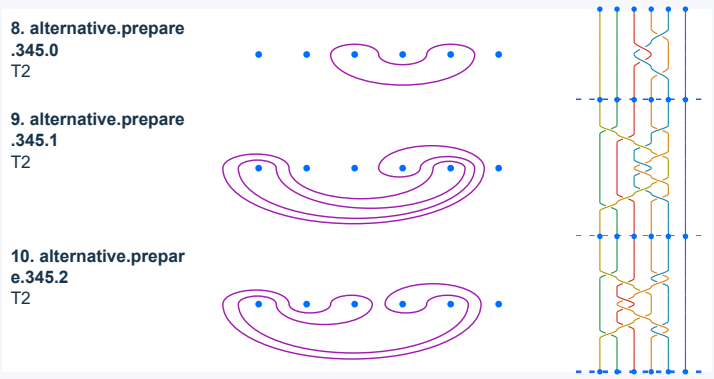
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



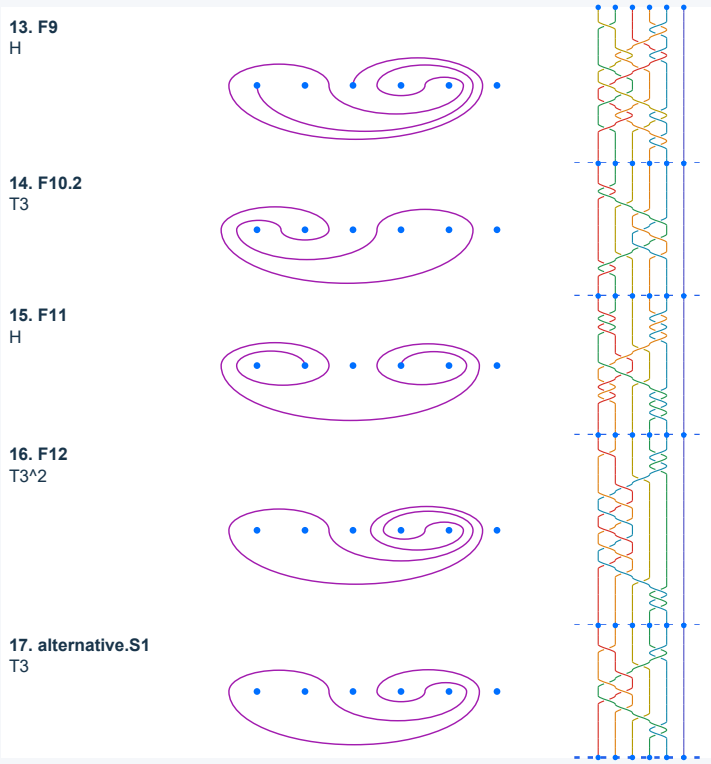
II - step 027

Remaining 124 root up over F9

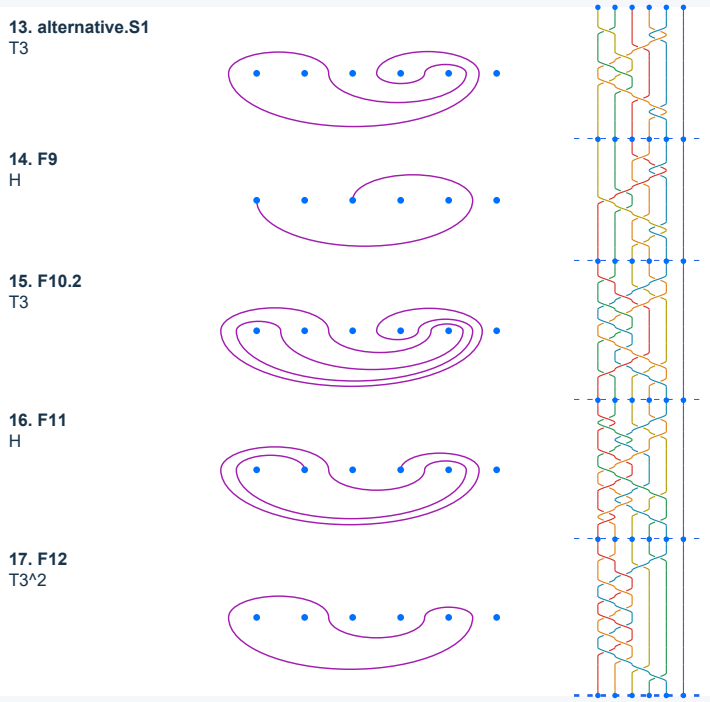
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



II - step 028

Carry P35 up over F3; a Q14 transport is still needed

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



II - step 029

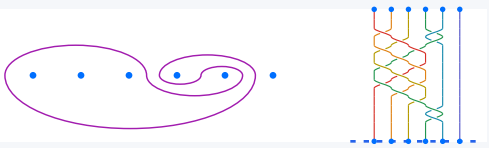
PC daisy lends its actual Q14 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 16 below. Only the changed interval is shown.

Before

1. alternative.O1234
T4



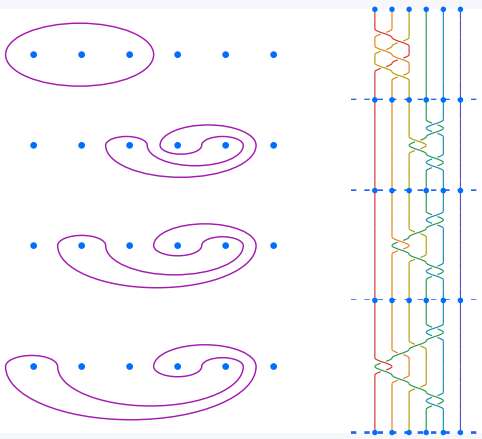
After

1.
alternative.qdaisy.0
T3

2.
alternative.qdaisy.1
T2

3.
alternative.qdaisy.2
T2

4.
alternative.qdaisy.3
T2



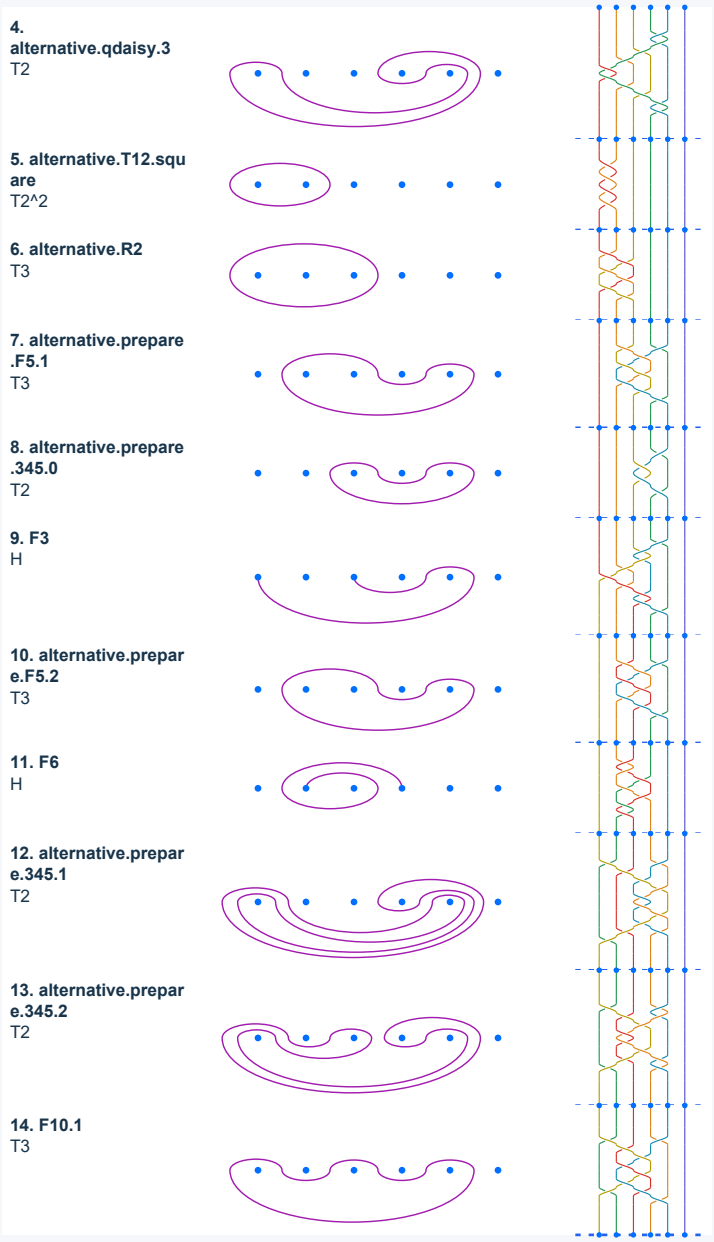
II - step 030

Carry Q14 down across F3 and the remaining 345 petals

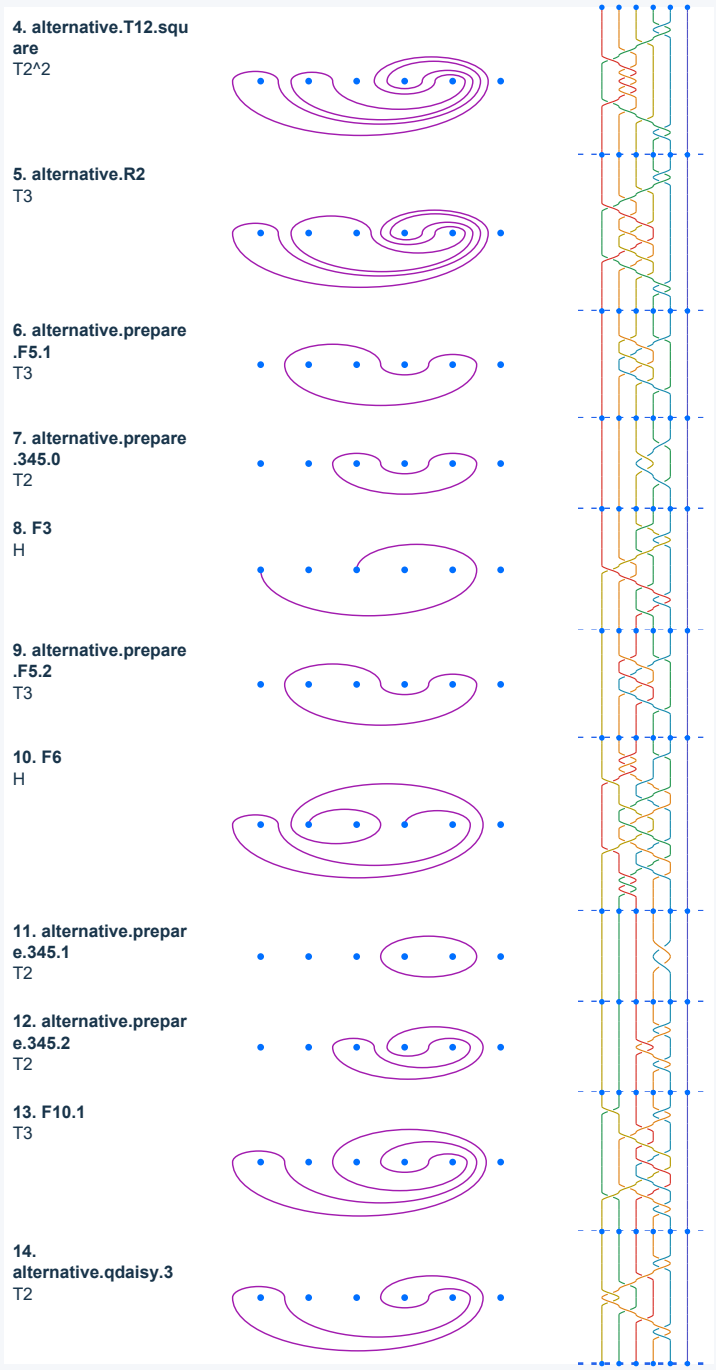
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 10.

Before



After



II - step 031

PC daisy contracts the ACTUAL transported petals to O1345

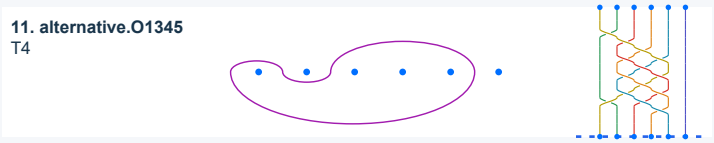
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



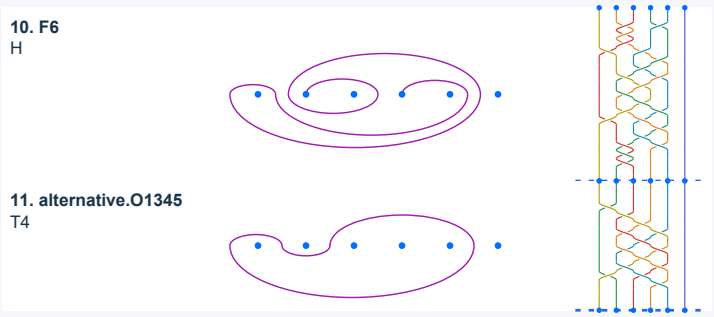
II - step 032

Carry O1345 up across F6 before the matching-half gather

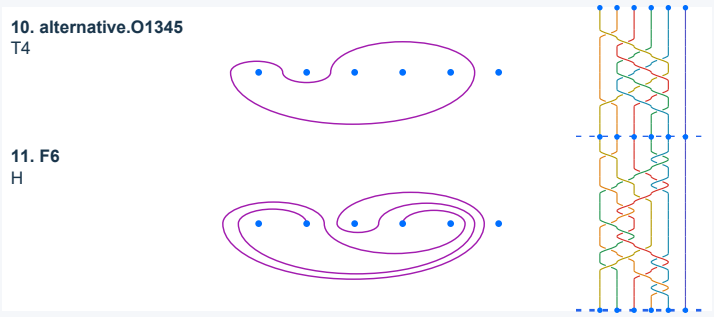
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



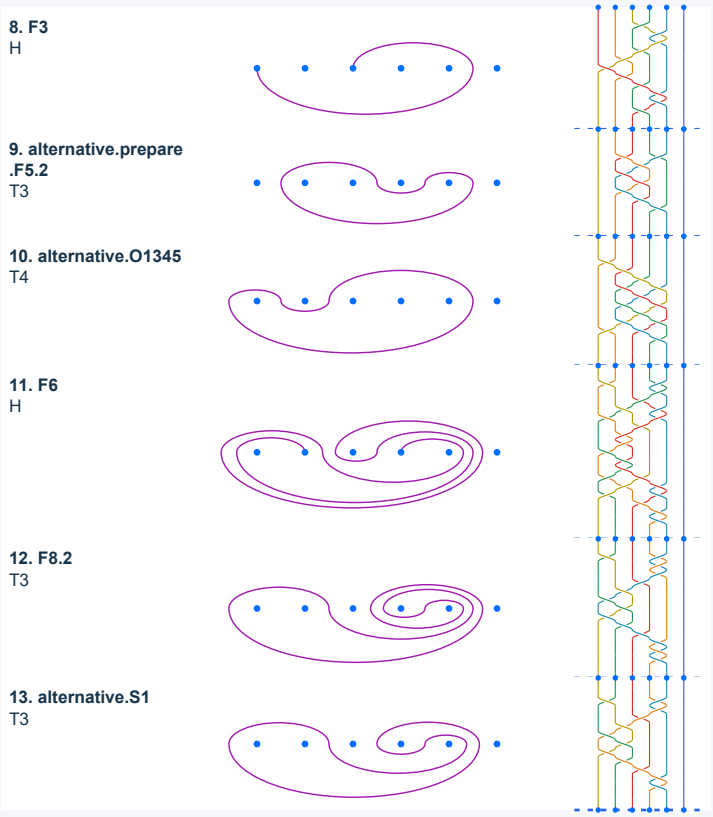
II - step 033

Carry the matched F3 down to F9; F6 unwinds

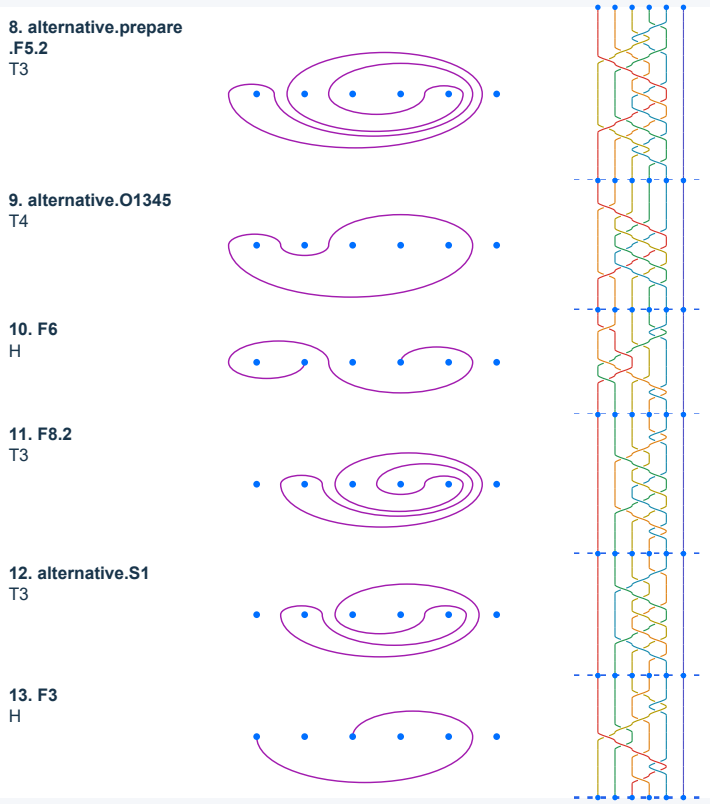
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



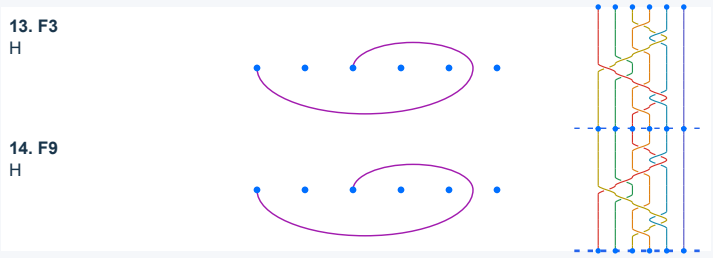
II - step 034

Combine the matching F3 and F9

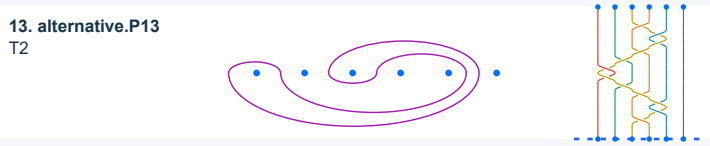
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



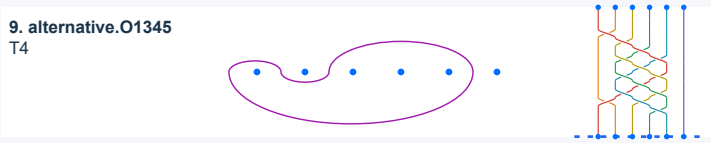
II - step 035

The O1345 daisy returns EXACTLY the borrowed Q14

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



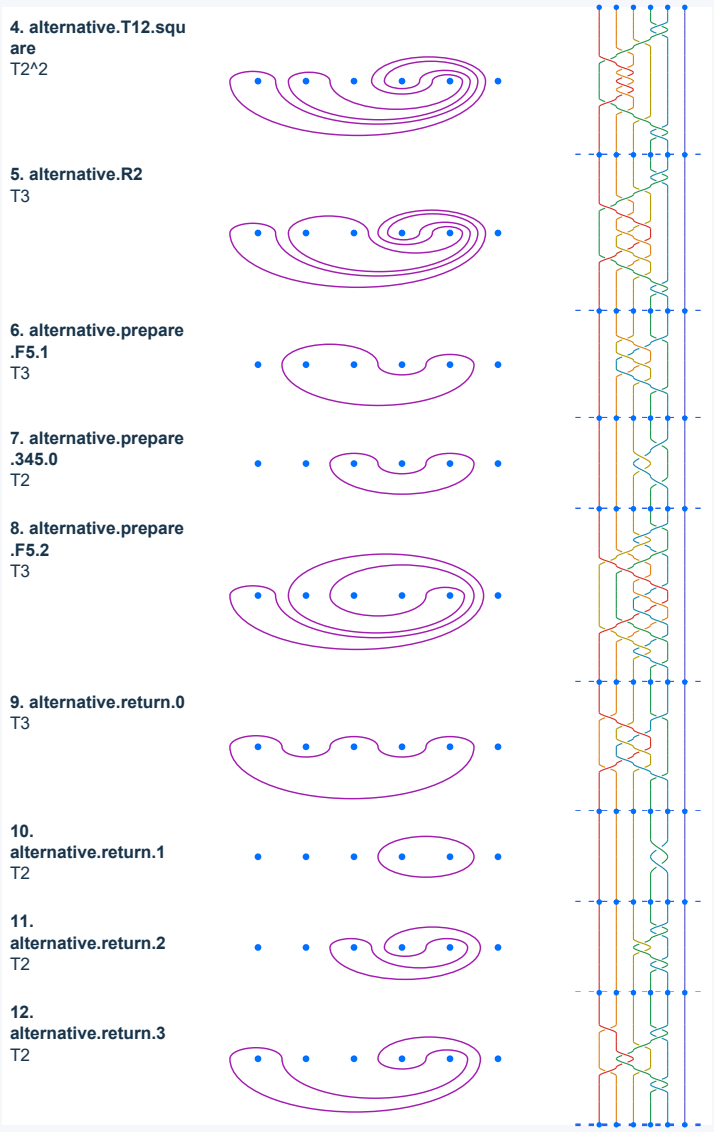
II - step 036

Return Q14 up to restore the round T12 square and T123

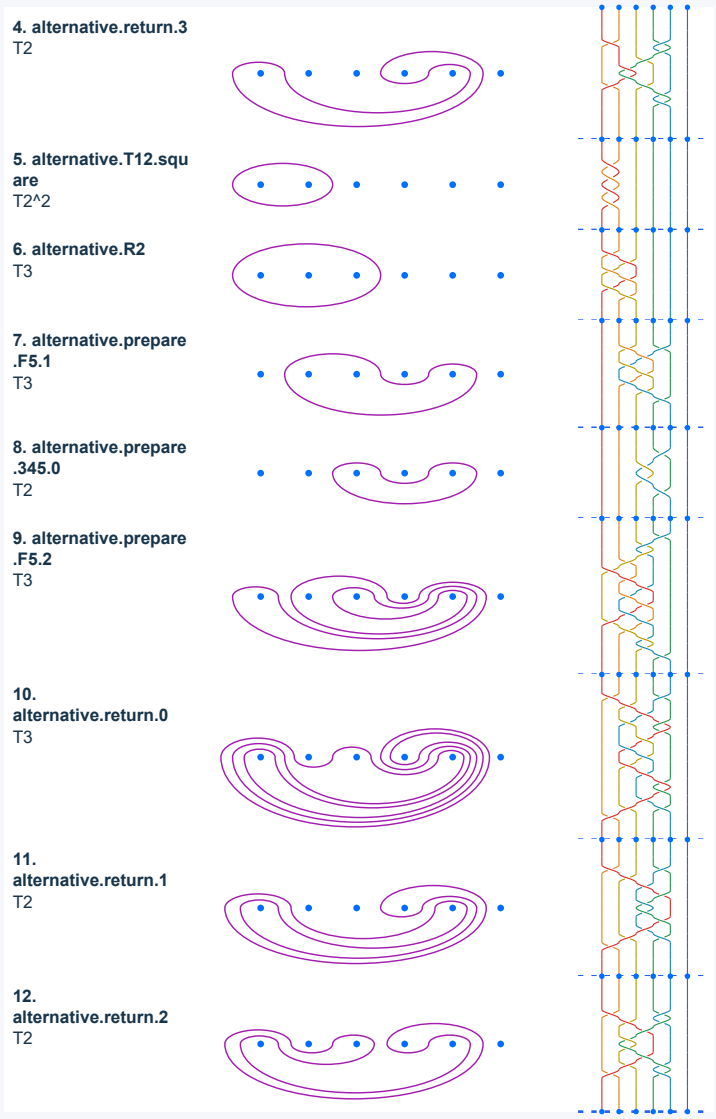
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



II - step 037

Close the original daisy after its actual Q14 returns

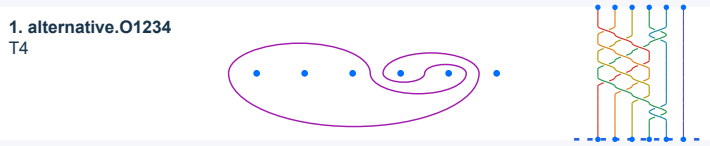
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown.

Before



After



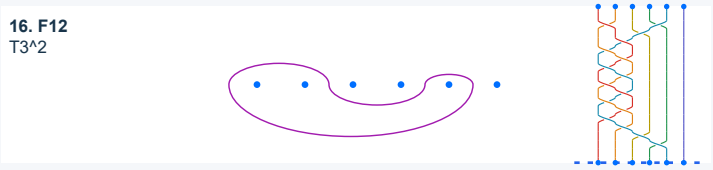
II - step 038

Split F12 into its two positive units

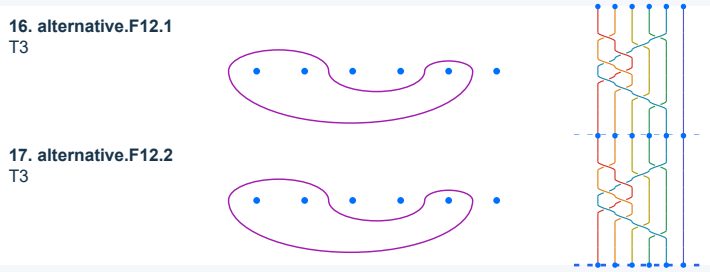
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



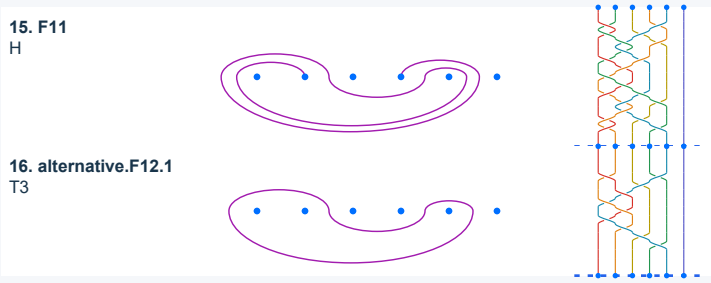
II - step 039

One C12 root shortens F11

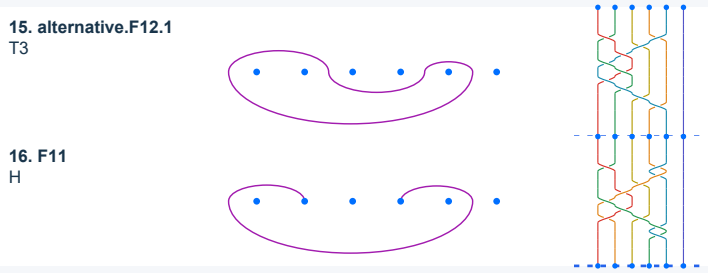
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



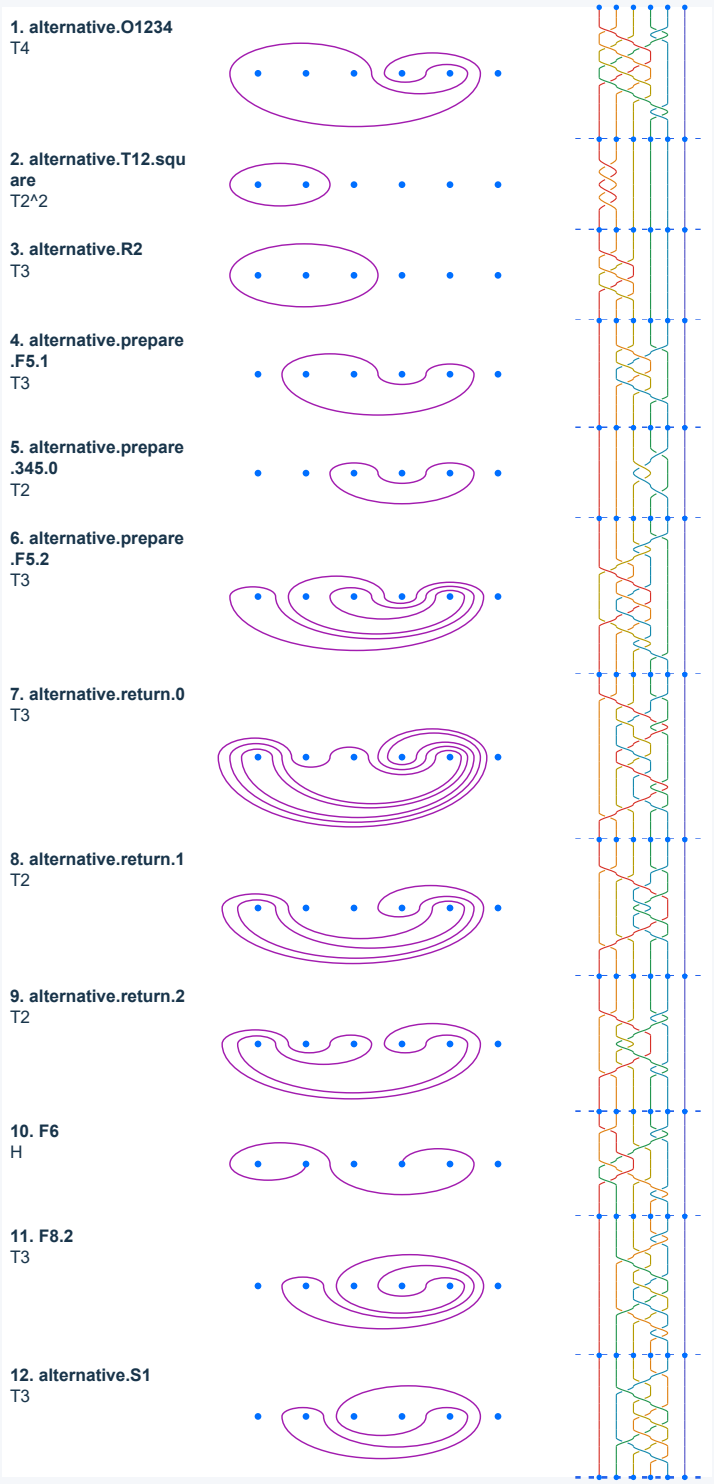
II - step 040

Cyclic start after BOTH the square and its commuting outer

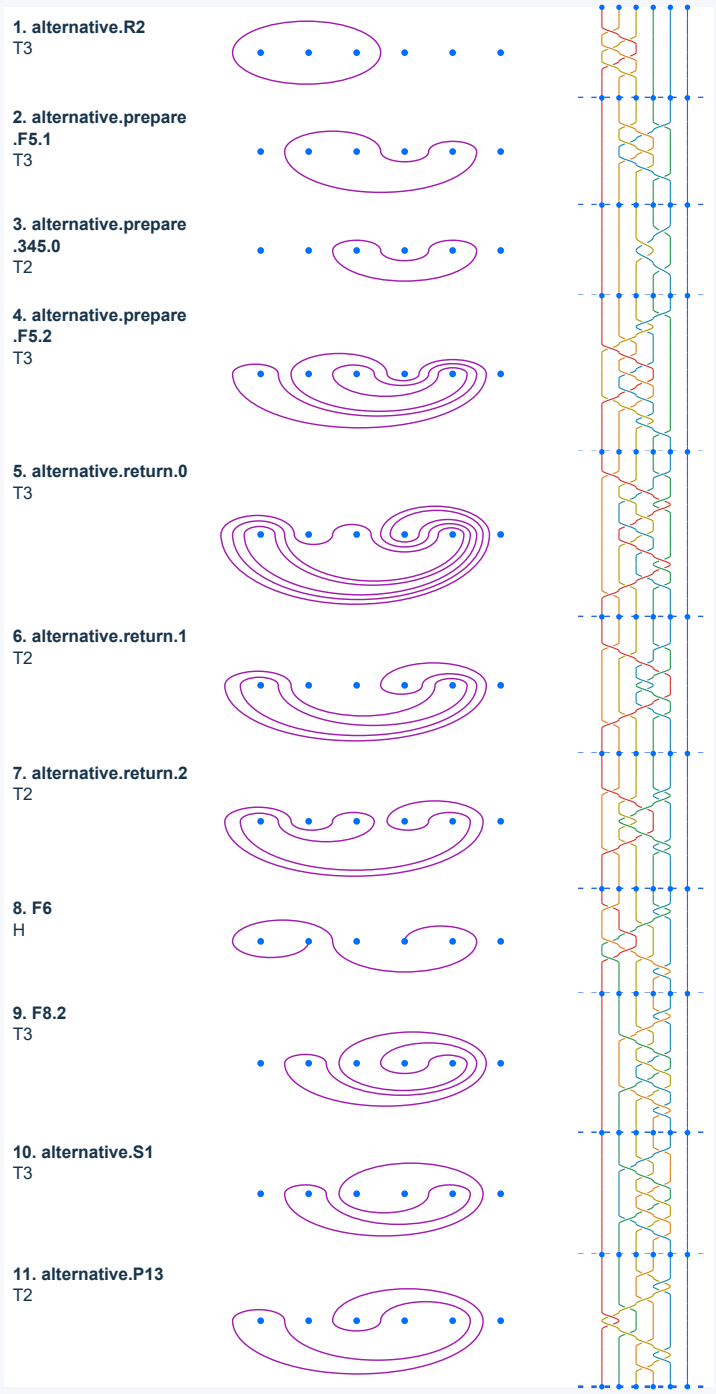
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



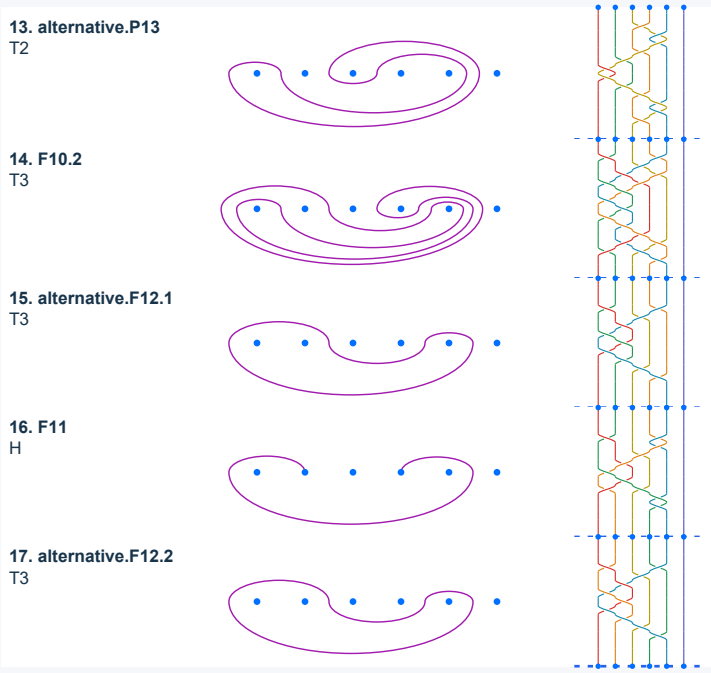
II - step 040 - continued

Cyclic start after BOTH the square and its commuting outer

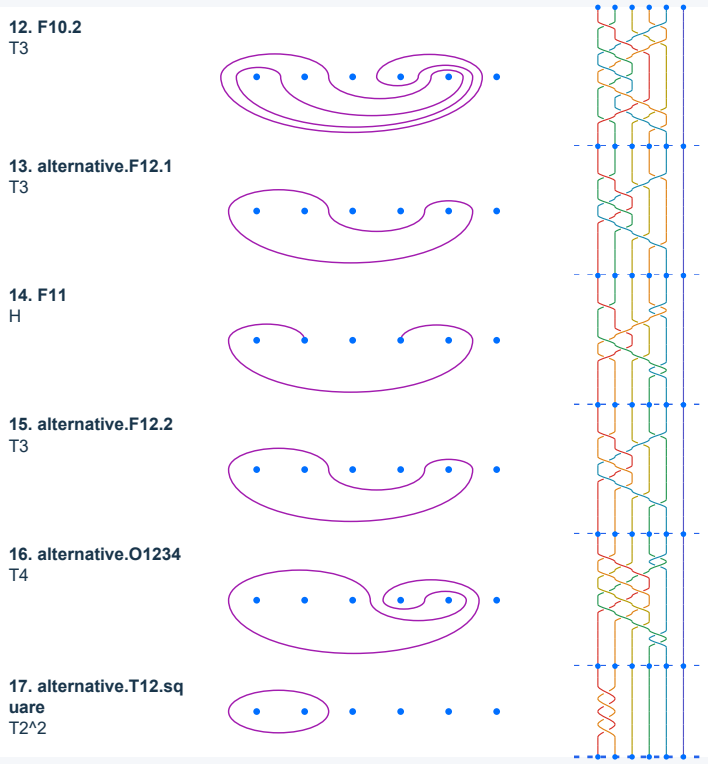
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



II - step 041

One short-side square drag includes its commuting outer

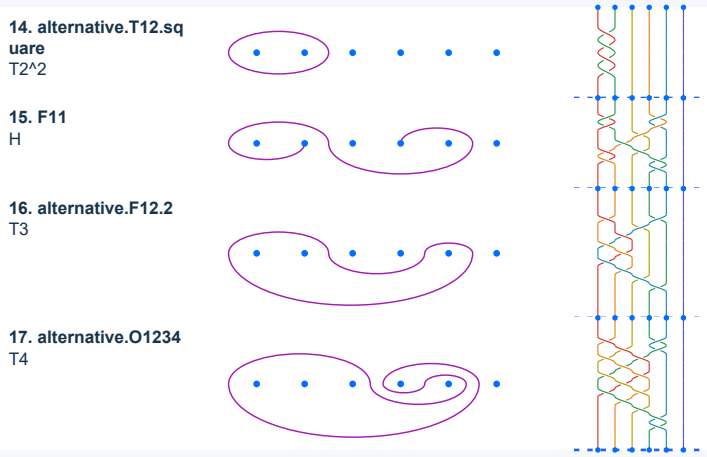
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



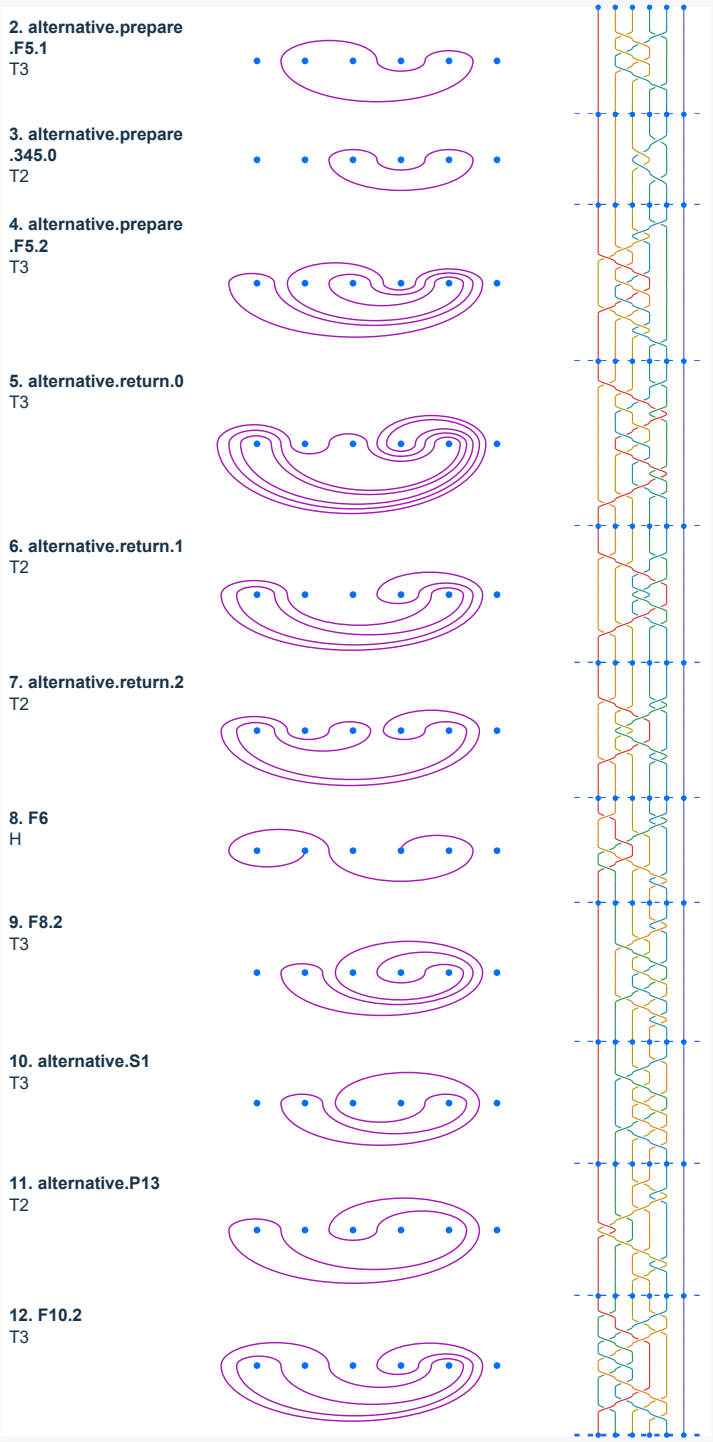
II - step 042

The same composed sigma4 common frame

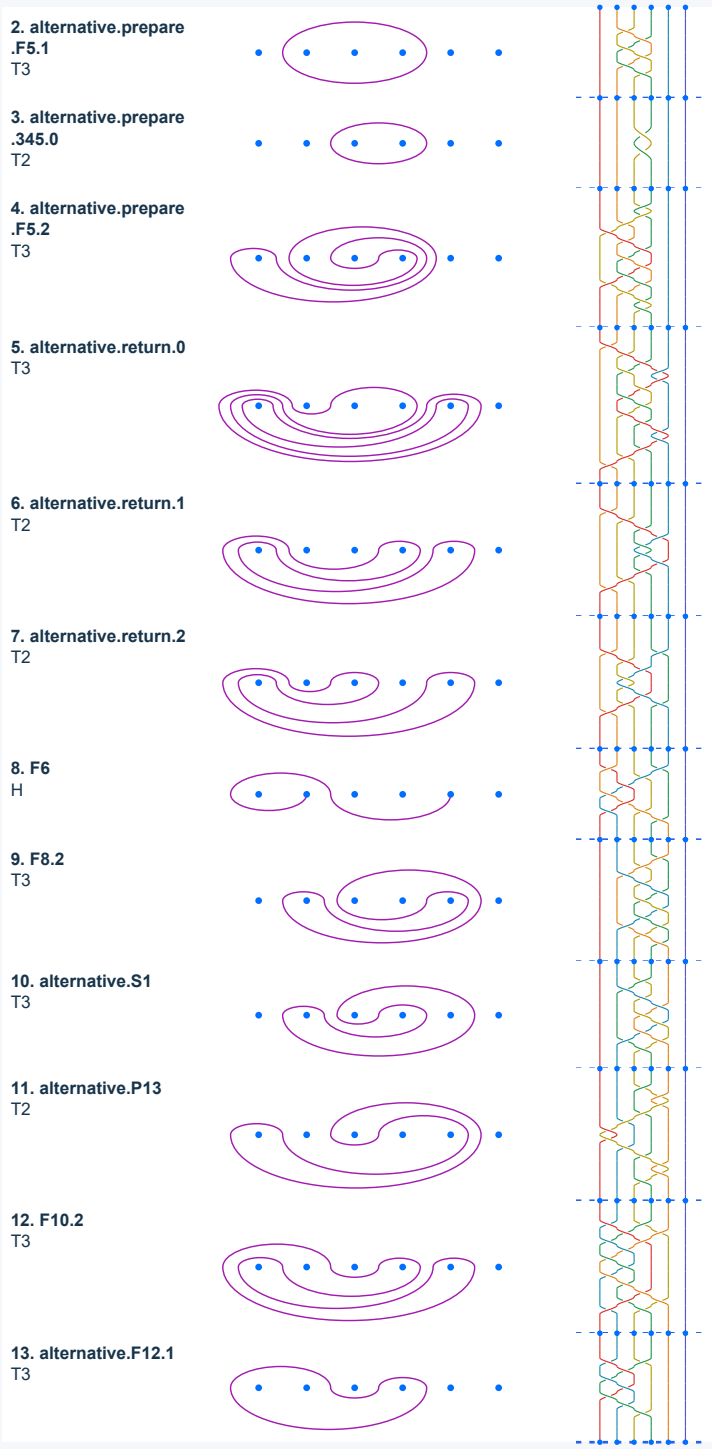
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



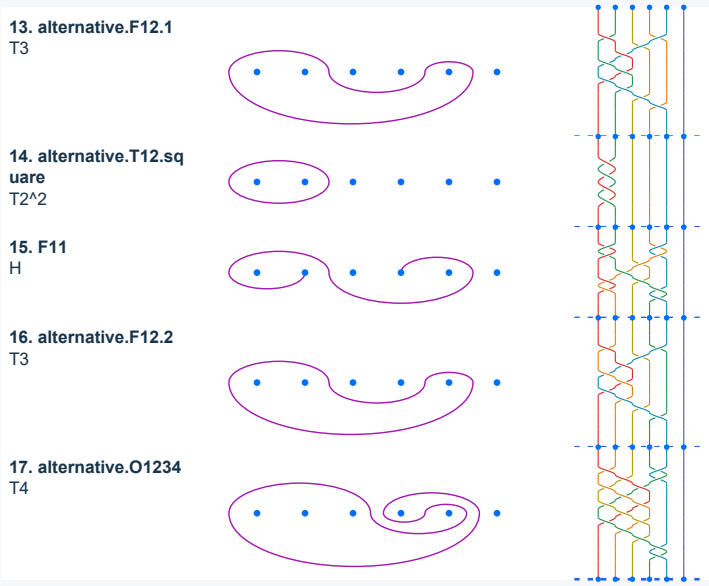
II - step 042 - continued

The same composed sigma4 common frame

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



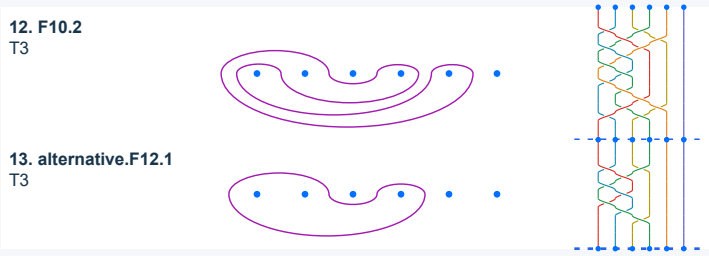
II - step 043

Reorder C12 before the matching-half gather

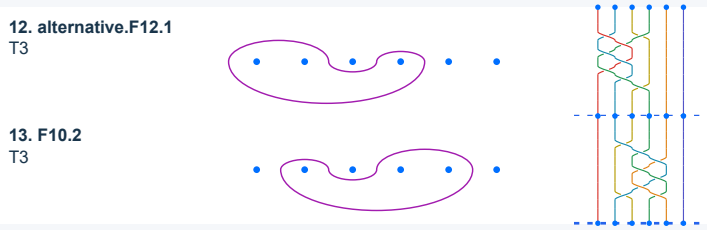
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



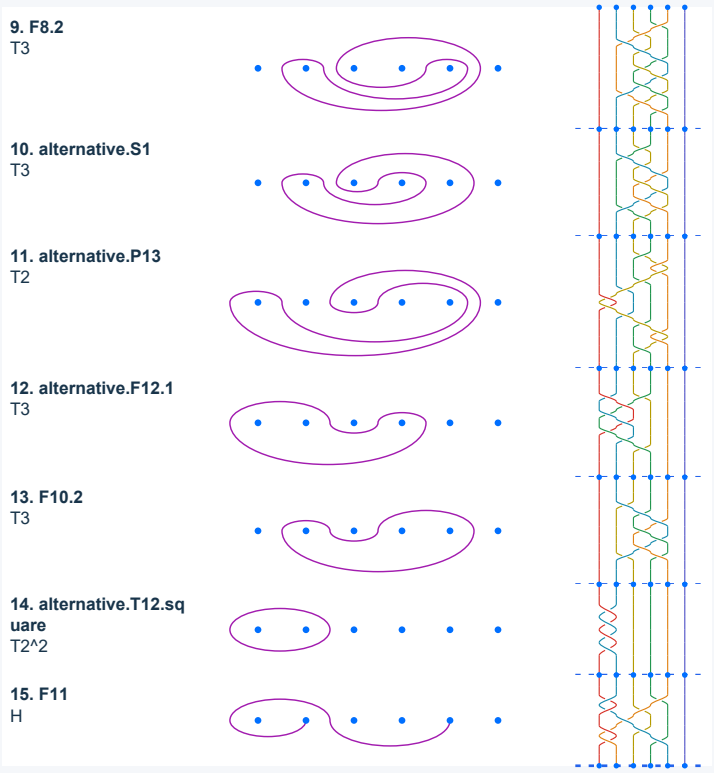
II - step 044

Gather the final matching original halves

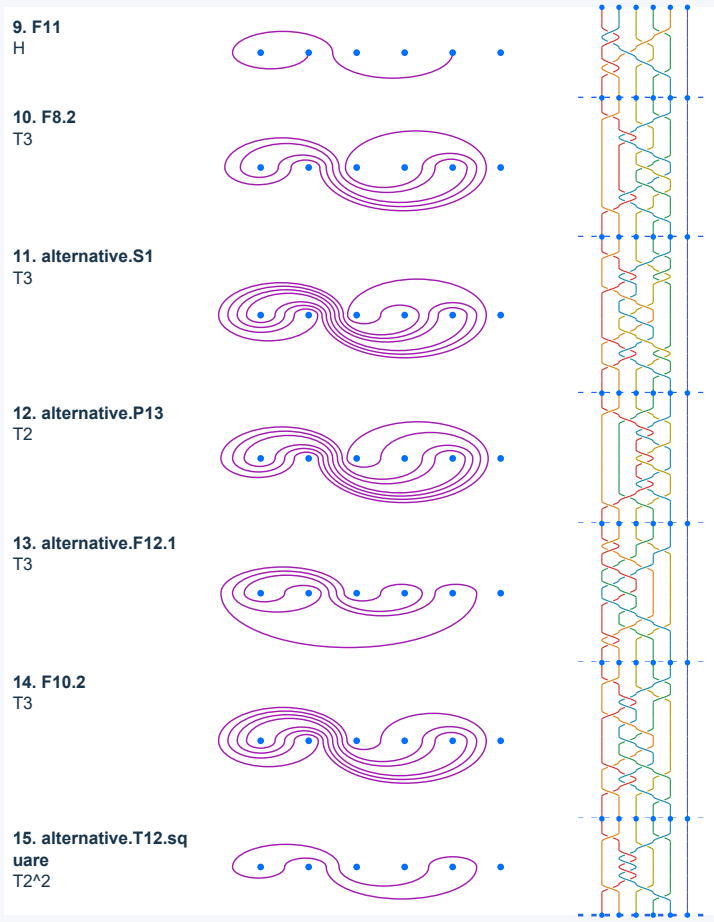
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



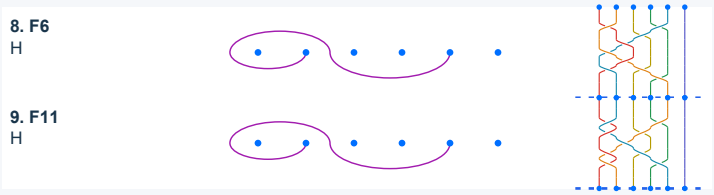
II - step 045

Combine the matching F6 and F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 8 below. Only the changed interval is shown.

Before



After



III. The first exterior and simpler inside daisy

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

III - step 046

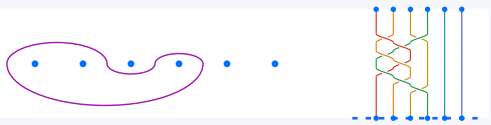
C12 daisy supplies the actual Q15

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 1 below. Only the changed interval is shown.

Before

15. alternative.F12.2
T3

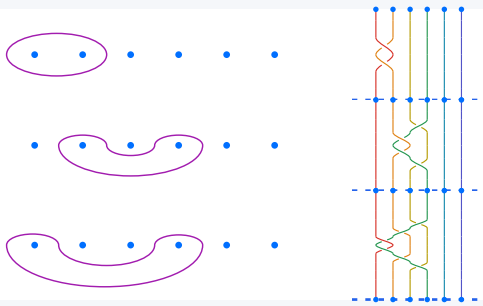


After

15. alternative.next.0
T2

16. alternative.next.1
T2

17. alternative.next.2
T2



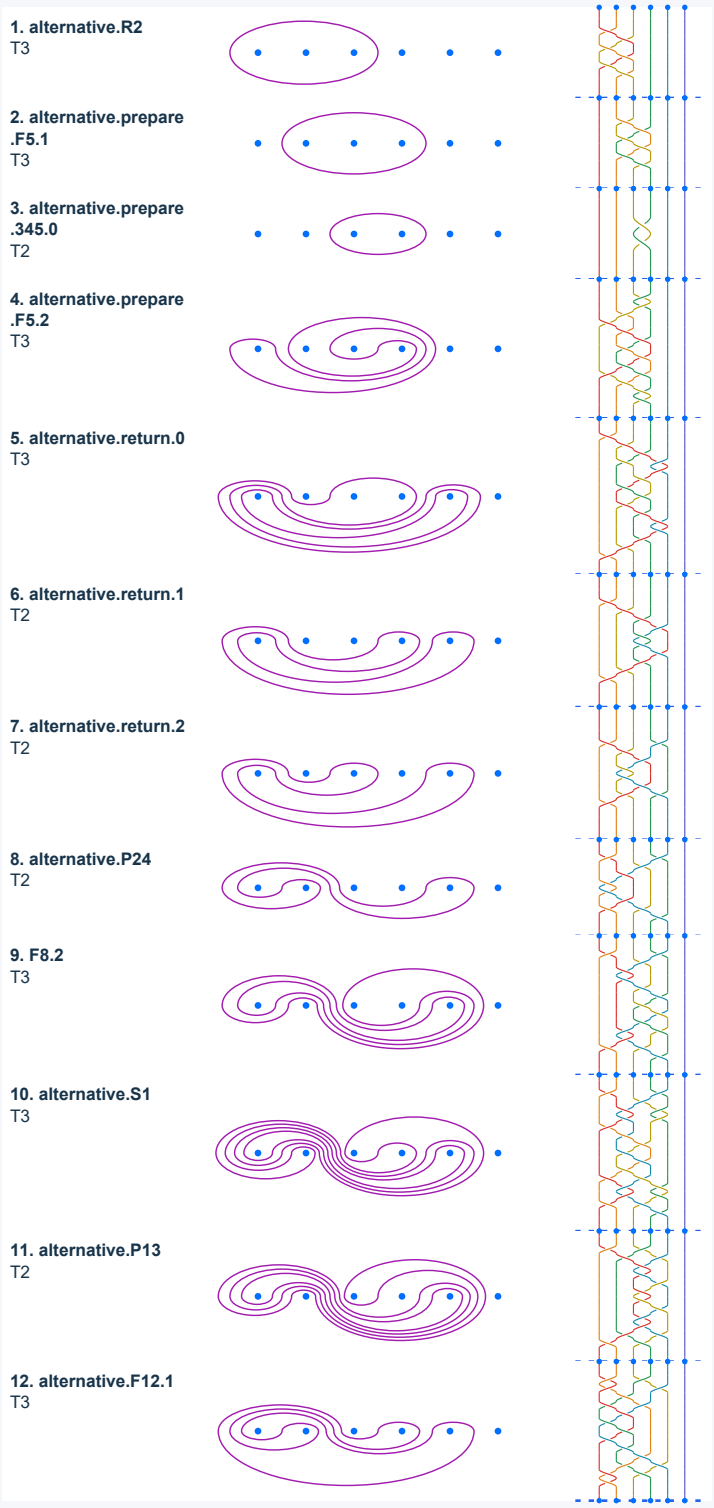
III - step 047

Cyclic start at Q15 REJOINS the unchanged route

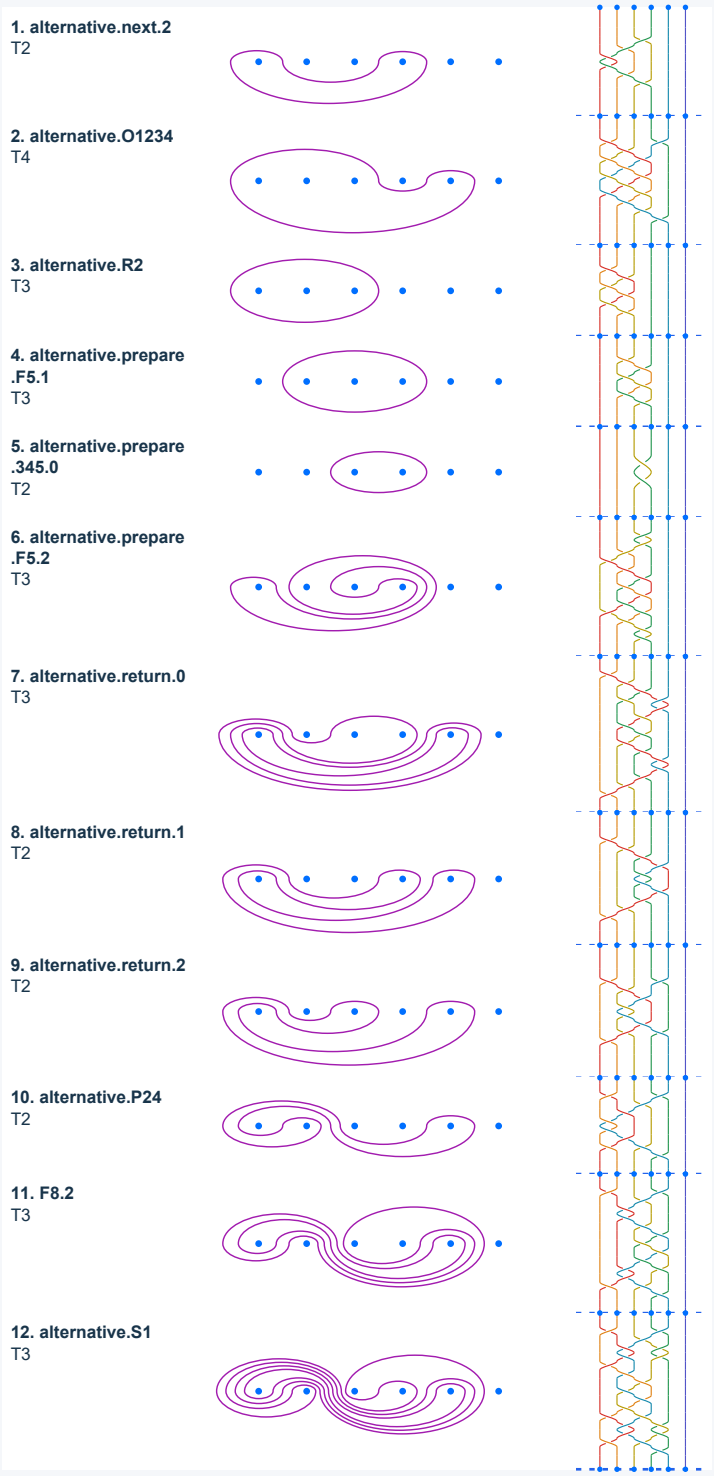
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



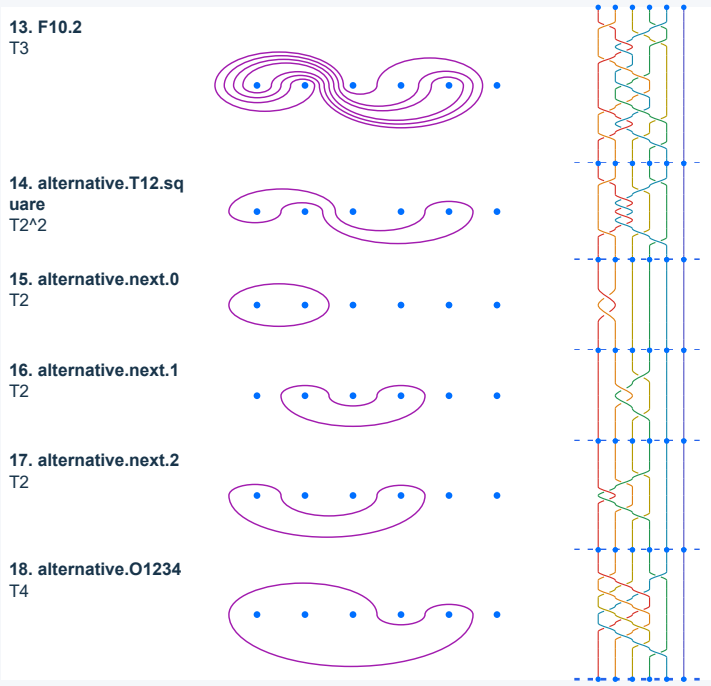
III - step 047 - continued

Cyclic start at Q15 REJOINS the unchanged route

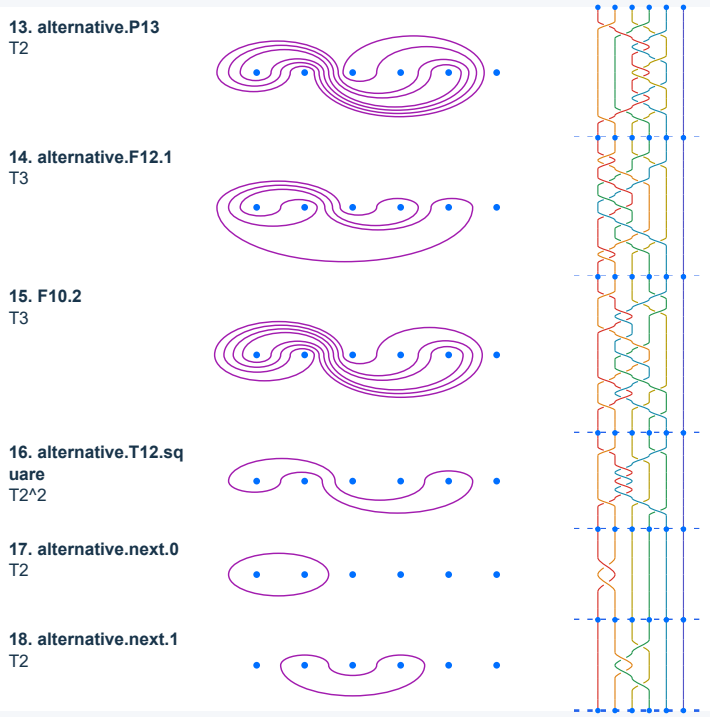
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



III - step 048

Carry Q15 through the combined four-point prefix and the two roots

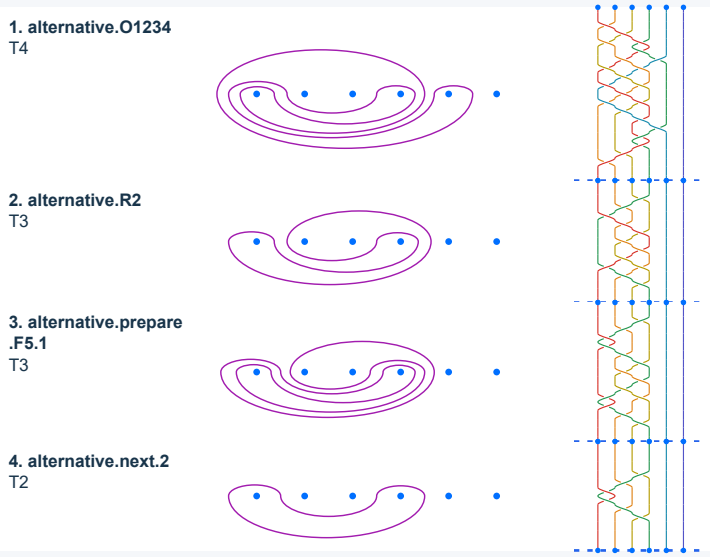
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



III - step 049

PC cluster lantern gives the second four-point twist

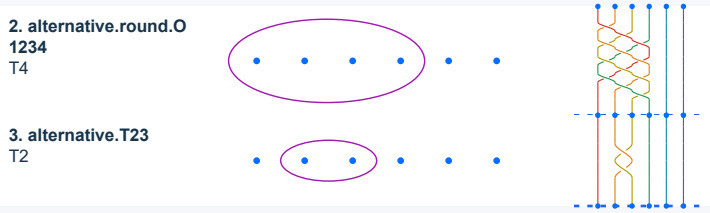
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



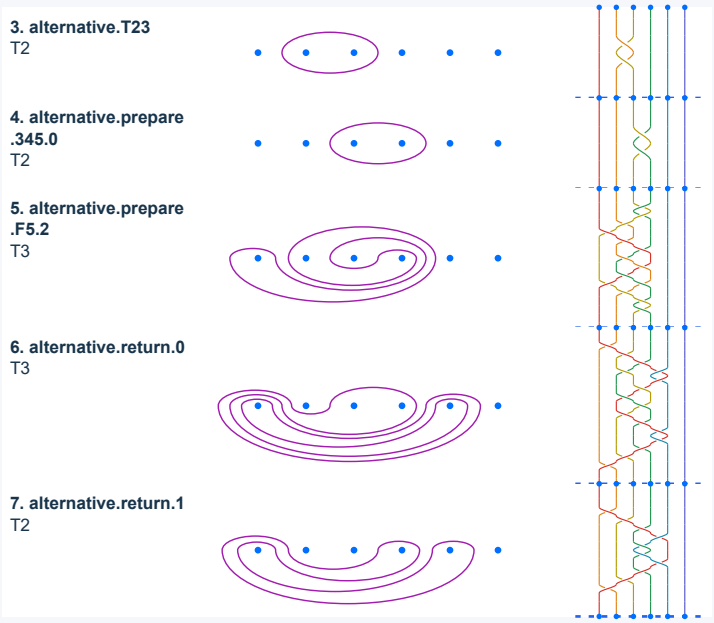
III - step 050

Gather the ACTUAL missing 45 petal by the short side

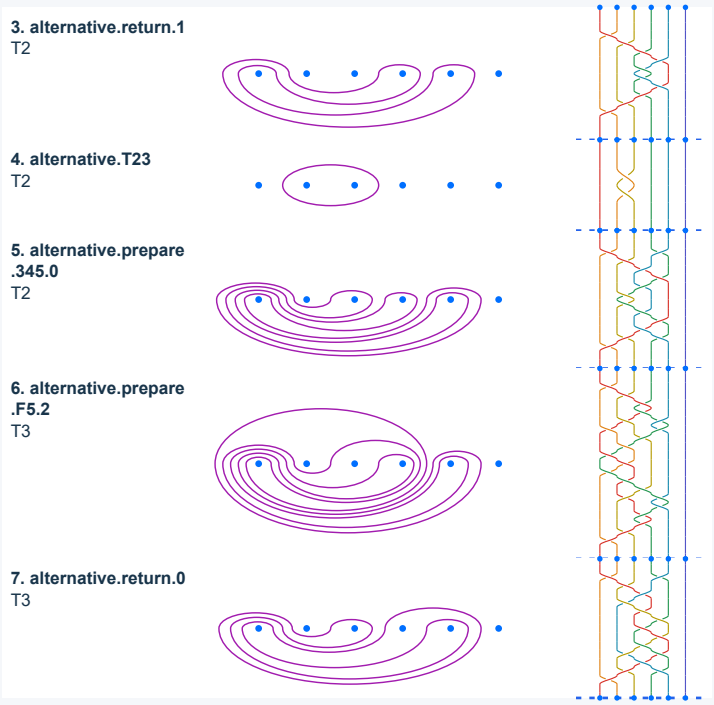
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



III - step 051

Cabled PC lantern on the ACTUAL two four-point curves and supplied 45 petal

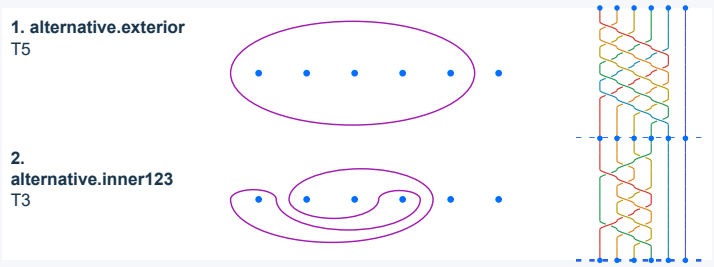
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



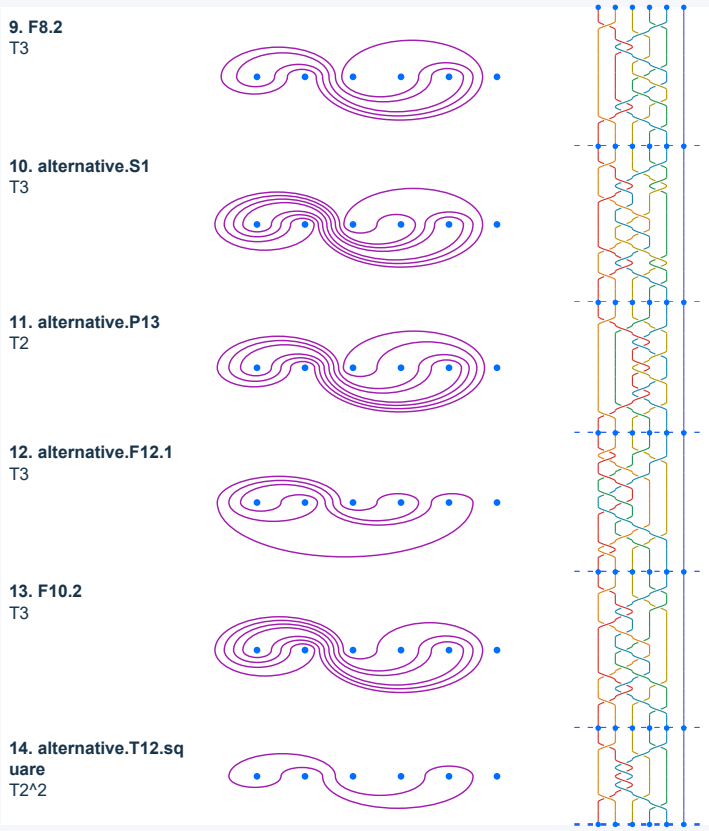
III - step 052

Unwind five actual residual curves with the fixed positive square

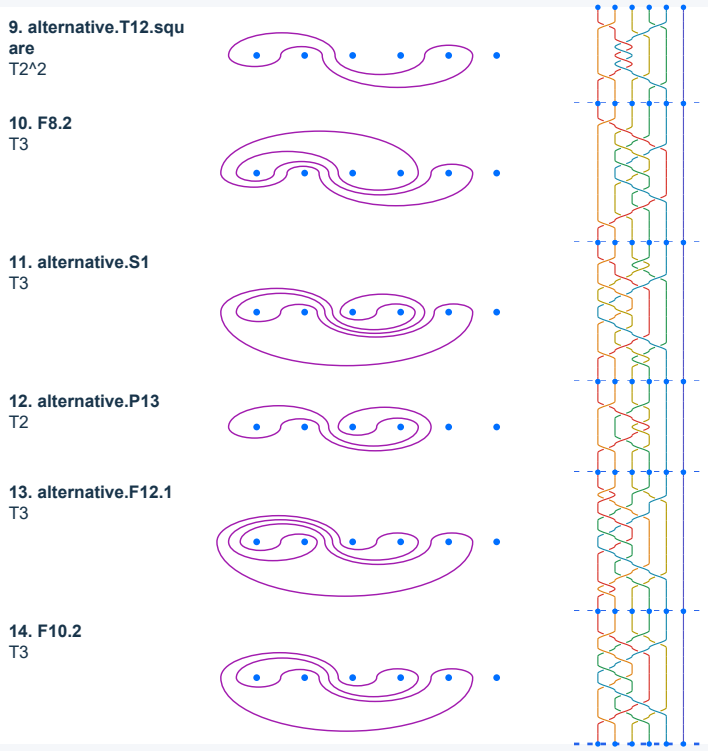
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



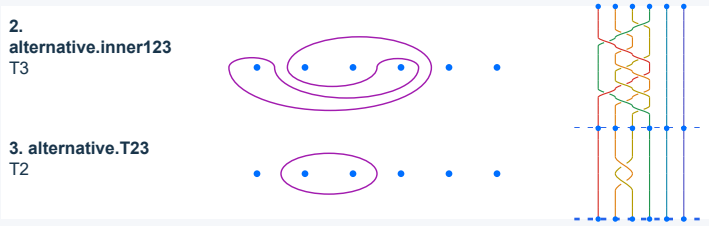
III - step 053

Commute the 123 root past T23 to its actual 34 and 124 companions

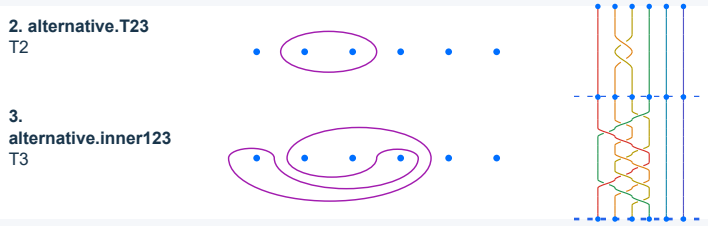
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



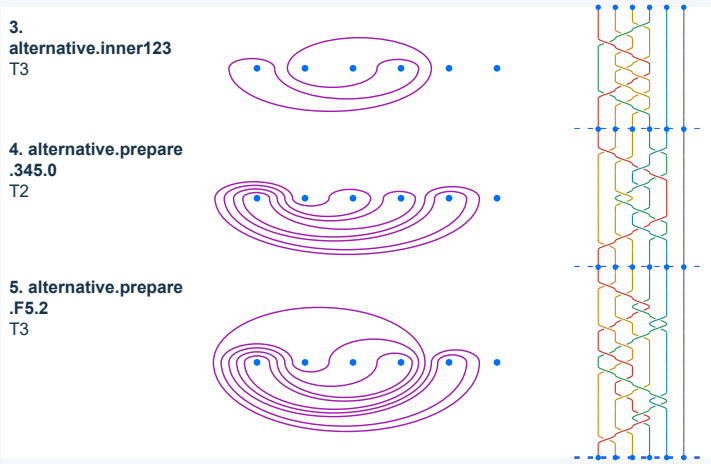
III - step 054

Cabled PC lantern contracts the ACTUAL 123, 34 and 124 curves

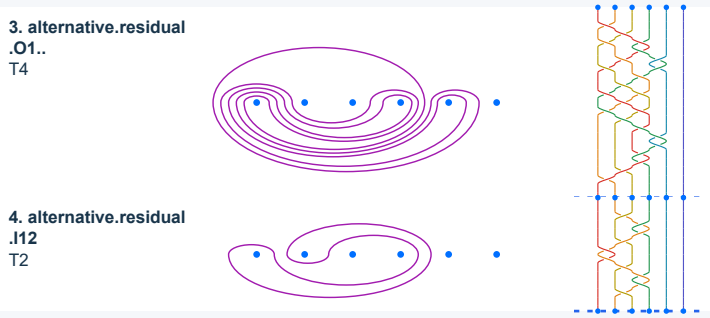
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



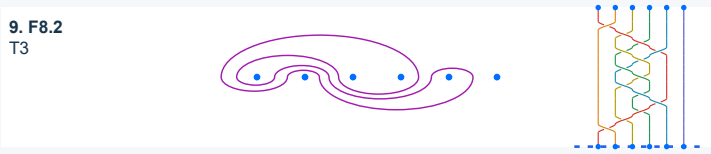
III - step 055

The actual 345 daisy lends its round 34 petal

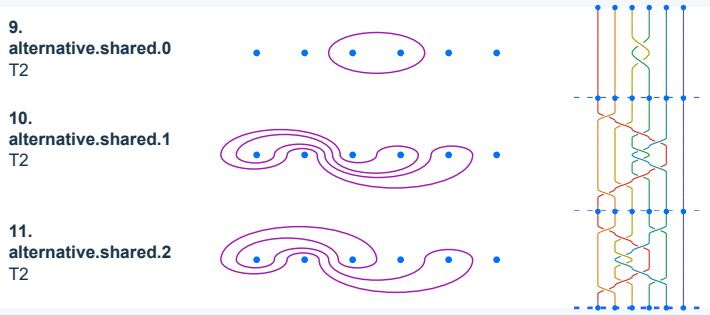
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



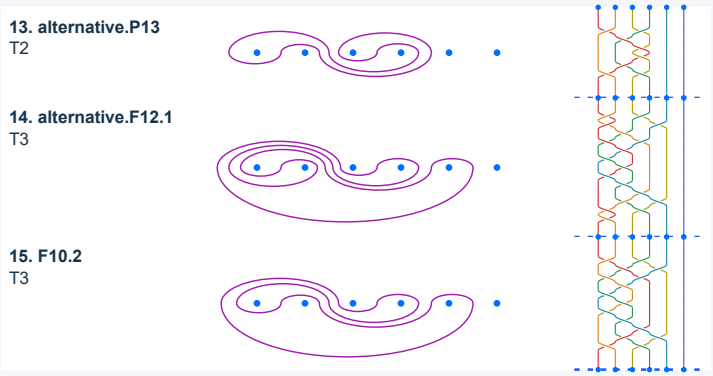
III - step 056

Carry the 245 root up across the 145 root and the 13 pair

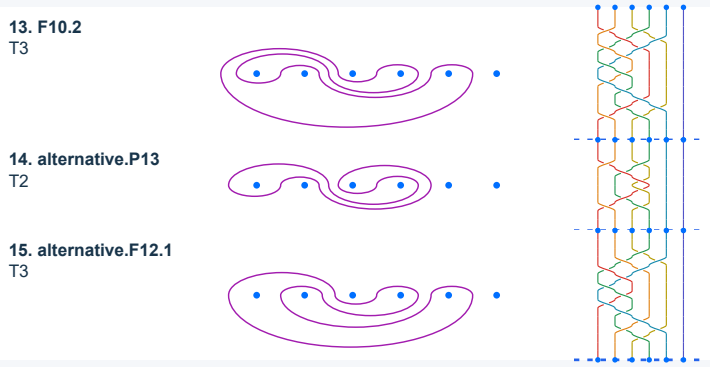
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



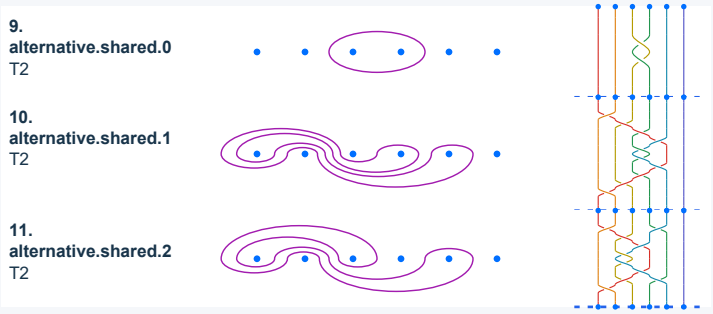
III - step 057

Carry round 34 down through its own two remaining petals

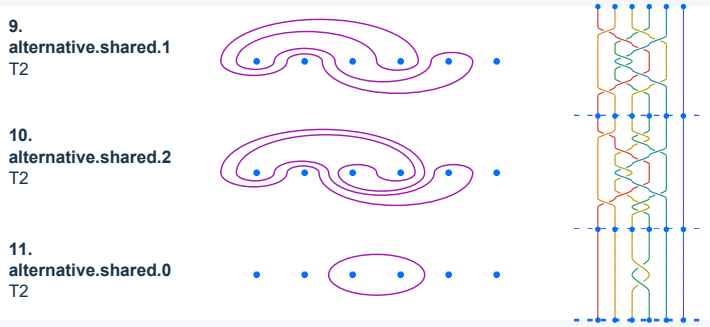
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



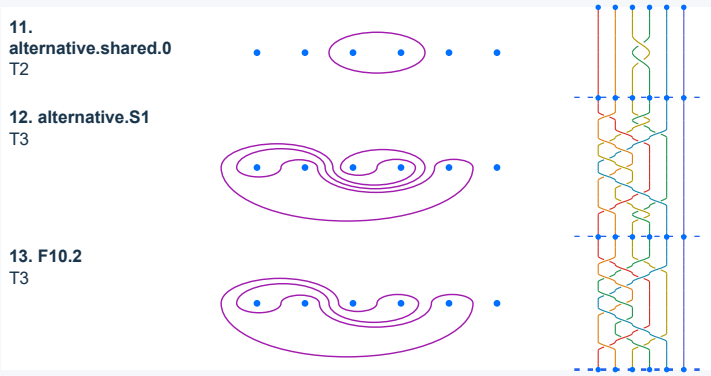
III - step 058

Cabled PC lantern closes the actual 34, 235 and 245 cluster

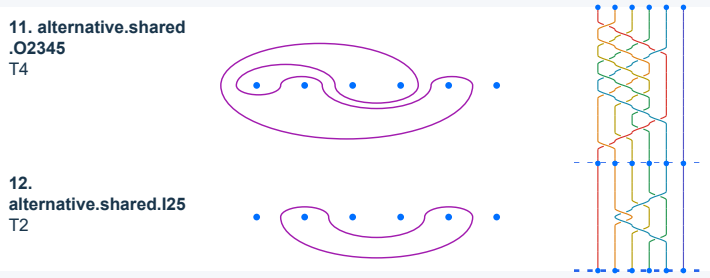
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



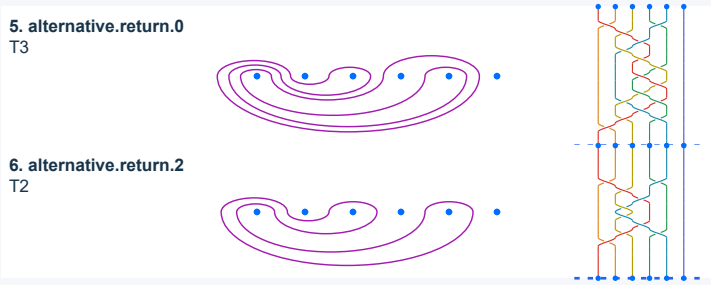
III - step 059

The actual 35 pair unwinds the 134 root to a short chart

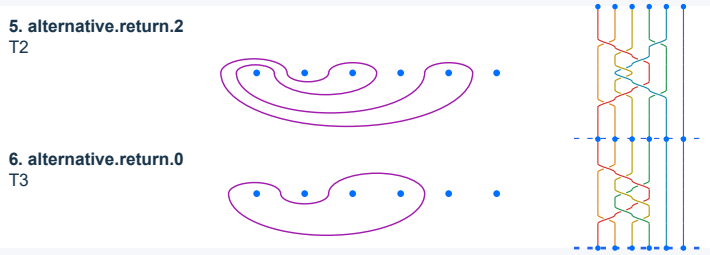
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



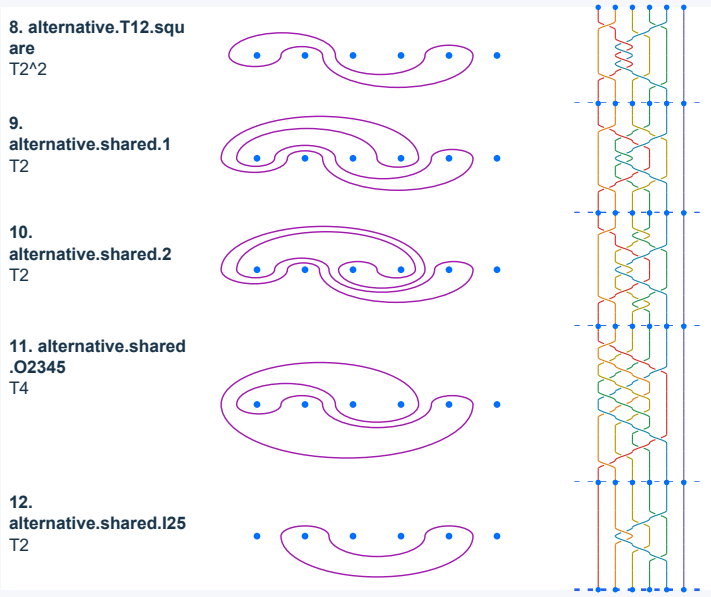
III - step 060

Carry the shared 25 pair up; unwind the square and its two petals

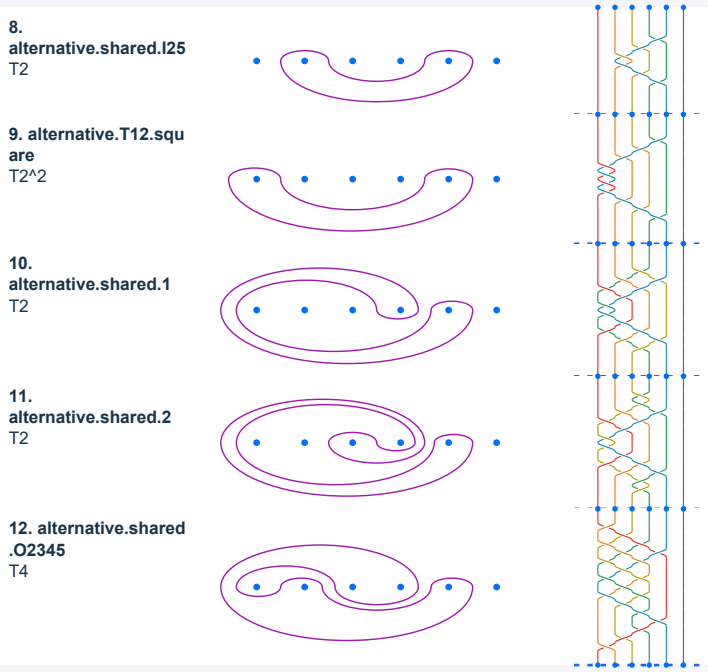
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



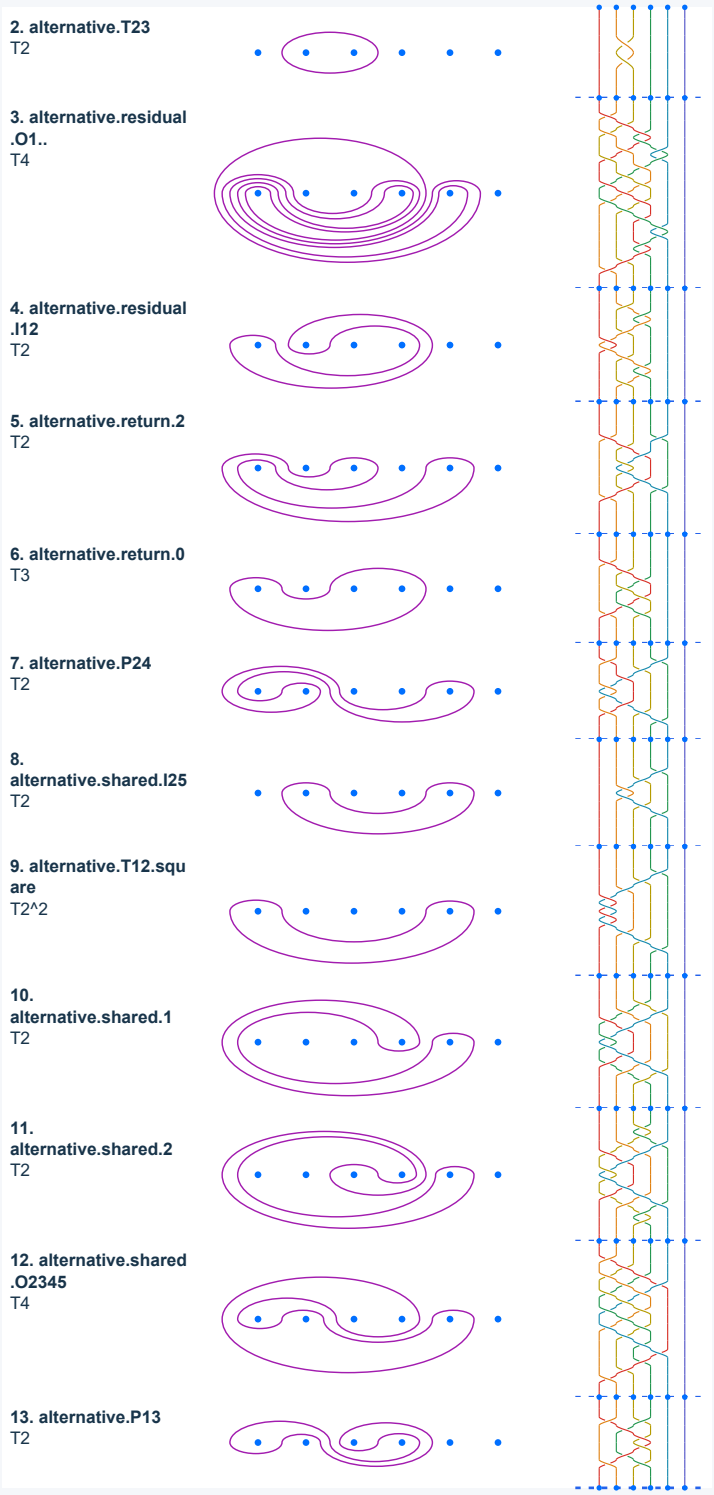
III - step 061

Common disk frame keeps the inside-four daisy readable

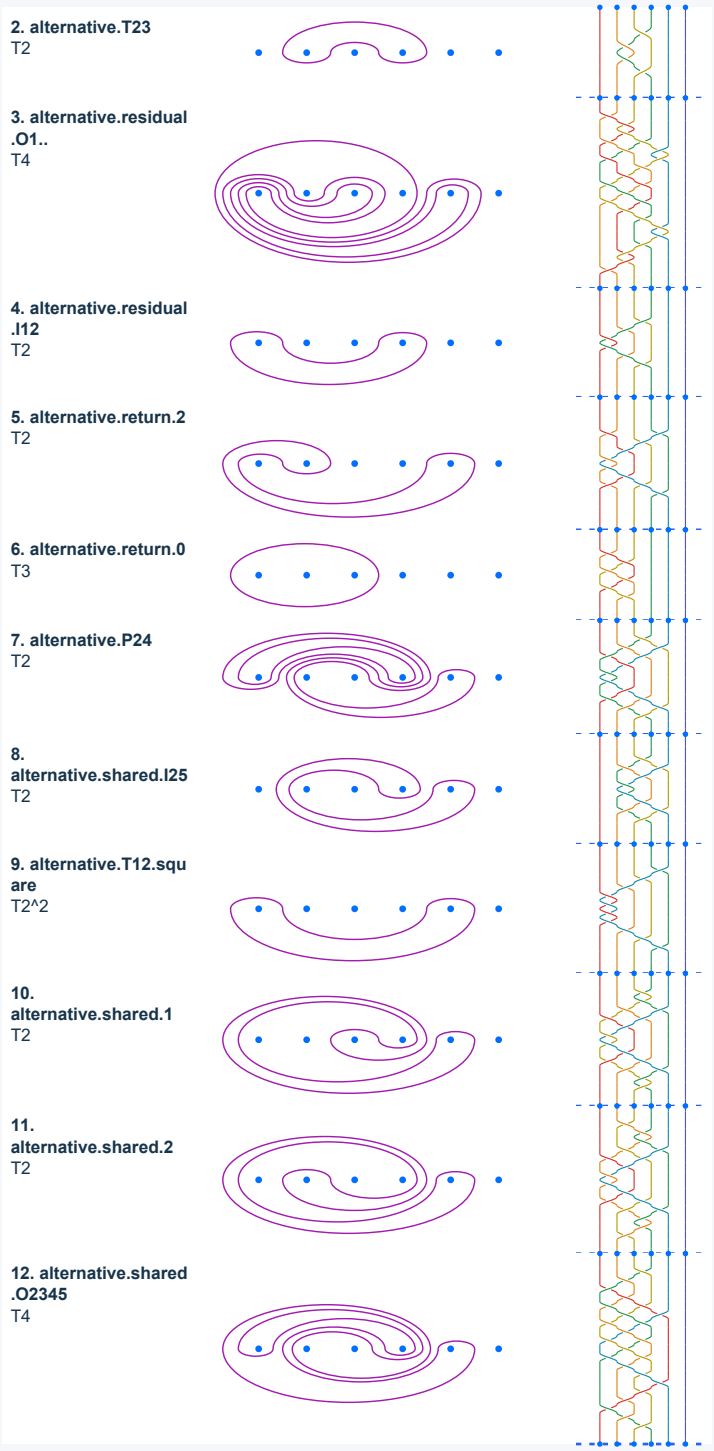
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



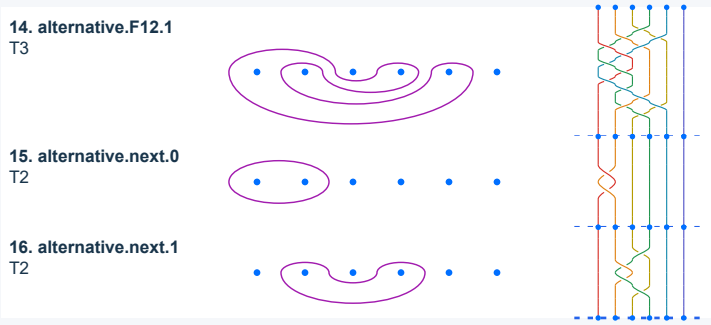
III - step 061 - continued

Common disk frame keeps the inside-four daisy readable

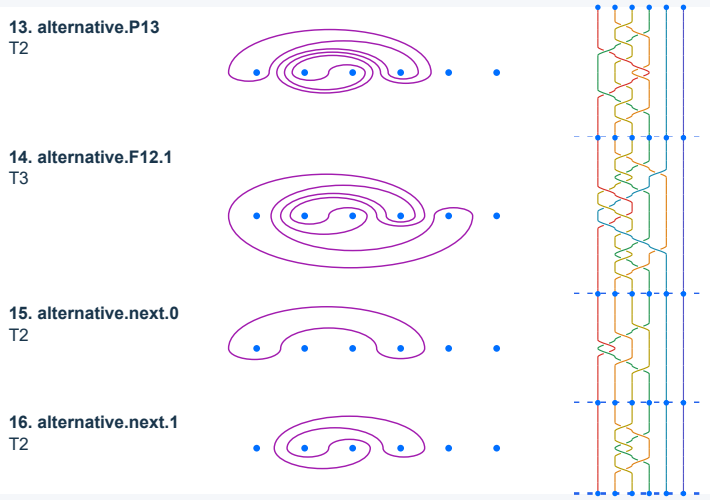
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



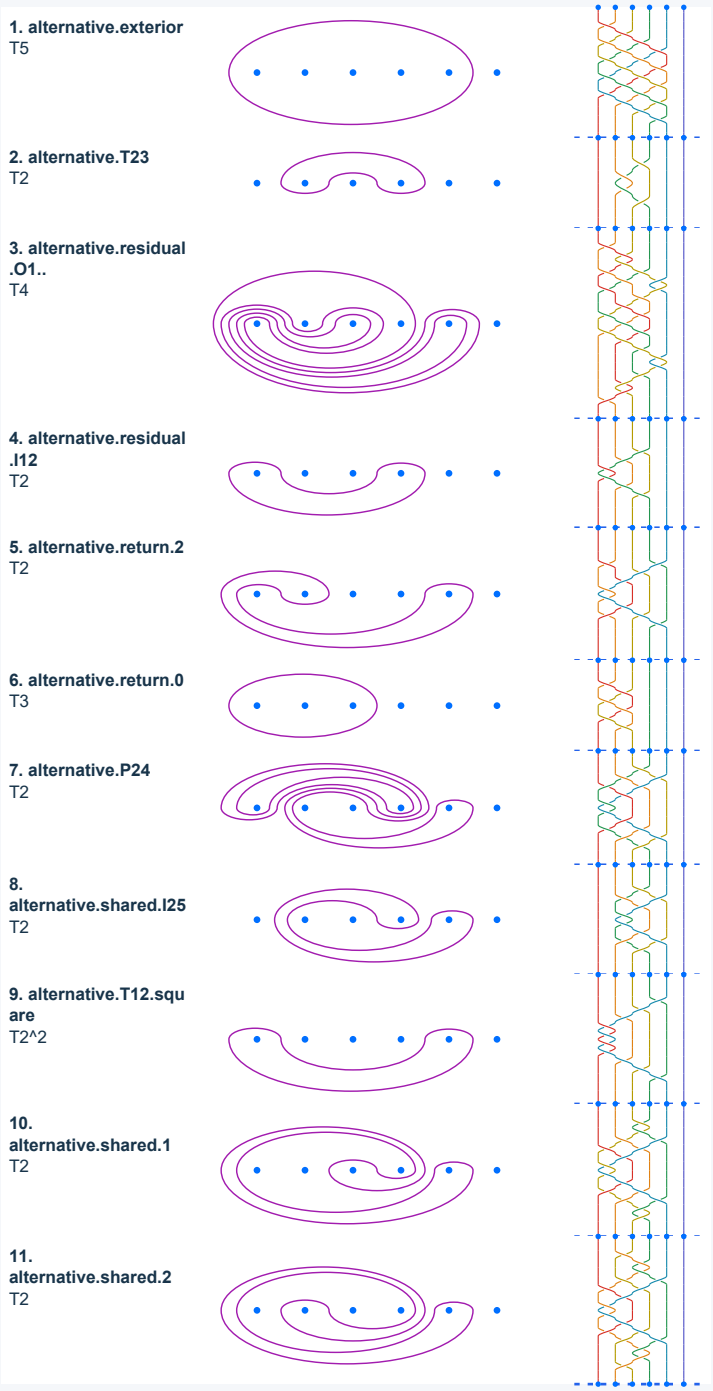
III - step 062

Choose the cyclic start at the actual three-point root

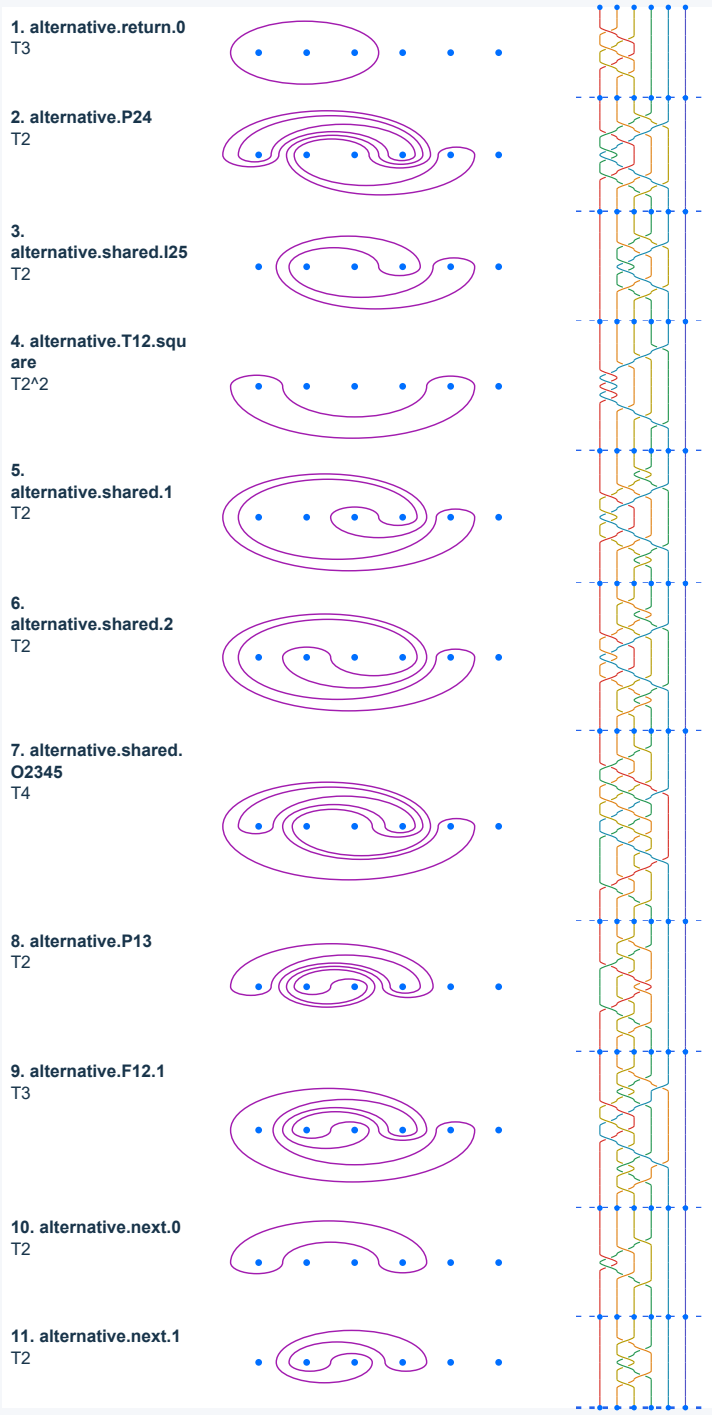
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



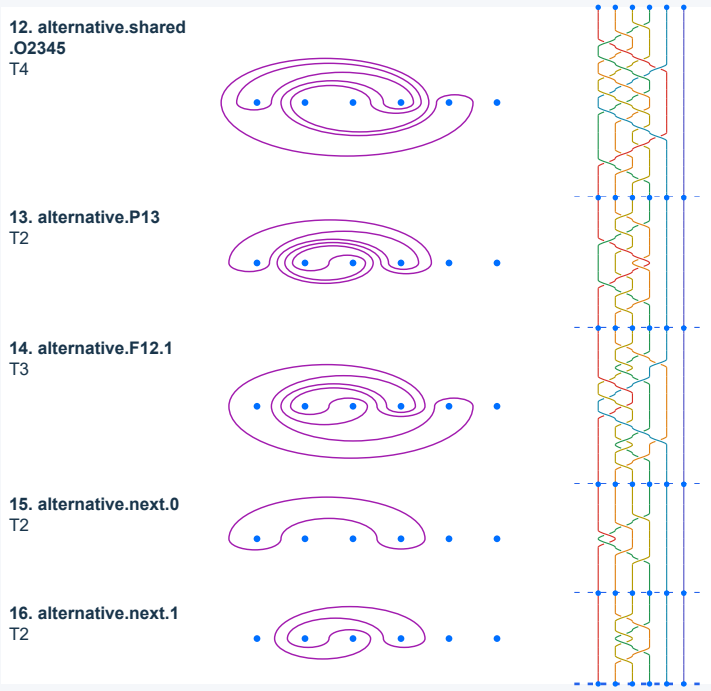
III - step 062 - continued

Choose the cyclic start at the actual three-point root

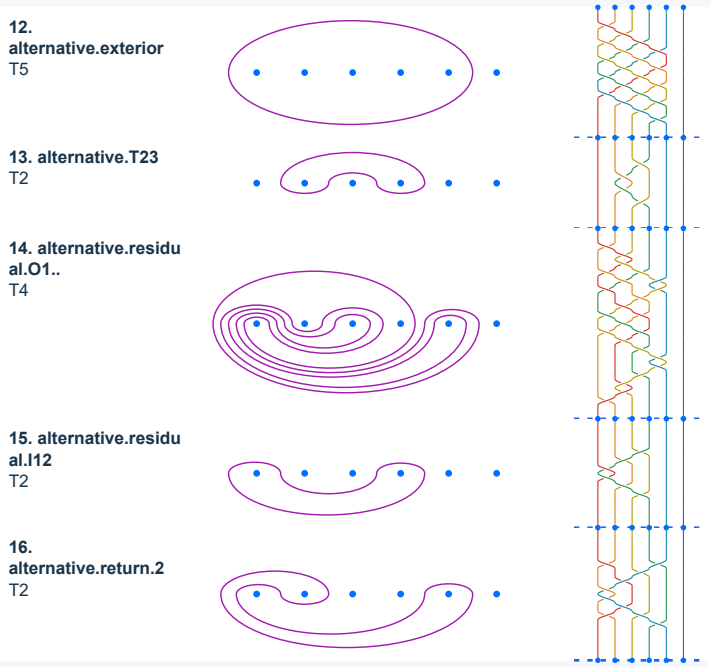
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



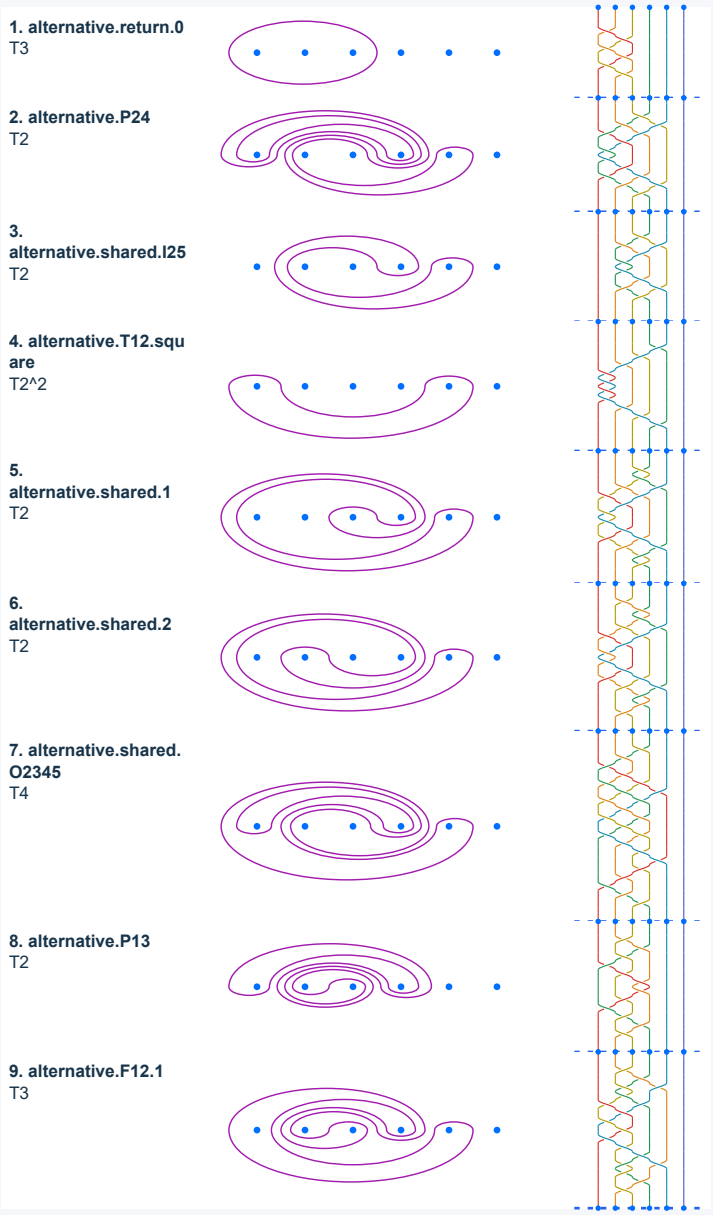
III - step 063

Carry the three-point root down through the eight residual factors

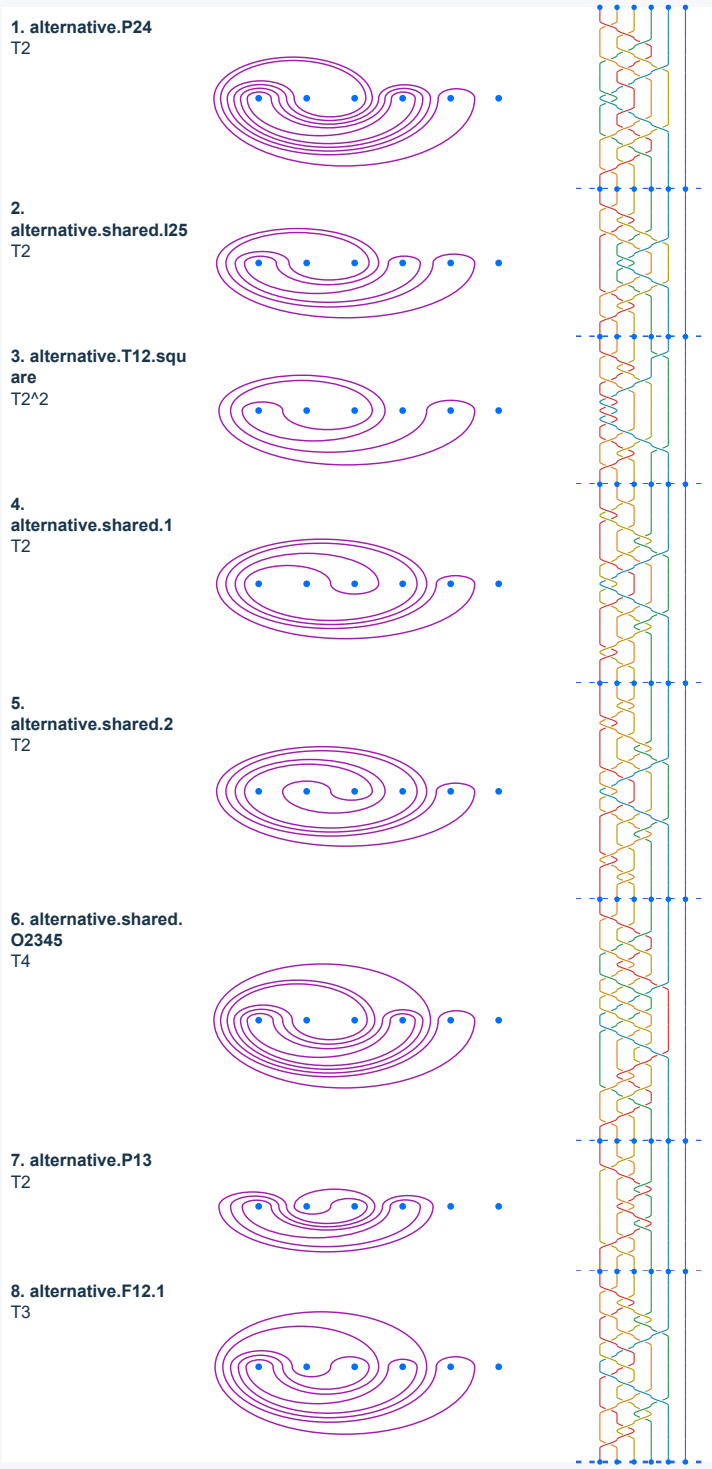
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



III - step 063 - continued

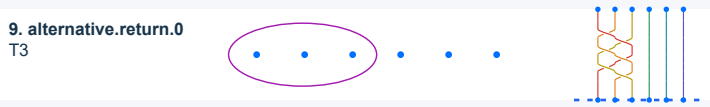
Carry the three-point root down through the eight residual factors

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before

After



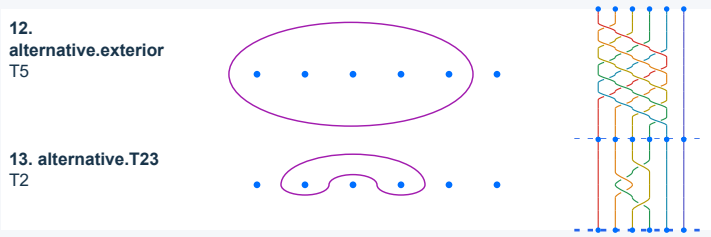
III - step 064

Commute the final pair petal past the visible exterior

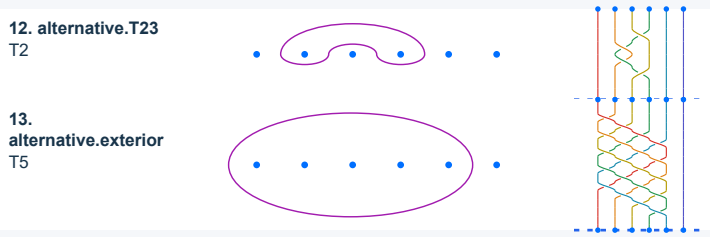
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



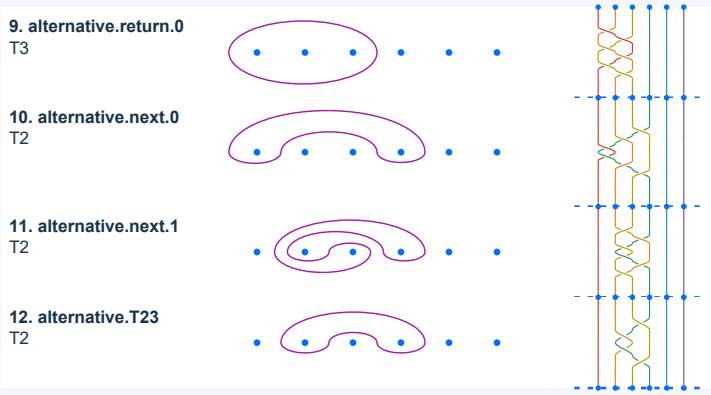
III - step 065

Transported PC daisy closes the actual root and its three pair petals

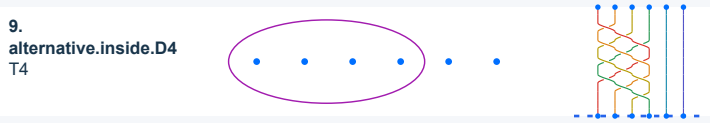
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



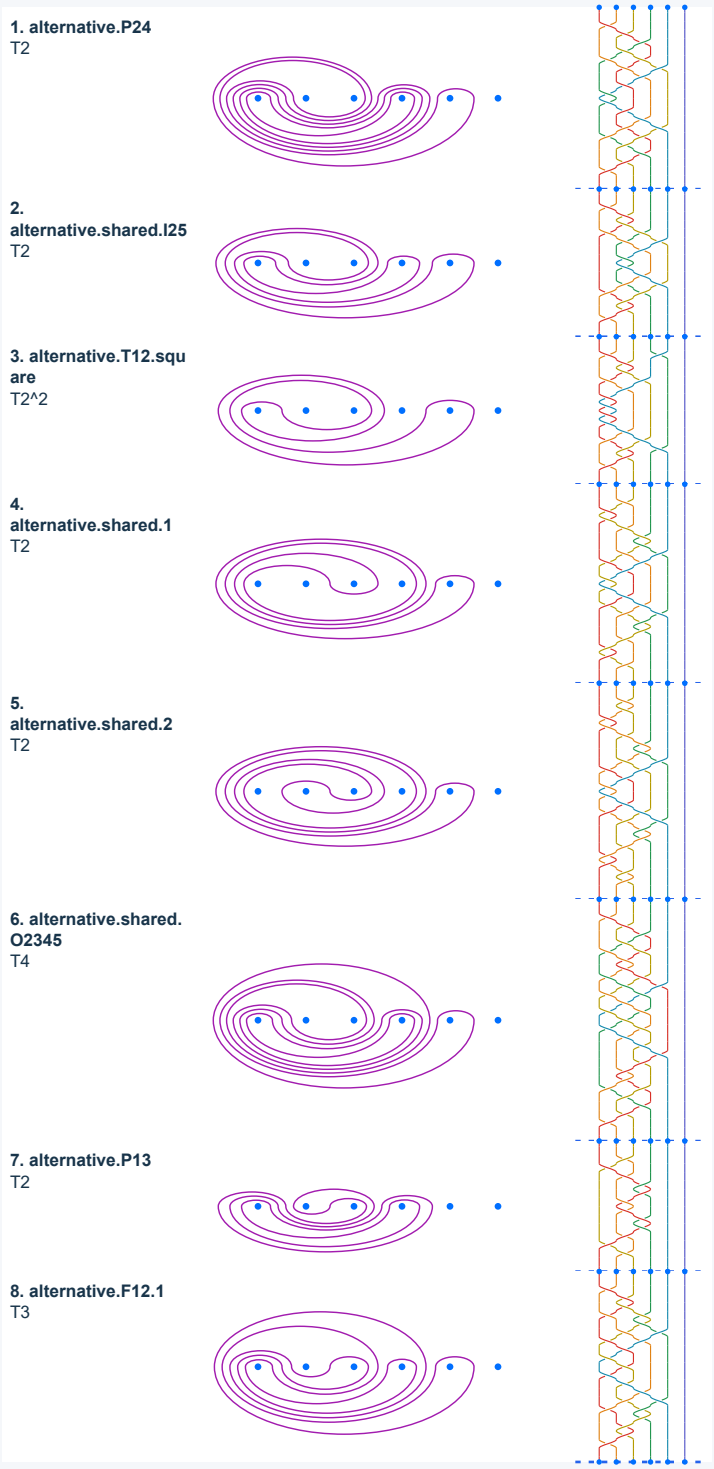
III - step 066

Carry the new round four-point twist up to unwind the residual block

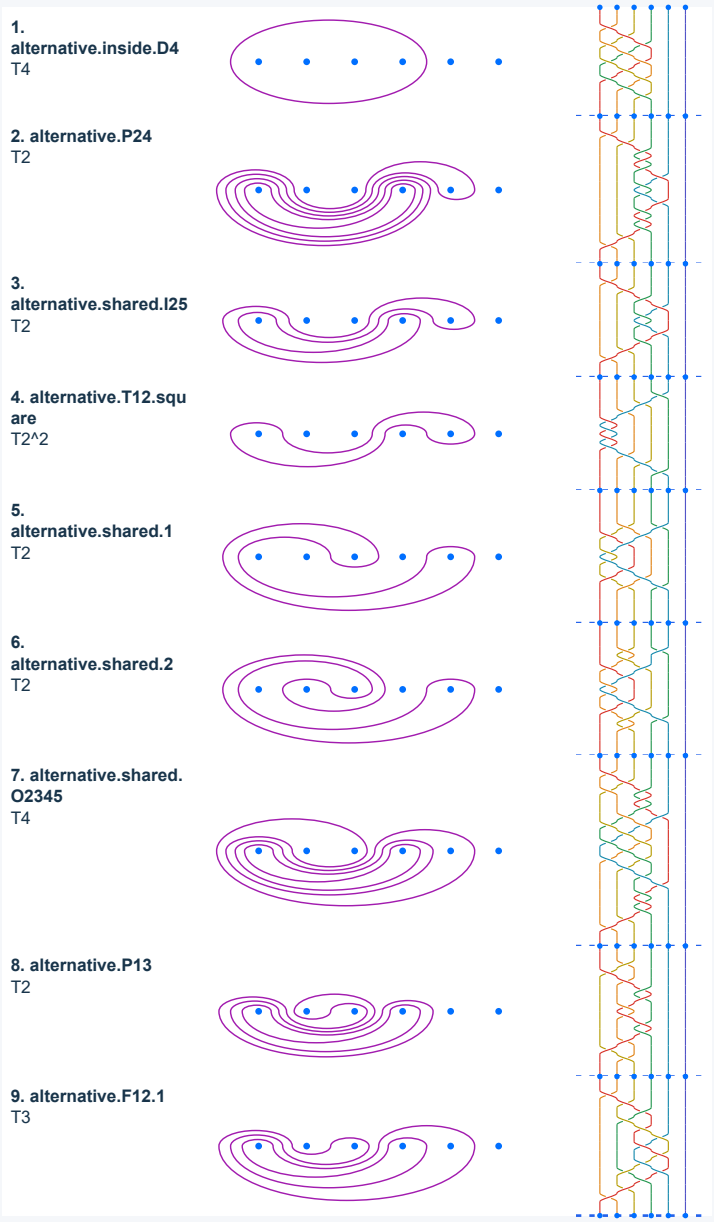
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



III - step 066 - continued

Carry the new round four-point twist up to unwind the residual block

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before

9.
alternative.inside.D4
T4



After

IV. Reach the second exterior twist

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

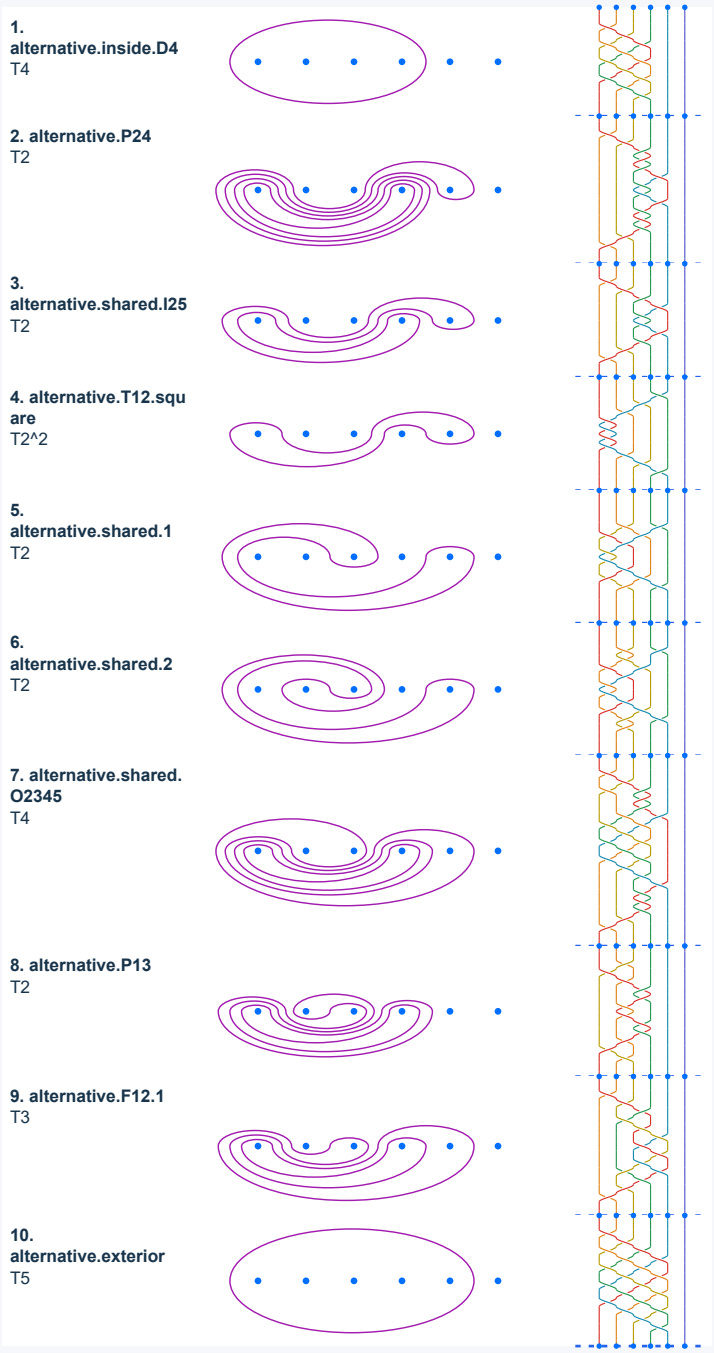
IV - step 067

Common disk frame keeps the second exterior lantern readable

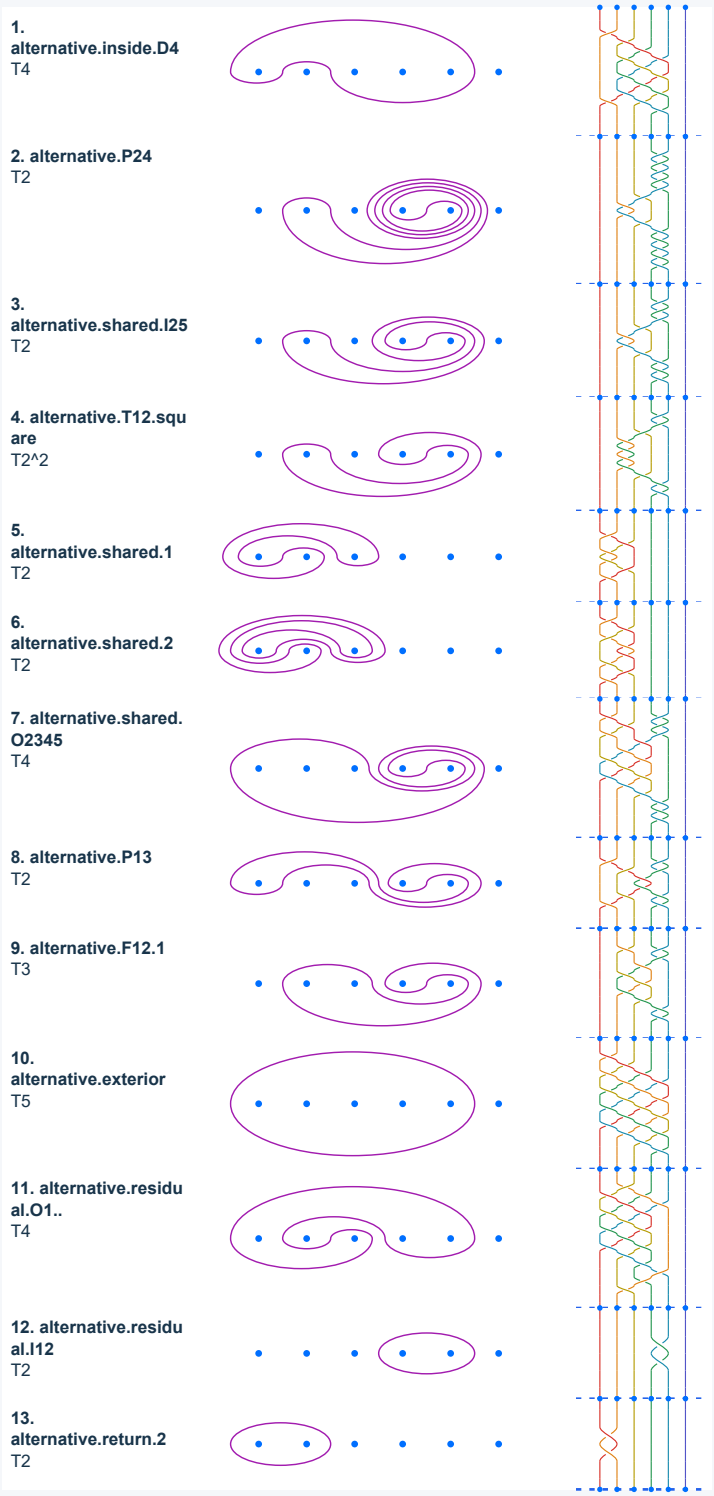
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



IV - step 067 - continued

Common disk frame keeps the second exterior lantern readable

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After

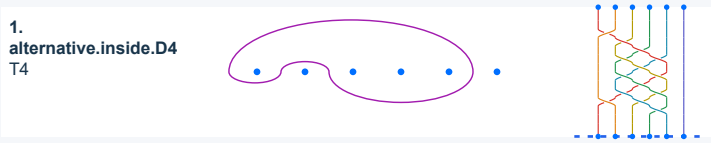
IV - step 068

The four-point PC daisy lends its actual three-point root

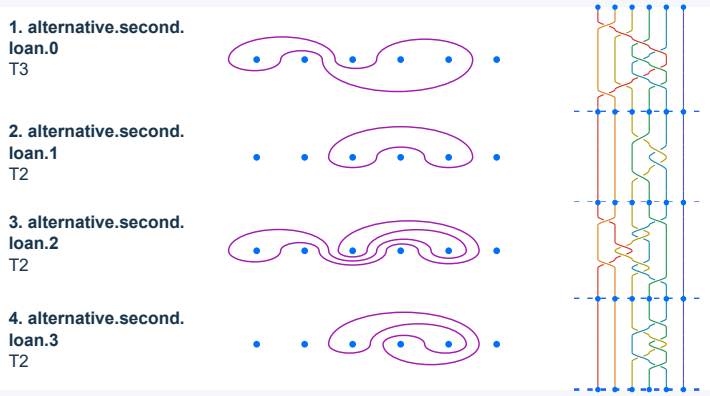
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



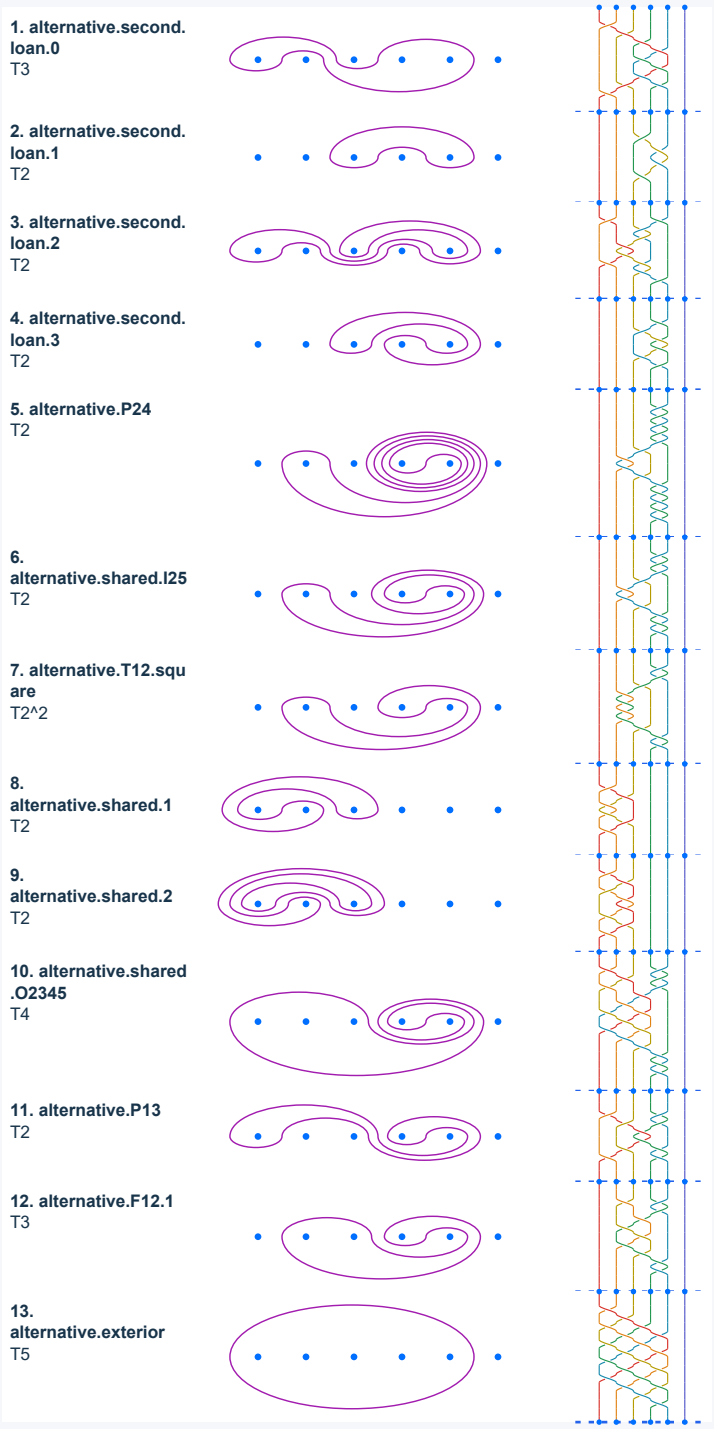
IV - step 069

Cyclic start at the actual pair that transports the borrowed root

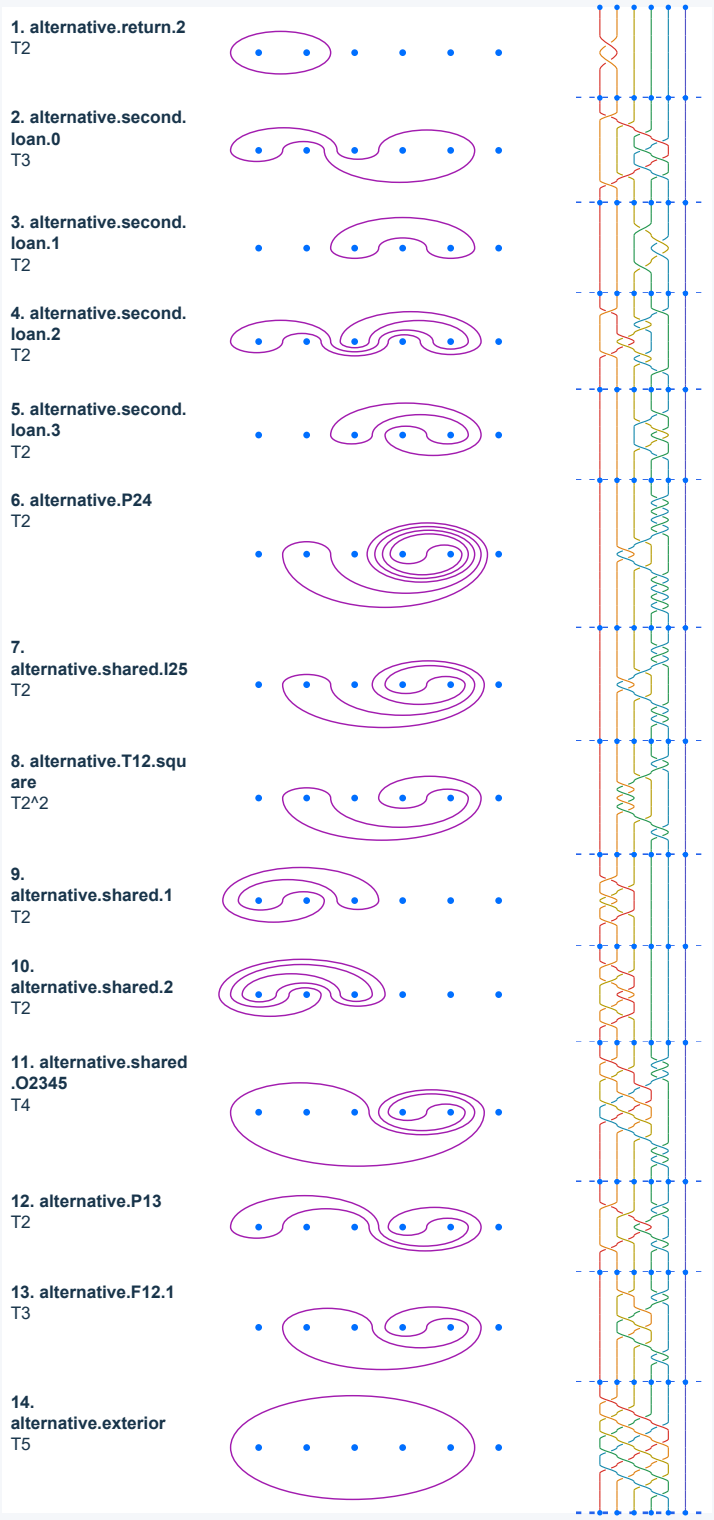
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



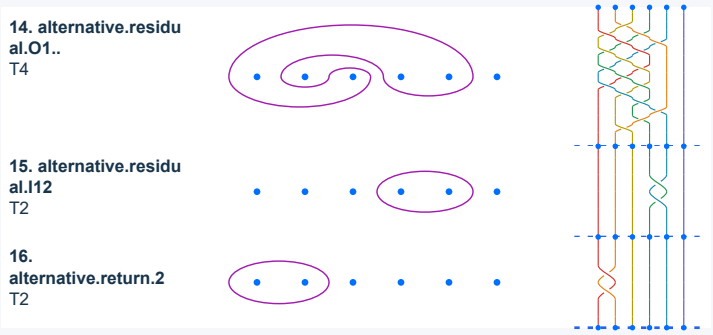
IV - step 069 - continued

Cyclic start at the actual pair that transports the borrowed root

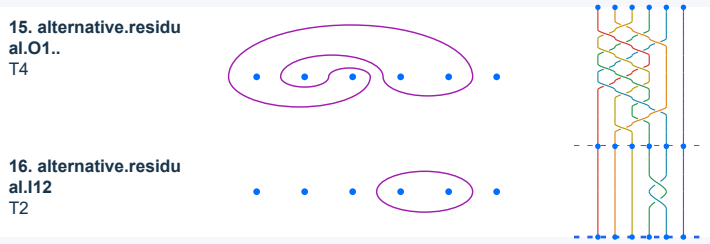
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



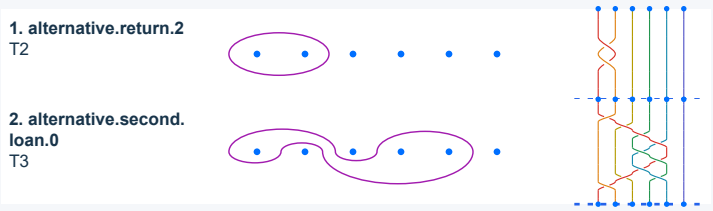
IV - step 070

The actual pair carries the borrowed root to the exact missing class

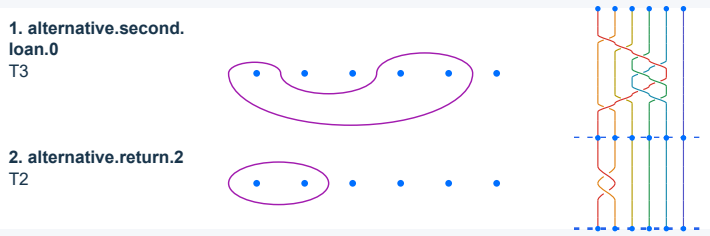
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



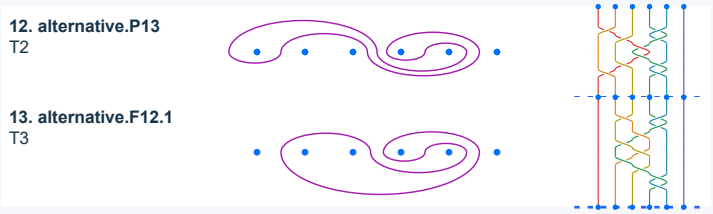
IV - step 071

Bring the three-point curve up across one neighboring pair

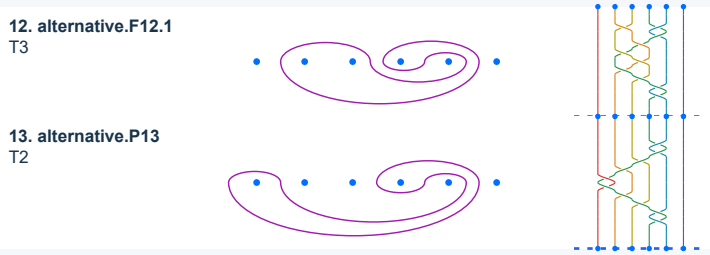
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



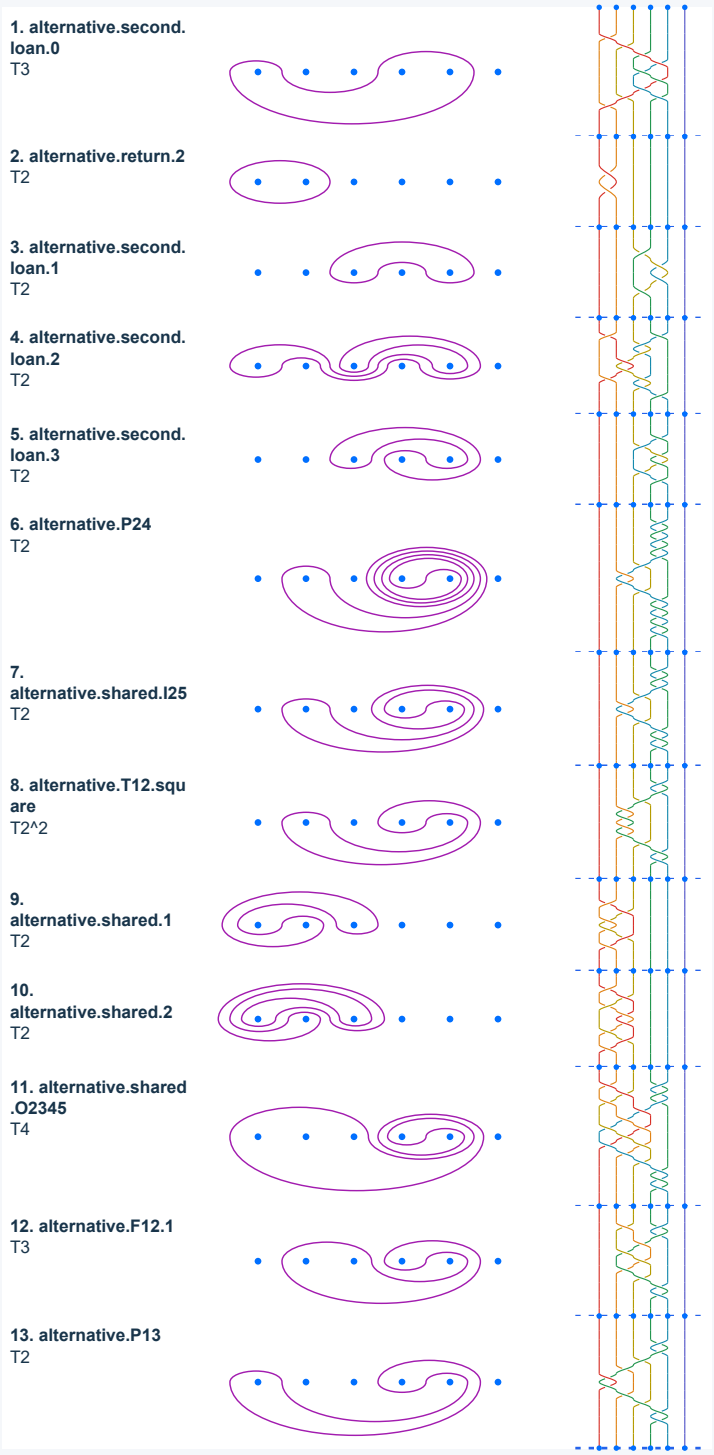
IV - step 072

Cyclic start at the matched four-point curve

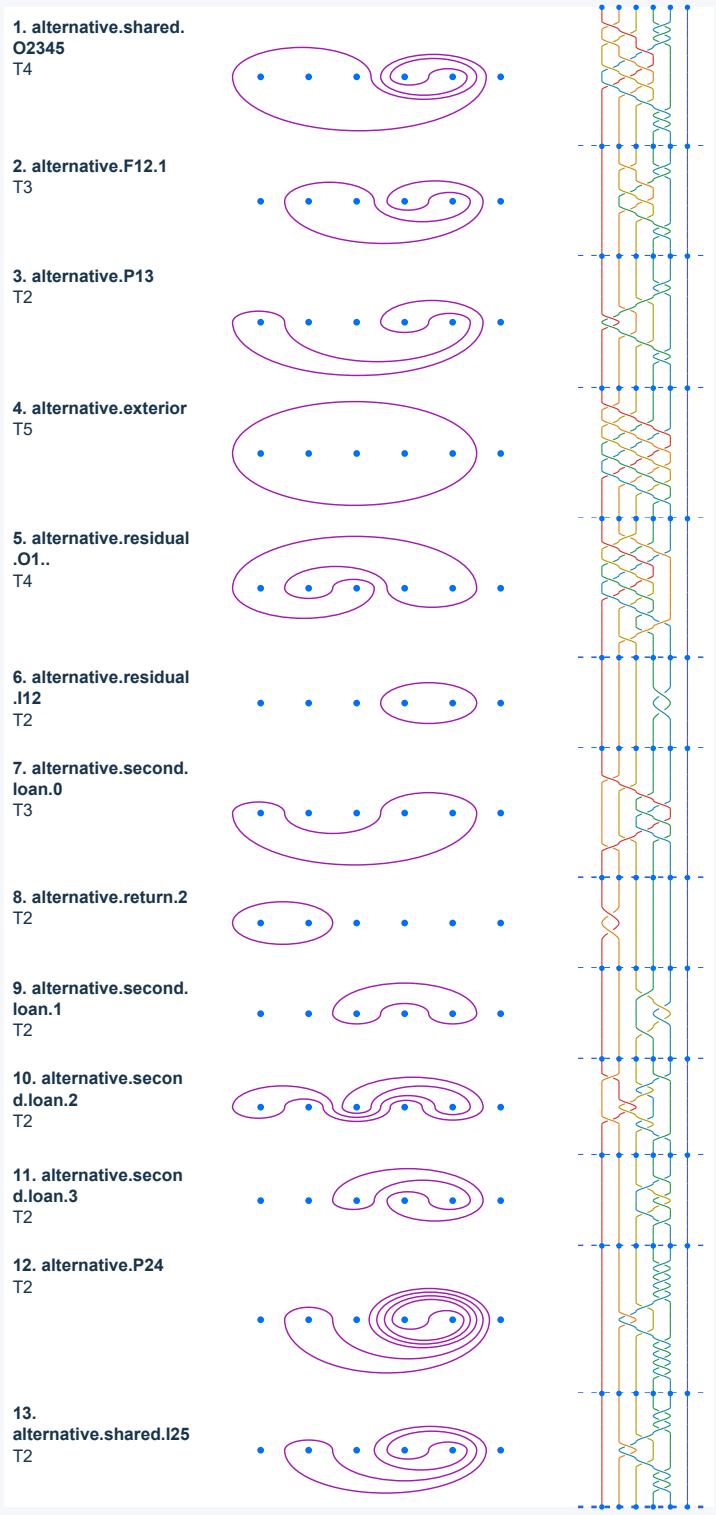
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



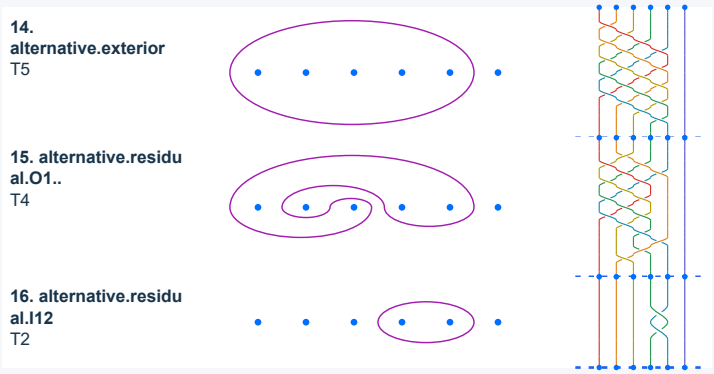
IV - step 072 - continued

Cyclic start at the matched four-point curve

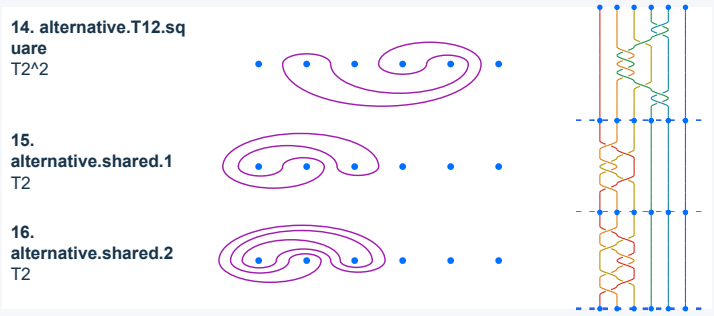
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



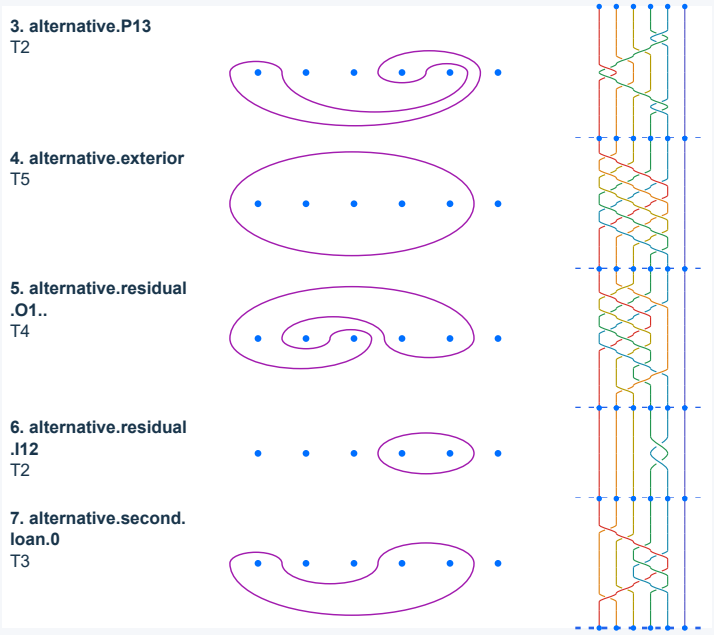
IV - step 073

Gather the borrowed root up the four-neighbor short side

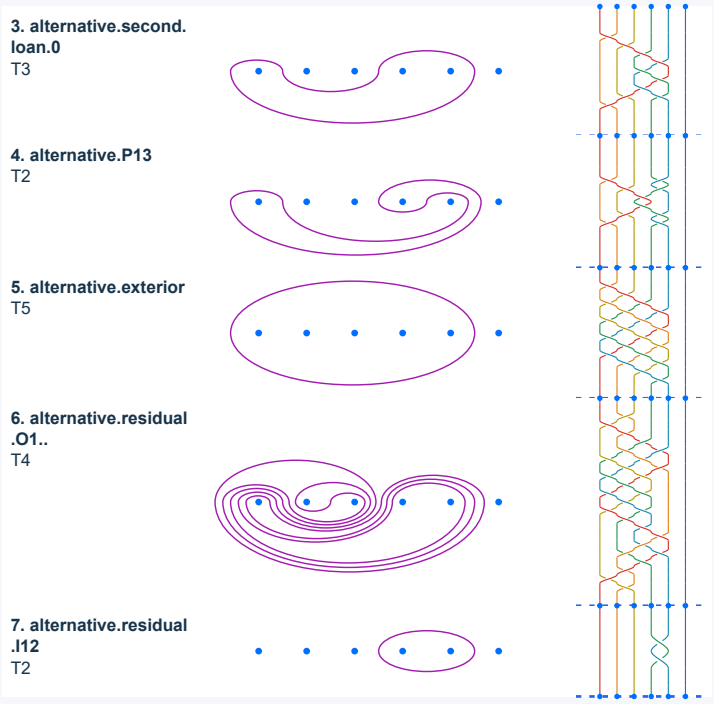
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



IV - step 074

Cabled PC lantern reaches a SECOND visible exterior twist

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



V. Return the loan as an expanded rose

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before Return the loan as an expanded rose

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. alternative.second.exte..
T5

2. alternative.second.l35
T2

3. alternative.second.l24
T2

4. alternative.P13
T2

5. alternative.exterior
T5

6. alternative.residual.O1..
T4

7. alternative.residual.l12
T2

8. alternative.return.2
T2

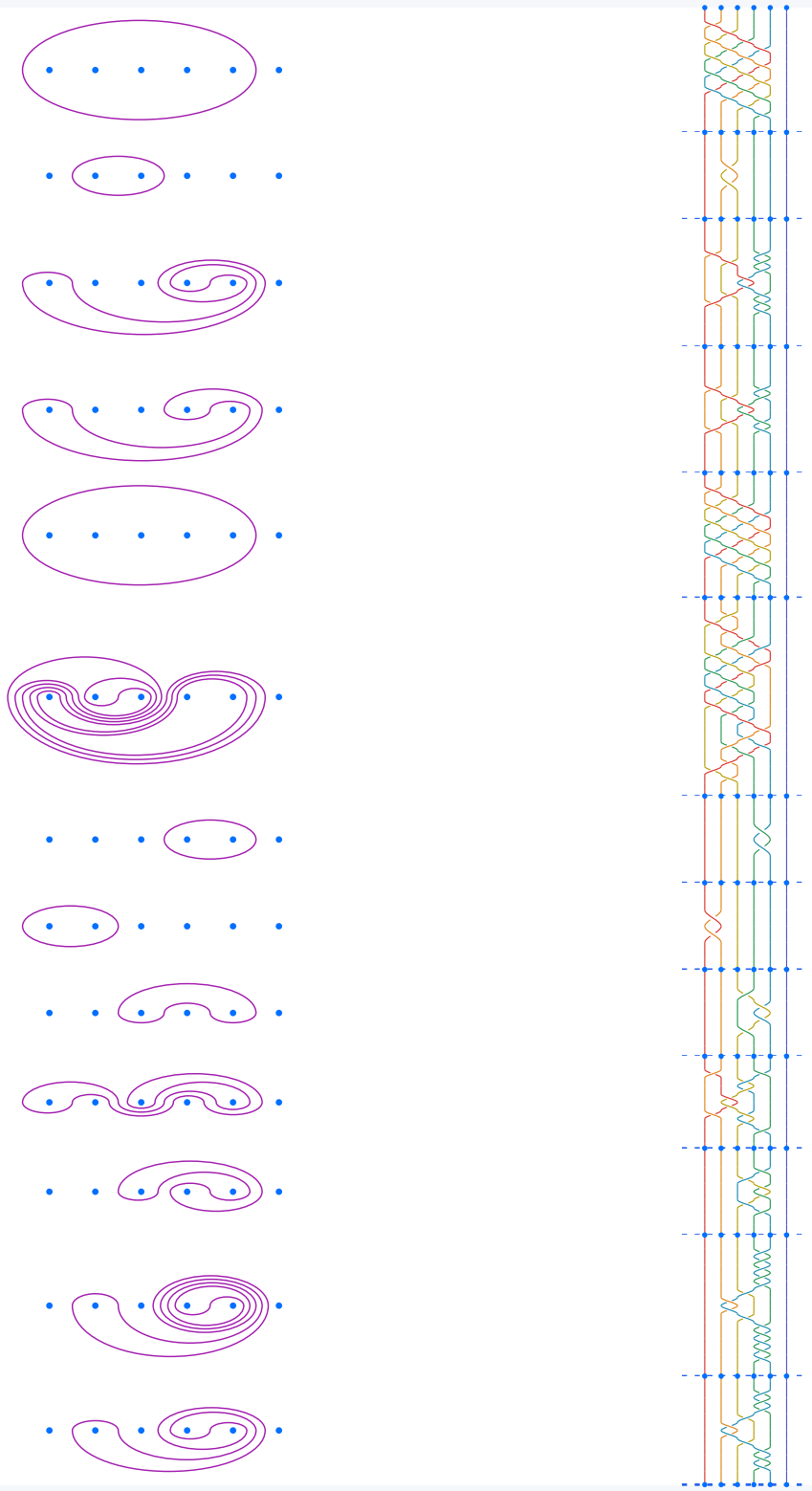
9. alternative.second.loan.1
T2

10. alternative.second.loan.2
T2

11. alternative.second.loan.3
T2

12. alternative.P24
T2

13. alternative.shared.l25
T2



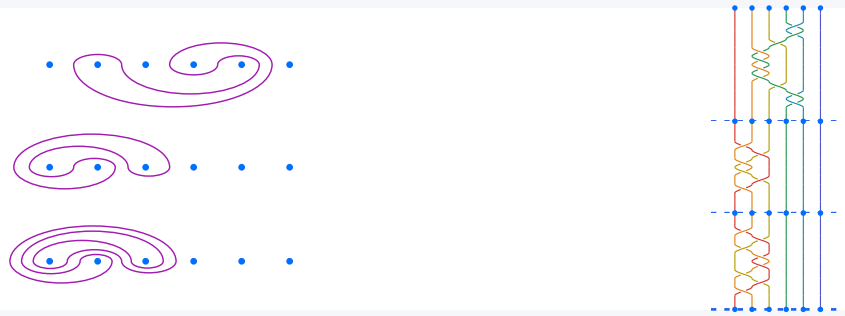
Complete factorization before Return the loan as an expanded rose - continued

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

14. alternative.T12.square
T2^2

15. alternative.shared.1
T2

16. alternative.shared.2
T2



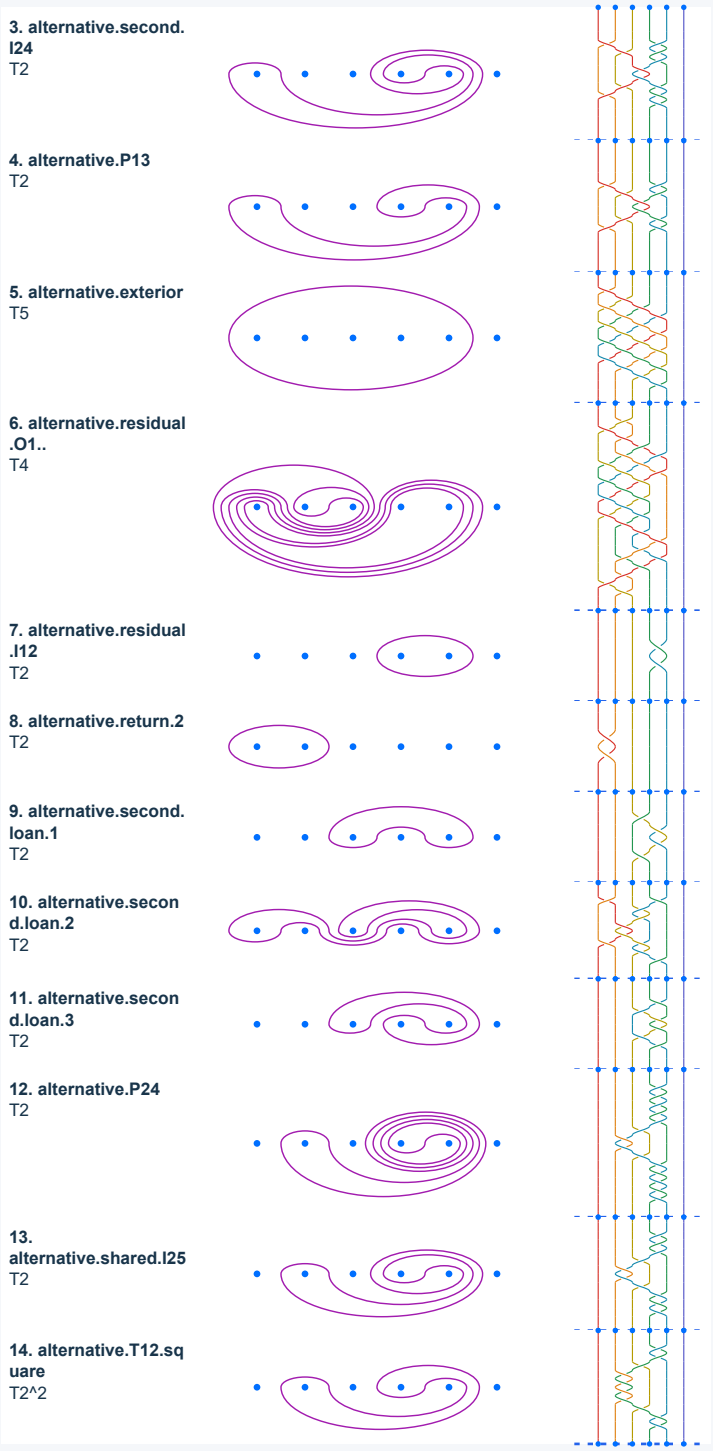
V - step 075

Positive T45 common frame bounds the unreordered loan return

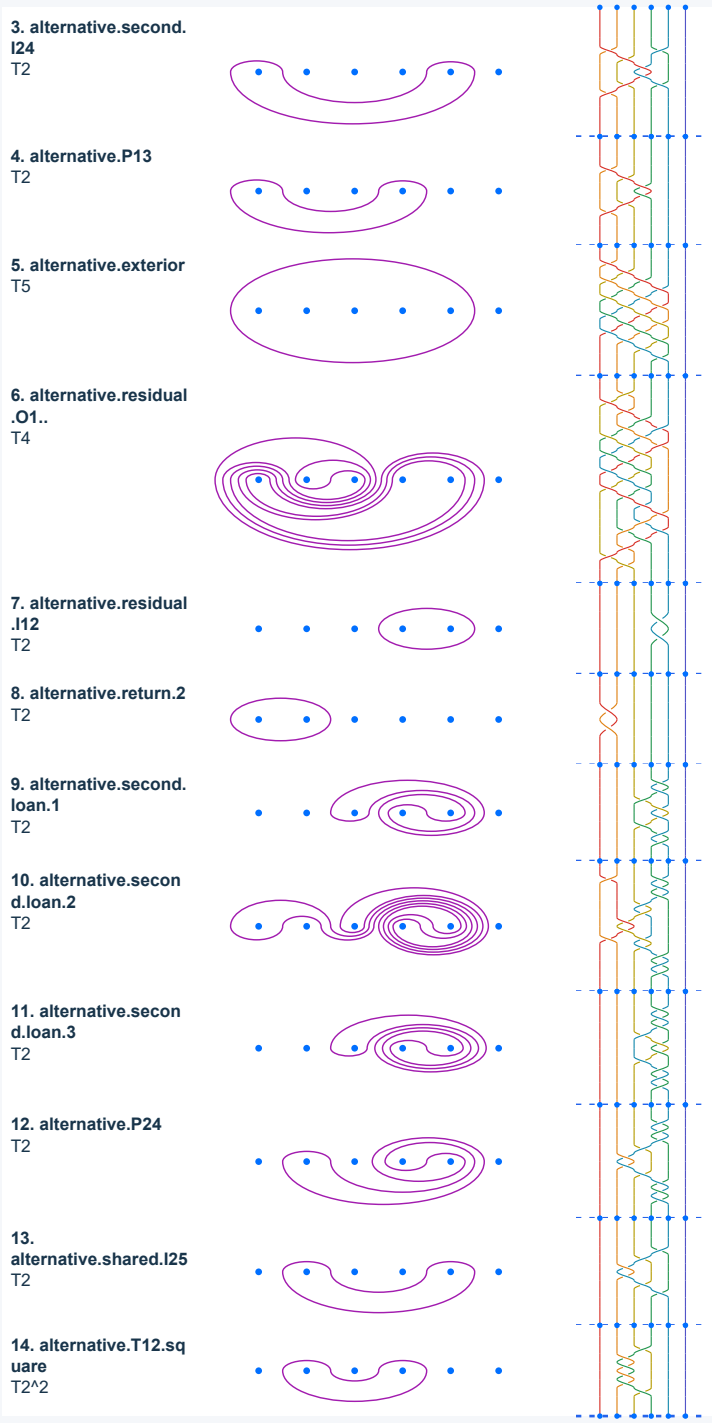
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



V - step 076

Commute the exterior out of the expanded loan

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 11 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



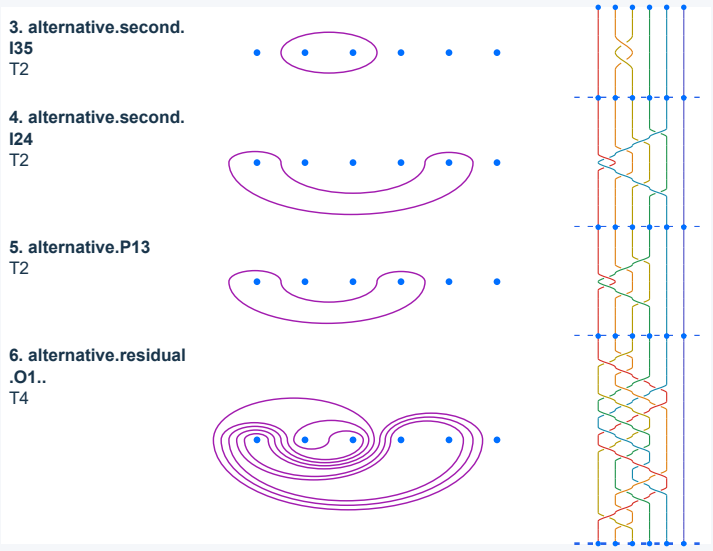
V - step 077

Inverse Hurwitz transport of the SAME outer past three petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 3. Inverse transport: the same outer moves upward and is conjugated at each crossed petal.

Before



After



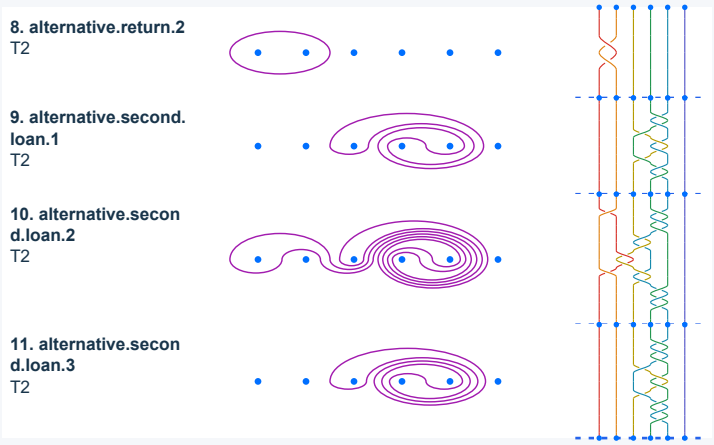
V - step 078

Return the12 petal through the three actual loan petals

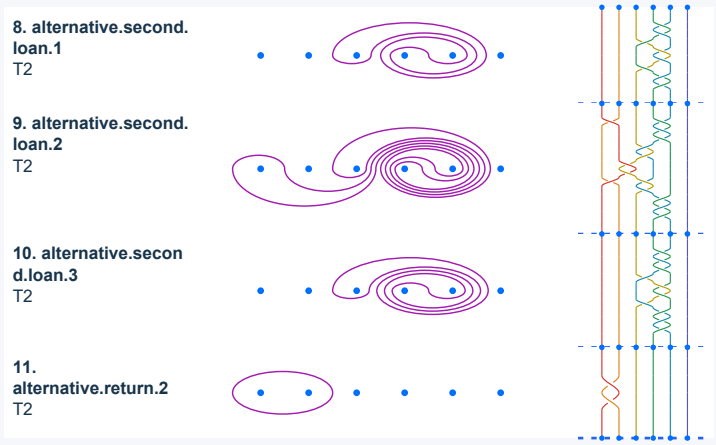
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



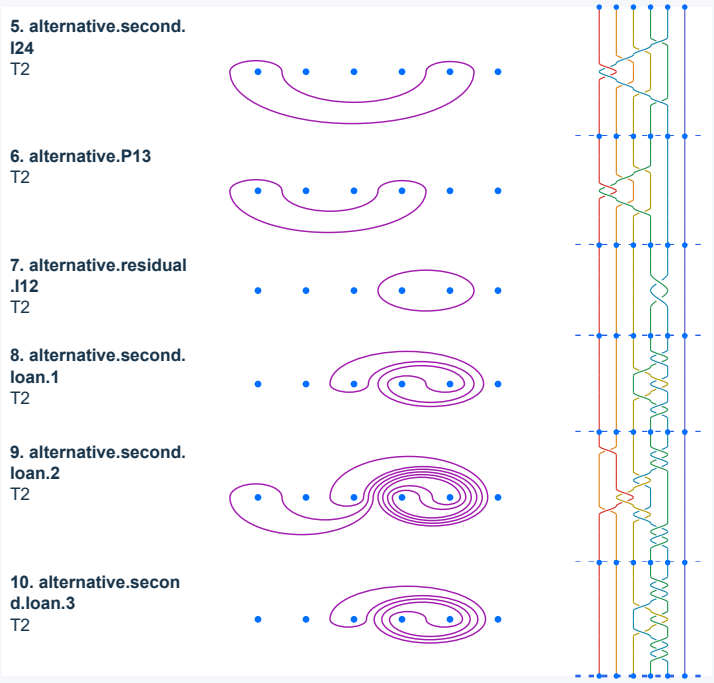
V - step 079

Expanded PC rose returns the loan WITHOUT a petal reorder

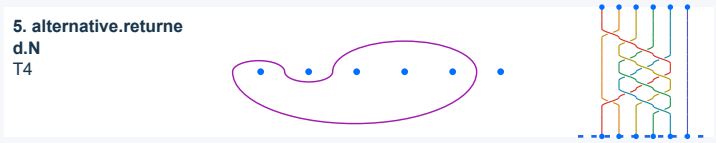
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



VI. Carry the first outer upward

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

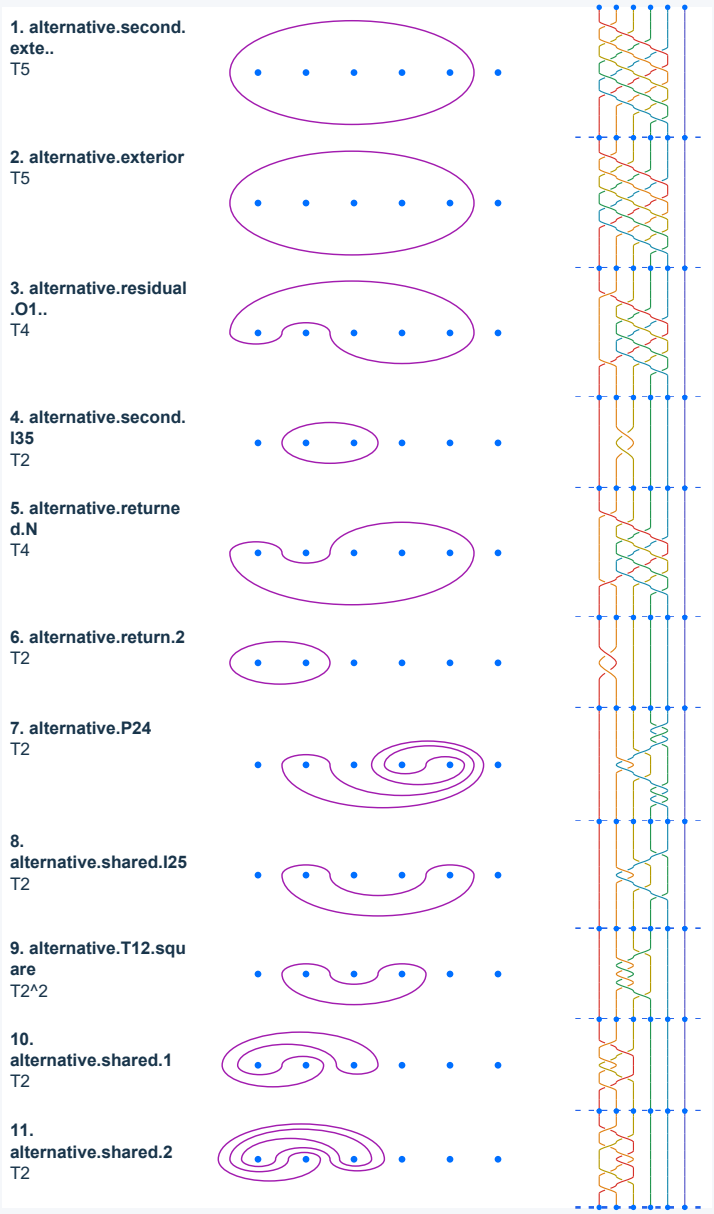
VI - step 080

Cyclic start BEFORE gathering the first daisy

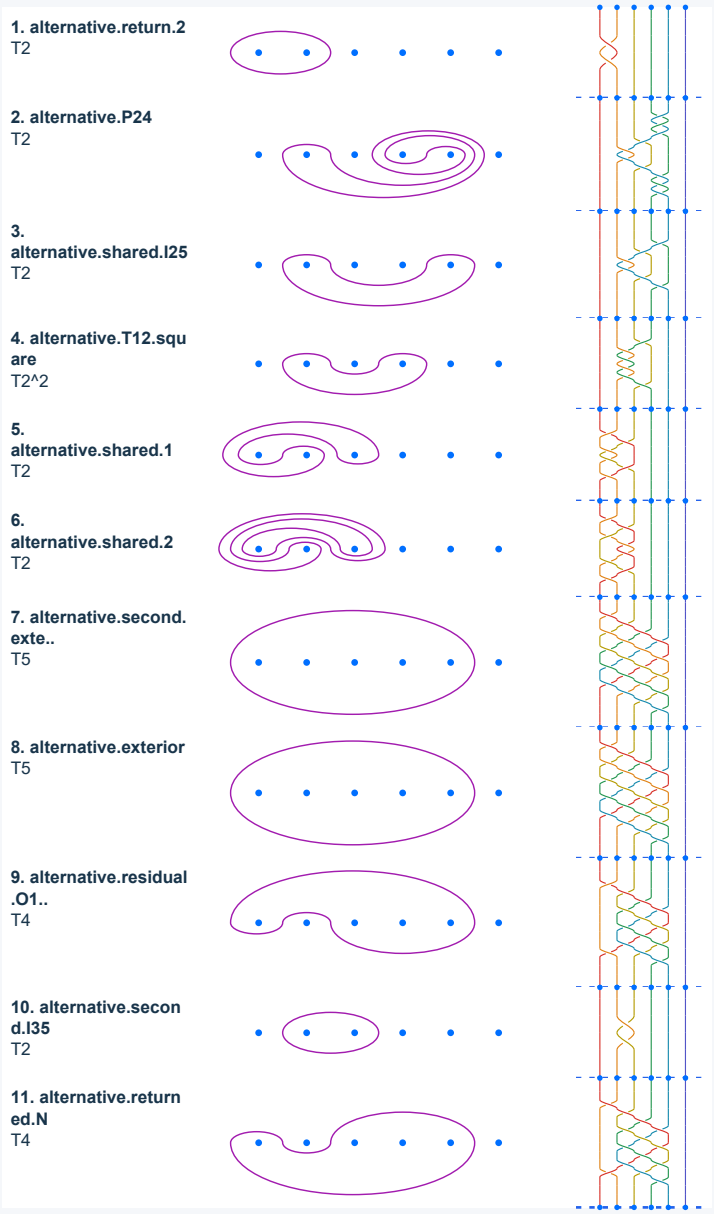
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



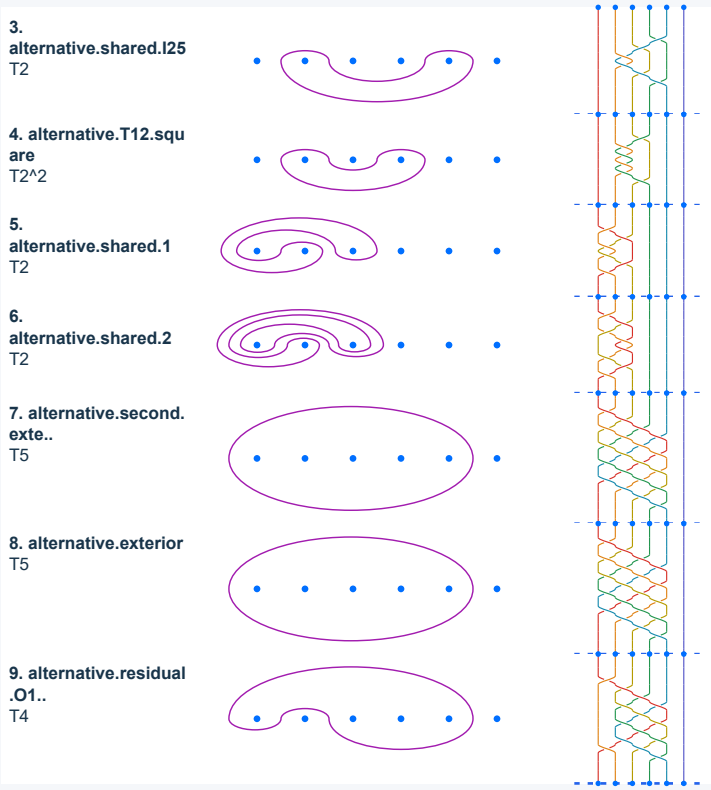
VI - step 081

Carry the SAME outer UP through its first-daisy petals

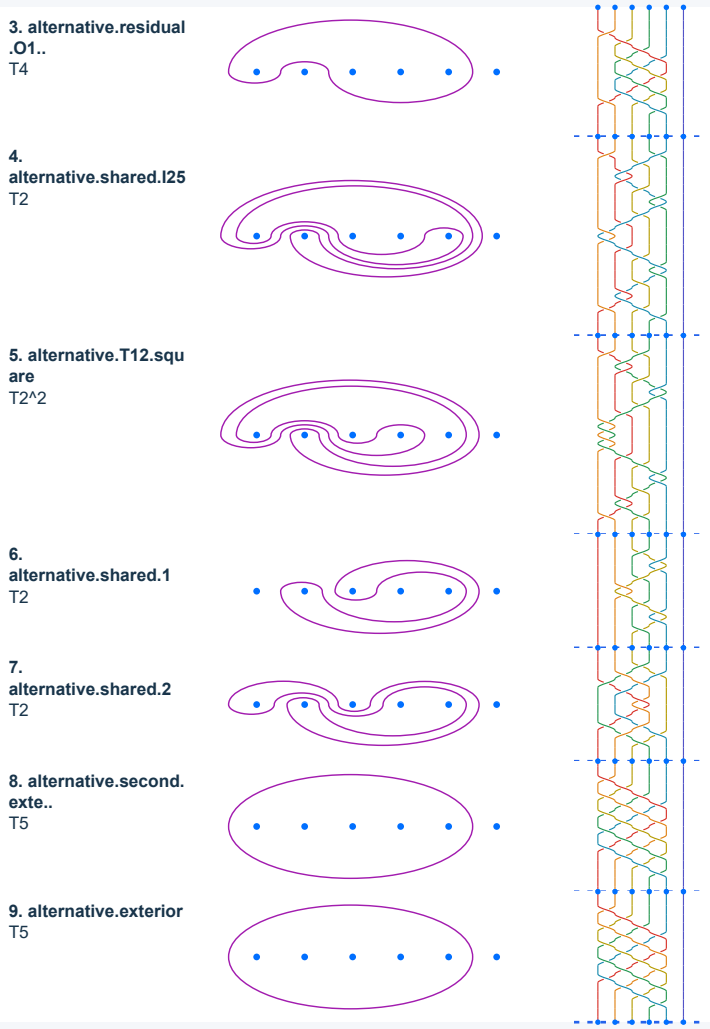
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



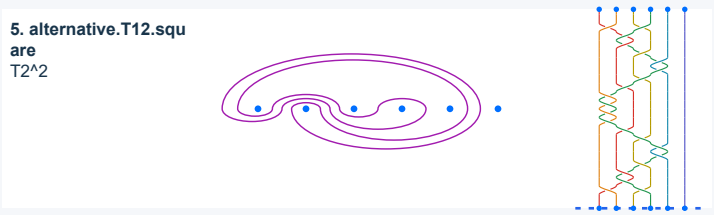
VI - step 082

Split alternative.T12.square into its two positive units

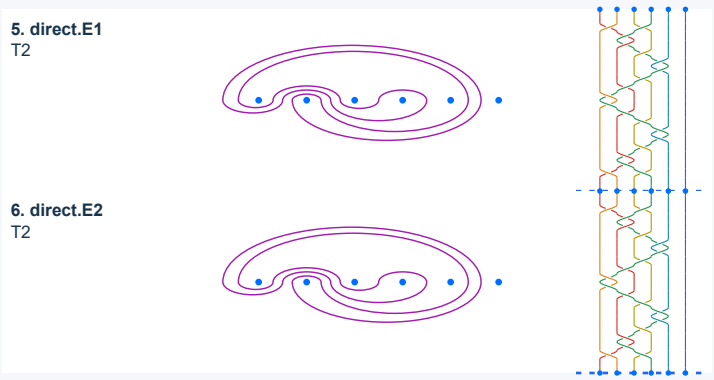
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



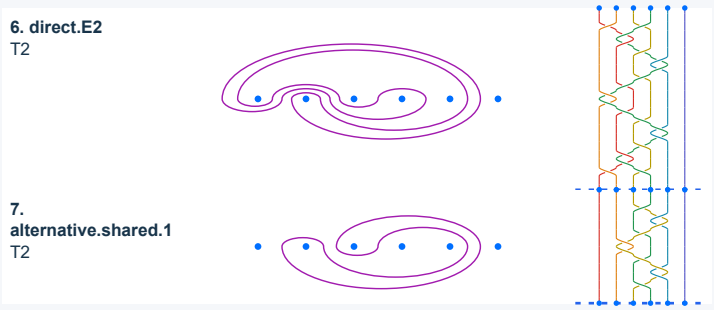
VI - step 083

Keep the23 petal fixed while returning the extra square unit

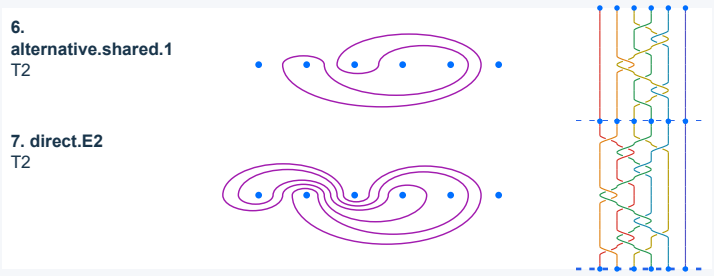
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



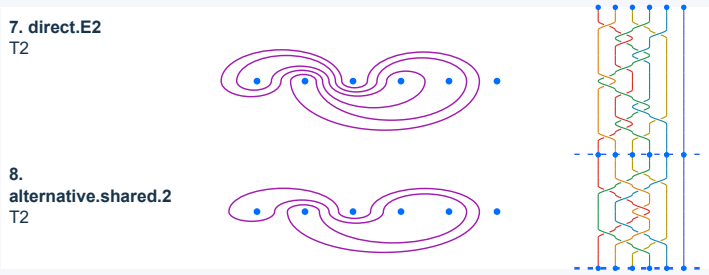
VI - step 084

Keep the12 petal fixed while returning the extra square unit

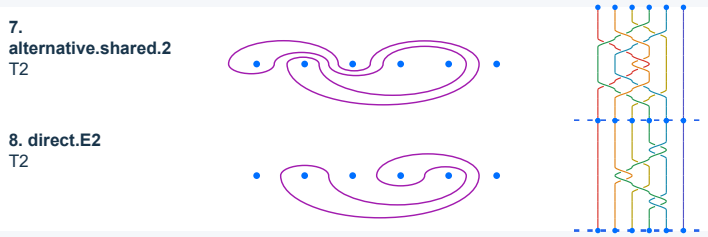
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



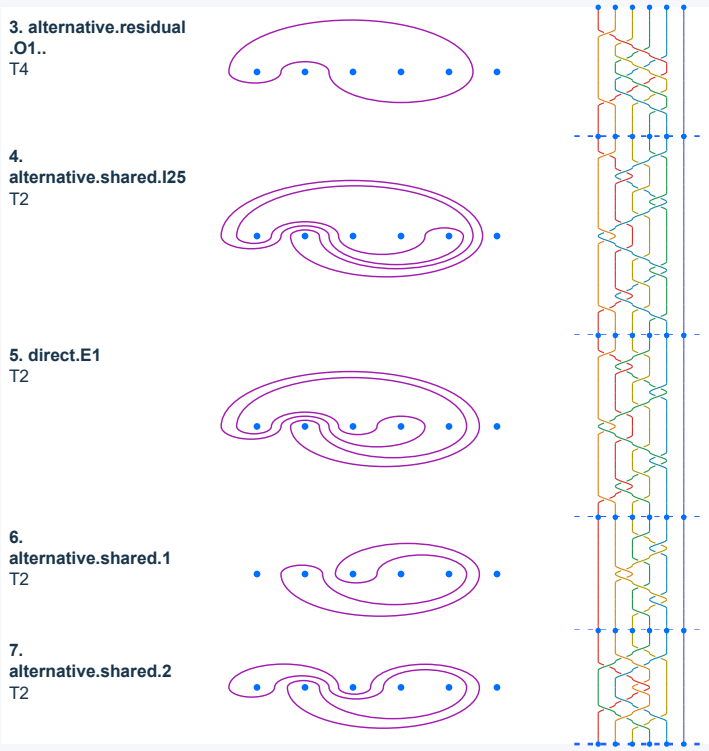
VI - step 085

FIRST actual PC daisy after the upward outer transport

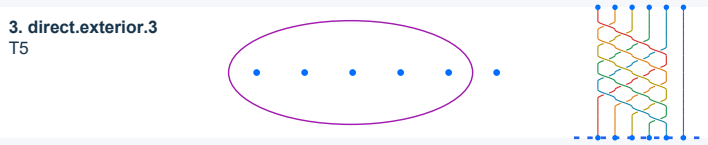
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



VII. Close the second daisy in cyclic order

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

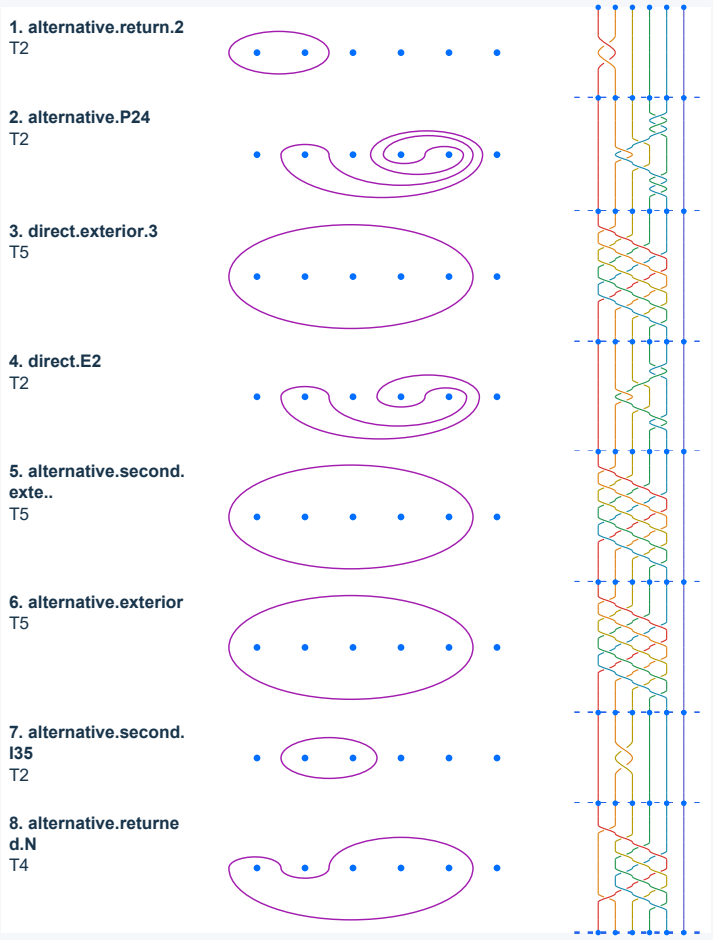
VII - step 086

Cyclic start at the NEW visible exterior

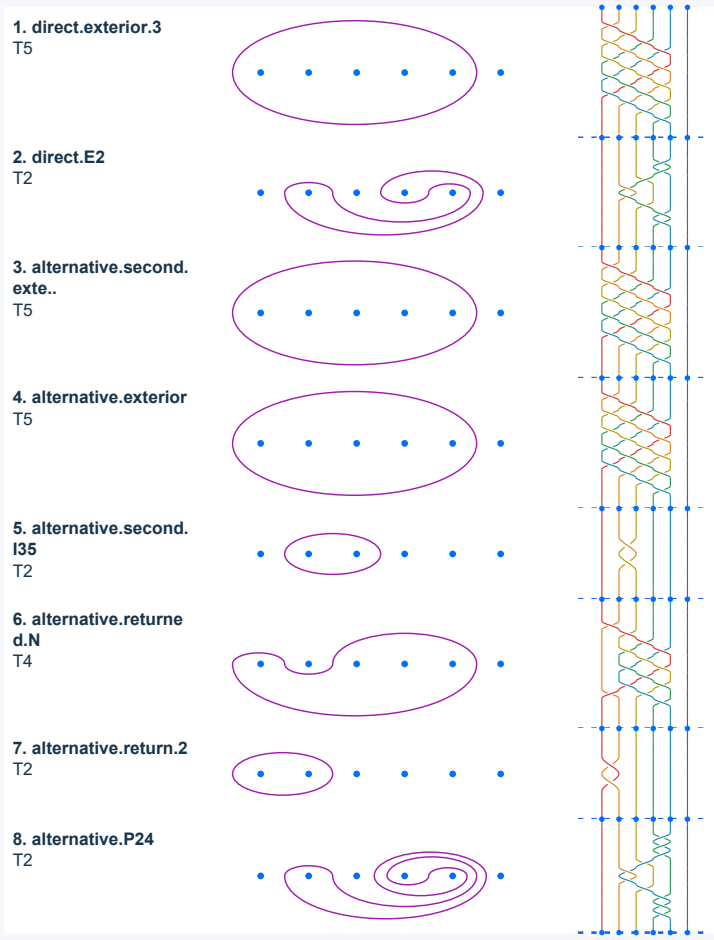
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



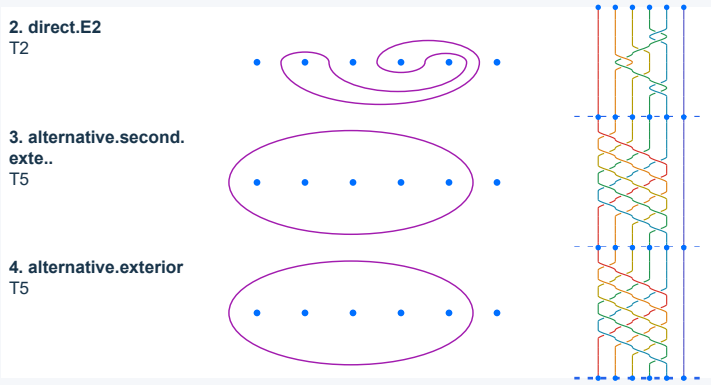
VII - step 087

The extra petal passes DOWN past two central exteriors

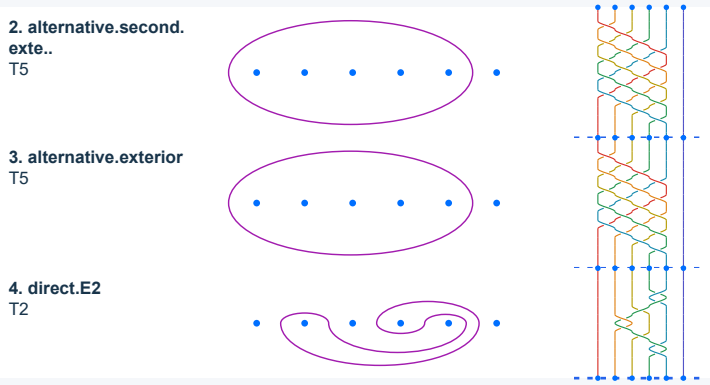
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



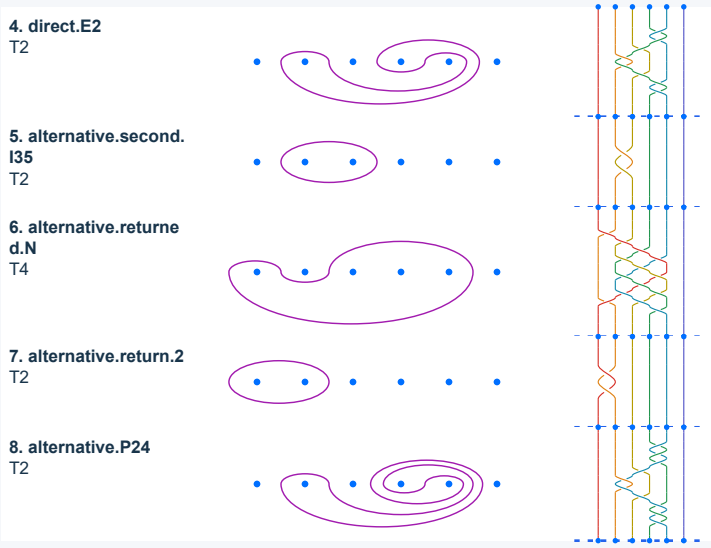
VII - step 088

SECOND actual PC daisy in its reached cyclic order

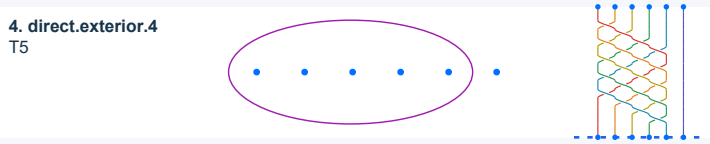
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 0 below. Only the changed interval is shown. Named PC daisy in its checked cyclic order; every input and the full internal conjugation are retained.

Before



After

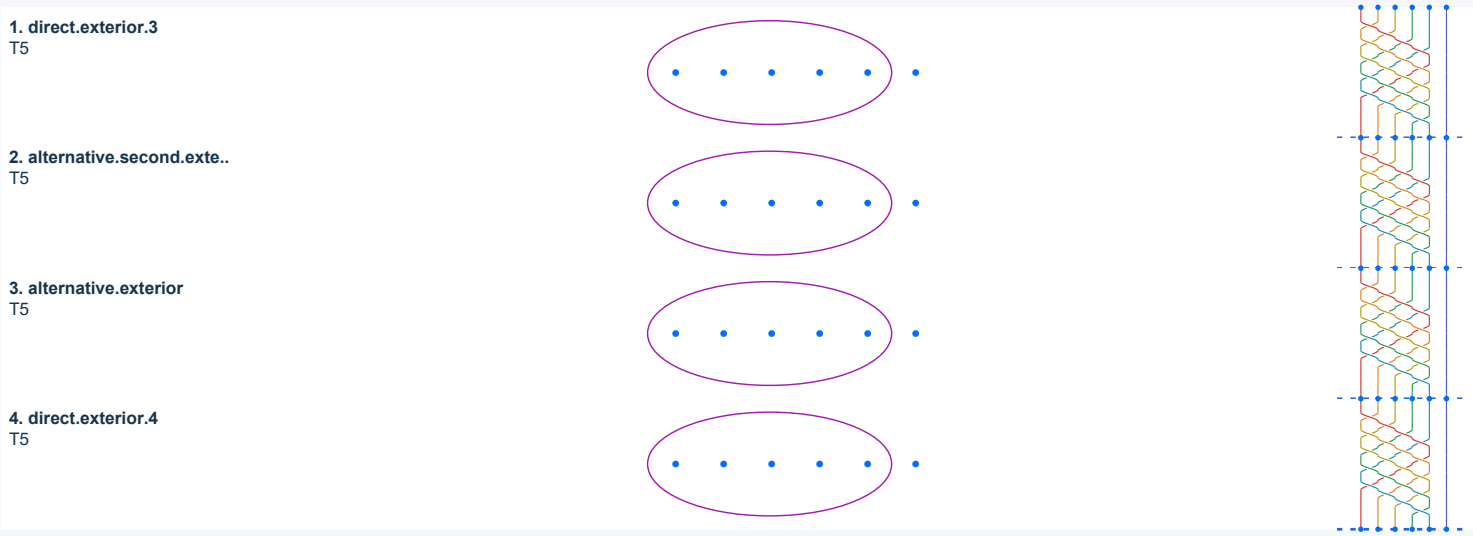


VIII. Four exteriors and the sphere cap

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before Four exteriors and the sphere cap

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.



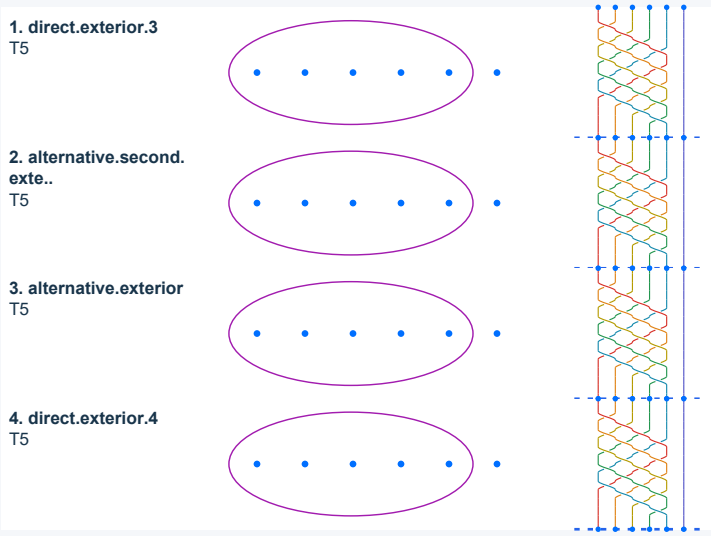
VIII - step 089

Combine the FOUR identical visible exterior twists

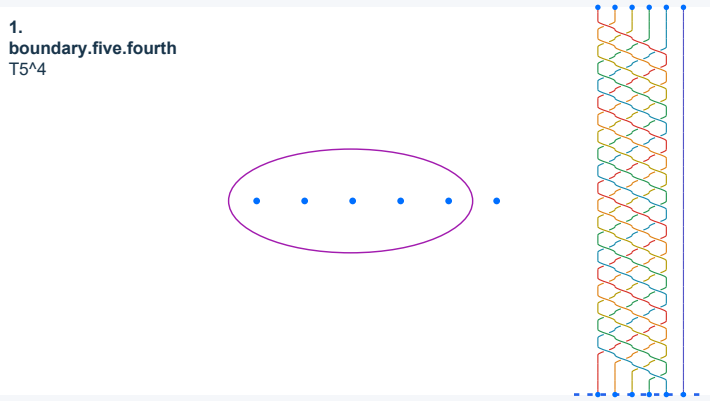
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



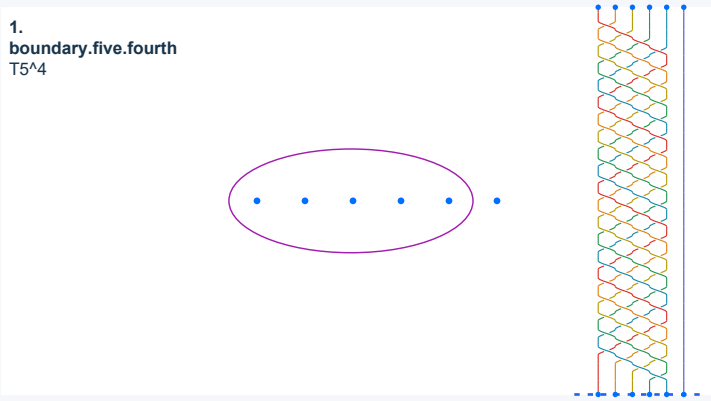
VIII - step 090

SPHERE MOVE: cap the FOURTH boundary power

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

IX. Sphere discrepancies and certificates

The following are ALL FOUR sphere operations in this named route. They are not disk braid equalities. Each discrepancy is written in signed generator indices; the exact sphere certificate and +I lift are retained online.

SPHERE MOVE at step 14

SPHERE LANTERN: absorb the matched pair and the two prefix roots

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (66 signed braid letters):

5 -4 2 3 2 3 2 3 4 -5 -4 -2 3 3 2 -5 -2 -3 4 2 3 2 3 2
3 -4 3 2 5 4 2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1
-1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 3 3 -5 -2 -3 -2

SPHERE MOVE at step 16

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (482 signed braid letters):

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 2 1 -4 2 3 2 3 2
3 2 3 2 3 2 3 4 2 3 2 5 -3 -3 -4 -5 -4 -4 -3 -2 -1 -2 -1 -2
-1 -2 -1 -2 -1 -2 -1 3 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 3 1
1 1 -4 -3 -2 -1 2 3 4 -1 -1 -1 4 4 4 1 1 2 -4 -3 -4 -3 -4 -3
-2 -1 -1 -4 -3 -4 -3 1 2 2 1 -4 -3 -2 -1 2 3 4 -1 -2 -2 -1 3 4
3 -4 -4 -3 -2 -1 -2 -1 -2 -1 3 4 4 1 -3 -2 -3 -2 -4 -3 -2 -3 -2 -3
-2 4 -3 -2 -1 4 -2 3 -4 -3 -3 2 -4 -3 -2 3 4 -2 3 3 4 -3 2 -4
-3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -1 -2 -4 -3 -3 -2 3 3 4 2 1
-2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -1 -1 -4 2 1 1 2 3 -4 -4 -3 -2
-1 -1 -2 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1

SPHERE MOVE at step 17

SPHERE MOVE: cap the SIX framing corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (120 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4

SPHERE MOVE at step 90

SPHERE MOVE: cap the FOURTH boundary power

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (80 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4

Certificates and exact scope

Complete named graphical route:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html?phase=namedfull

Full raw certificates:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/certificate.json

The independent six-point sphere cut comparison has six marked points and five consecutive-point arcs. It is separately jointly certified; the individual factor supports in this book are not joint cut complexes.

The older23-operation closure remains available as a separate chapter. A positive T45 frame bounds the unreordered six-pair rose. An earlier cyclic cut lets the first outer pass upward through its own petals, avoiding the second outer. The second actual daisy is matched in cyclic order, with its full internal common conjugation and four nonabelian petal kernels retained. This saves one operation and two external adjacent checks from91. The new fourteen-operation closure stays within support maximum18; the whole disk tail stays within20. Four actual exterior twists remain visible. The under-twenty complete derivation remains open.

The archived 125-page/177-step PDF is preserved separately. This edition contains the new 90-step named proof, with no action-only boundary-recognition step.

Independent jointly certified six-point sphere cut comparison

